A Split-Decomposition Approach to the Bandelt–Dress Quartet Distance Bound

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Abstract

Let X be a set of N taxa. The quartet distance between two binary leaf-labelled phylogenetic trees on X counts the four-subsets on which the two induced quartet trees differ. Bandelt and Dress conjectured that its maximum is asymptotic to

$$\left(\frac{2}{3}+o(1)\right)\binom{N}{4}.$$

We prove the conjectured upper bound. The argument has two stages. First, we form an auxiliary functional f_1 from squared 2×2 split determinants and ordered-adjacent-edge corrections. A local decomposition uses row atoms R_i and column atoms C_k. Write $a_{i,k}=|R_i\cap C_k|/N$, $c_k=|C_k|/N$, and let $C_{\bar{k}}$ denote the opposite-side column block, with $a_{i,\bar{k}}=|R_i\cap C_{\bar{k}}|/N$ and $c_{\bar{k}}=|C_{\bar{k}}|/N$. Every retained atom is nonempty, and the opposite side of a relevant principal edge is nonempty; hence $c_k, c_{\bar{k}}>0$. The symmetric coefficient

$$r_i^k=\frac{1}{2}\left(\frac{a_{i,k}}{c_k}+\frac{a_{i,\bar{k}}}{c_{\bar{k}}}\right)$$

reduces f_1 to four families of nonnegative quadratic forms. The only nonlocal step in this reduction is a bijective reversal of ordered adjacent-edge bridges: it exchanges the two endpoint blocks while preserving the middle block.

Second, we expand the same functional as a sum over X^4. The resulting antisymmetric kernel takes only the values $-1,0,1$. The 24 orderings of a four-subset contribute 4 if the two trees display the same quartet and -2 otherwise, while all repeated-label tuples contribute $O(N^3)$. Hence writing $\mathrm{Same}(T_1,T_2)$ and $\mathrm{Diff}(T_1,T_2)$ for the numbers of four-subsets on which the displayed quartets agree and differ, respectively,

$$N^4 f_1 = 4\,\mathrm{Same}(T_1,T_2)-2\,\mathrm{Diff}(T_1,T_2)+O(N^3),$$

and $f_1\geq 0$ yields the Bandelt–Dress bound.

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1 Introduction

Let X be a finite set of taxa. A binary leaf-labelled phylogenetic tree on X induces exactly one quartet tree on every four-subset of X. The quartet distance between two such trees is the number of four-subsets on which these induced quartet trees differ. It is a basic combinatorial distance in phylogenetics. We denote this number by $\mathrm{Diff}(T_1,T_2)$ and put $\mathrm{Same}(T_1,T_2):=\binom{|X|}{4}-\mathrm{Diff}(T_1,T_2)$.

We use the standard language of splits. A split is a bipartition $A\mid B$ of X, and it displays $ab\mid cd$ when a,b lie on one side and c,d on the other. Edge splits of a tree form a compatible split

system. This viewpoint is also used in quartet reconstruction, QNet, and weighted quartet systems [2, 3, 4].

For $N = |X|$, put

$$M_N = \max_{T_1,T_2} \mathrm{Diff}(T_1, T_2),$$

where the maximum is over all binary leaf-labelled phylogenetic trees on X. Bandelt and Dress introduced the symmetric quartet distance in their work on reconstructing tree shape from dissimilarity data [1]. Their conjecture states that

$$M_N \le \left(\frac{2}{3} + o(1)\right)\binom{N}{4}.$$

The matching lower bound follows by independently and uniformly labelling two fixed binary tree shapes. Alon, Naves and Sudakov proved the conjectured upper bound for caterpillar trees and the general bound $(0.69 + o(1))\binom{N}{4}$ [5]. A partial-information version was studied in [7].

Our main result is the conjectured upper bound. Equivalently, any two binary leaf-labelled phylogenetic trees share at least

$$\left(\frac{1}{3} - o(1)\right)\binom{N}{4}$$

displayed quartet trees. More precisely, for $|X| = N$ we prove

$$\frac{\mathrm{Diff}(T_1, T_2)}{\binom{N}{4}} \le \frac{2}{3} + O\left(\frac{1}{N}\right).$$

Together with the random-labelling lower bound this gives

$$M_N = \left(\frac{2}{3} + o(1)\right)\binom{N}{4}.$$

The proof is organized around one auxiliary functional. We first describe the four steps of the argument; the definitions and calculations follow in the subsequent sections.

1. Split determinants. Given two splits $P \mid P^c$ and $Q \mid Q^c$, consider their normalized intersection table

	Q	Q^c
P	a	b
P^c	c	d

The determinant square $(ad - bc)^2$ is a squared covariance of centered split indicators. We combine these squares into f_1. Its four terms correspond to using an ordinary edge split or an ordered-adjacent correction in each of the two trees. The coefficients $1, -1/2, -1/2, 1/4$ correct the two orientation multiplicities.

2. Local symmetric decomposition. A caterpillar has a linear order of adjacent intervals; a general tree does not. We replace that order by

$$r_i^k = \frac{1}{2}\left(\frac{a_{i,k}}{c_k} + \frac{a_{i,\bar{k}}}{c_{\bar{k}}}\right).$$

Here i is a row atom, k is a column branch atom, $a_{i,k}$ is their normalized intersection mass, c_k is the mass of the column atom, k^+ is its whole same-side block, and $\bar{k}$ is the opposite-side block. Only

nonempty atoms are retained, and the principal edge is relevant, so $c_k, c_{\bar{k}} > 0$ and both displayed ratios are defined. The two coefficients $1/2$ are algebraic symmetry weights. They give

$$\sum_i r_i^k = 1$$

where the sum ranges over all row atoms of the fixed chart, and the coefficient bound

$$4c_k c_l r_i^k r_j^l \geq \frac{c_k c_l}{c_{\bar{k}} c_{\bar{l}}} a_{i,\bar{k}} a_{j,\bar{l}}.$$

This inequality controls the negative terms left by the completion of squares.

3. Global bridge reversal and positivity. For a principal chart χ (a pair of relevant principal edges, formally defined in Subsection 4.1), denote the four determinant levels by $U_\chi, R_\chi, C_\chi, B_\chi$. After multiplying by four, the expression to be decomposed is

$$4U_\chi - 2R_\chi - 2C_\chi + B_\chi.$$

The row coordinate is decomposed while the column term is held fixed. The remaining column coefficients are grouped by the four-sector partition defined in Subsection 4.4. For an ordered adjacent edge pair, let (K, H, L) denote, respectively, the selected endpoint block, the middle block, and the opposite endpoint block created by deleting the two edges. Bridge reversal gives the different principal chart with blocks (L, H, K). This fixed-point-free involution matches the second endpoint square coefficient in each chart with the first endpoint square coefficient in its reversed chart only after the complete chart sum is taken. The result is

$$f_1 = \sum_\chi \left(\frac{1}{4}\mathcal{S}_{1,\chi} + \frac{1}{2}\mathcal{S}_{2,\chi} + \frac{1}{2}\mathcal{S}_{3,\chi} + \mathcal{S}_{4,\chi} \right).$$

Here $\mathcal{S}_{1,\chi}, \ldots, \mathcal{S}_{4,\chi}$ denote the four chart-indexed square-family polynomials declared explicitly in Subsection 4.5 and derived in Appendix A. The first two families are explicit sums of completed squares. The last two are squares multiplied by a coefficient $\Psi_{ij}^{\chi}(u)$; the displayed coefficient bound proves that this multiplier is nonnegative. Thus $f_1 \geq 0$.

4. Quartet counting. Give every edge term weight 1 and every ordered-adjacent correction weight $-1/2$. Their antisymmetric side-pair functions define a kernel K_T on X^4 satisfying

$$N^4 f_1(T_1, T_2) = \frac{1}{4} \sum_{\boldsymbol{q} \in X^4} K_{T_1}(\boldsymbol{q}) K_{T_2}(\boldsymbol{q}).$$

For either positional pairing, the separating edges form an empty path or a path of signed weight 1. This identifies $K_T = \lambda_T(13 \mid 24) - \lambda_T(14 \mid 23)$ and hence $K_T \in \{-1, 0, 1\}$, even when labels repeat. Exactly $6N^3 - 11N^2 + 6N$ ordered tuples contain a repeated label, and each contributes at most $1/4$ in absolute value. For each four-subset with distinct labels, a direct count of its 24 orderings gives total kernel contribution 4 when the displayed quartets agree and -2 when they differ. Consequently

$$N^4 f_1 = 4\,\mathrm{Same}(T_1, T_2) - 2\,\mathrm{Diff}(T_1, T_2) + O(N^3).$$

Since $f_1 \geq 0$, the desired quartet bound follows.

Sections 2–5 develop these four steps in order. Section 6 explains the scope of the result for general X-trees. Appendix A contains the coefficient calculation proving positivity; Appendix B records the related distance-covariance identity; and Appendix C works through the complete exact-arithmetic calculation for two non-caterpillar binary trees.

2 Preliminaries

We distinguish the general objects needed for the local algebra from the binary leaf-labelled trees to which the quartet-distance theorem applies. Let X be a finite set, $|X| = N$. An *X-tree* is a finite tree T together with a vertex labelling

$$v \in V(T) \longmapsto X_v \subseteq X$$

such that the nonempty sets X_v form a partition of X and $X_v \neq \varnothing$ whenever v has degree at most two. Thus a taxon may be attached to an internal vertex, and a vertex may carry more than one taxon. The condition on vertices of degree at most two is the usual reducedness condition; in particular, every leaf carries a taxon, so deleting any edge produces two nonempty taxon sets.

A *leaf-labelled phylogenetic tree on X* is the special case in which each nonempty X_v is a singleton and the labelled vertices are exactly the leaves. In the main theorem the tree has no degree-two vertices and is *binary*, meaning that every internal vertex has degree three. The coefficient calculation is written for the more general X-tree notation, but the quartet-counting conclusion is used only in this binary leaf-labelled setting.

For a leaf-labelled tree and $Y \subseteq X$, let $T|Y$ denote the tree obtained from the minimal subtree of T connecting the leaves labelled by Y by suppressing all unlabelled degree-two vertices. If $|Y| = 4$ and T is binary, then $T|Y$ is a quartet tree. We write $ab \mid cd$ for the quartet tree in which the path from a to b is edge-disjoint from the path from c to d.

A *split* of X is a bipartition $A \mid B$ with $A, B \neq \varnothing$. We usually write $B = A^c$. Every edge e of an X-tree induces a split $\sigma_e = A_e \mid A_e^c$ by deleting e: the two components of $T - e$ carry the taxa in A_e and A_e^c, respectively. In the leaf-labelled case this is the usual edge split. The set

$$\Sigma(T) = \{\sigma_e : e \in E(T)\}$$

is the split system associated with T. We also use the quartet-relevant edge set

$$E_2(T) = \{e \in E(T) : |A_e| \geq 2 \text{ and } |A_e^c| \geq 2\},$$

and the corresponding quartet-relevant split system

$$\Sigma_2(T) = \{\sigma_e : e \in E_2(T)\}.$$

Only splits in $\Sigma_2(T)$ can display a resolved quartet tree on four distinct taxa. A split $A \mid B$ *displays* a quartet tree $ab \mid cd$ if $a, b \in A$ and $c, d \in B$, or vice versa.

Let $\mathcal{Q}(X)$ be the set of quartet trees with leaf set a four-subset of X. A binary leaf-labelled phylogenetic tree T displays the quartet set

$$\mathcal{Q}(T) = \{T|Y : Y \in \binom{X}{4}\} \subseteq \mathcal{Q}(X).$$

For two binary leaf-labelled phylogenetic trees T_1, T_2, define

$$\mathrm{Same}(T_1, T_2) = |\mathcal{Q}(T_1) \cap \mathcal{Q}(T_2)|$$

and

$$\mathrm{Diff}(T_1, T_2) = \binom{N}{4} - \mathrm{Same}(T_1, T_2).$$

Equivalently, $\mathrm{Diff}(T_1, T_2)$ is the number of four-subsets $Y \in \binom{X}{4}$ for which $T_1|Y \neq T_2|Y$.

3 Split determinants and the auxiliary functional

This section constructs f_1 in three steps. We first associate a nonnegative determinant square with two splits, then extend the same formula to the side-pairs obtained from adjacent edges, and finally combine the four ordinary/corrected choices in the two tree coordinates.

3.1 One pair of splits

Let

$$\sigma = P \mid P^c, \qquad \tau = Q \mid Q^c$$

be two splits of X. Define the normalized 2×2 table

	Q	Q^c
P	a	b
P^c	c	d

where

$$a = \frac{|P \cap Q|}{N}, \quad b = \frac{|P \cap Q^c|}{N}, \quad c = \frac{|P^c \cap Q|}{N}, \quad d = \frac{|P^c \cap Q^c|}{N}.$$

Set

$$D(\sigma, \tau) = (ad - bc)^2.$$

This determinant square is the leading normalized contribution of the pair of splits to the expression $4\,\mathrm{Same} - 2\,\mathrm{Diff}$.

3.2 Indicator form of the split determinant

We use the uniform probability measure on X. For $A \subseteq X$, let

$$\mathbf{1}_A(x) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A, \end{cases} \qquad \mu(A) = \mathbb{E}\mathbf{1}_A = \frac{|A|}{N},$$

and put

$$z_A = \mathbf{1}_A - \mu(A).$$

For a logical condition P, we also use $\mathbf{1}_{\mathsf{P}} \in \{0, 1\}$ for its truth indicator. This scalar notation is used later to turn endpoint or residual cases on and off. For a split $\sigma = A \mid A^c$, replacing A by A^c changes z_A to $-z_A$. Hence the square of any inner product involving z_A is independent of the chosen side of the split.

For $\sigma = P \mid P^c$ and $\tau = Q \mid Q^c$, a direct calculation gives

$$\mathbb{E} z_P z_Q = \mathbb{E}(\mathbf{1}_P - a - b)(\mathbf{1}_Q - a - c) = a - (a + b)(a + c).$$

Using $a + b + c + d = 1$, this becomes

$$a - (a + b)(a + c) = a(1 - a - b - c) - bc = ad - bc.$$

Therefore

$$D(\sigma, \tau) = (\mathbb{E} z_P z_Q)^2 .$$

The expression is unchanged if one reverses either split, because this only changes the sign of the corresponding centered indicator.

Thus the local determinant is a squared covariance. The role of the adjacent-pair terms is not to change this basic quantity, but to combine it across neighboring edges in a way that both admits a nonnegative decomposition and counts each displayed quartet with the desired weight.

3.3 Adjacent-pair correction operator

Let $E(\mathcal{T})$ denote the edge set of a tree $\mathcal{T}$. For $e, e' \in E(\mathcal{T})$, write $e \sim e'$ if the two edges are adjacent. Let

$$\overrightarrow{A}_2(\mathcal{T}) = \{(e, e') \in E_2(\mathcal{T})^2 : e \sim e'\}$$

be the set of ordered adjacent pairs of quartet-relevant edges.

We now define the adjacent correction. A *side-pair* is an ordered pair (A, B) of disjoint nonempty subsets of X; unlike a split, it need not cover X. For two side-pairs (A, B) and (C, D) define

$$T[(A, B), (C, D)] := \left(\frac{|A \cap C|}{N} \frac{|B \cap D|}{N} - \frac{|A \cap D|}{N} \frac{|B \cap C|}{N} \right)^2.$$

Reversing either side-pair only changes the sign of the determinant before squaring. If $A \cup B = X$ and $C \cup D = X$, this is the ordinary split determinant $D(A \mid B, C \mid D)$.

Let $(e, e') \in \overrightarrow{A}_2(\mathcal{T})$, and let v be the common endpoint of e and e'. Removing the two adjacent edges gives three taxa blocks. Let $K_{e,e'}$ be the component beyond e' away from v, and let $L_{e,e'}$ be the component beyond e away from v. The adjacent-pair correction $C_{e,e'}$ is the side-pair

$$C_{e,e'} := (K_{e,e'}, L_{e,e'}).$$

Thus, for a split $\tau = Q \mid Q^c$ in a second tree,

$$T[C_{e,e'}, \tau] := T[(K_{e,e'}, L_{e,e'}), (Q, Q^c)],$$

and the definitions of $T[\sigma, C_{f,f'}]$ and $T[C_{e,e'}, C_{f,f'}]$ are analogous. In the local cluster notation of Appendix A, this side-pair determinant is exactly the neighboring contrast squared in equation (A.18); equations (A.20)–(A.25) then expand and complete its mixed coefficient.

Indeed, let $C_{e,e'} = (K, L)$ and let $\tau = Q \mid Q^c$. Put

$$A = \frac{|K \cap Q|}{N}, \qquad C = \frac{|L \cap Q|}{N}, \qquad c_K = \frac{|K|}{N}, \qquad c_L = \frac{|L|}{N}.$$

For the centered split-side indicator $Z = z_Q$, one has

$$x_{Z,K} = A - c_K \mu(Q), \qquad x_{Z,L} = C - c_L \mu(Q).$$

Therefore

$$\begin{aligned} T[(K, L), (Q, Q^c)] &= \big(A(c_L - C) - (c_K - A)C\big)^2 \\ &= (c_L A - c_K C)^2 = (c_K x_{Z,L} - c_L x_{Z,K})^2. \end{aligned}$$

Thus the side-pair determinant is precisely the local neighboring contrast used in Appendix A.

For four distinct taxa, the adjacent side-pair has the expected quartet indicator $I^{\pi}_{C_{e,e'}} = I^{\pi}_e I^{\pi}_{e'}$. The counting argument requires the positional version of this identity on all of X^4, including repeated labels. That version, with every indicator defined explicitly, is proved in Lemma 5.1.

For an edge $e \in E_2(\mathcal{T})$, define the one-sided adjacent correction by

$$T[\sigma^+_e, \tau] := \sum_{e' : (e,e') \in \overrightarrow{A}_2(\mathcal{T})} T[C_{e,e'}, \tau].$$

For $e \in E_2(T_1)$ and $f \in E_2(T_2)$, the remaining plus notations mean formal sums over indexed occurrences:

$$T[\sigma, \tau_f^+] := \sum_{f' : (f,f') \in \overrightarrow{A}_2(T_2)} T[\sigma, C_{f,f'}],$$

$$T[\sigma_e^+, \tau_f^+] := \sum_{e' : (e,e') \in \overrightarrow{A}_2(T_1)} \sum_{f' : (f,f') \in \overrightarrow{A}_2(T_2)} T[C_{e,e'}, C_{f,f'}].$$

Thus every determinant is squared before the indexed occurrences are summed; no linearity of the squared determinant itself is being asserted. This is the convention used throughout the paper for neighboring corrections.

Definition 3.1 (Auxiliary functional)**.** *Define the auxiliary functional by summing only over quartet-relevant edge splits:*

$$f_1(T_1, T_2) = \sum_{e \in E_2(T_1)} \sum_{f \in E_2(T_2)} \Big(T[\sigma_e, \tau_f] - \frac{1}{2} T[\sigma_e^+, \tau_f] - \frac{1}{2} T[\sigma_e, \tau_f^+] + \frac{1}{4} T[\sigma_e^+, \tau_f^+] \Big),$$

where σ_e and τ_f are the splits induced by e and f, and where adjacent-pair corrections are taken over $\overrightarrow{A}_2(T_1)$ and $\overrightarrow{A}_2(T_2)$. Splits with a side of size one—and hence pendant splits in the binary leaf-labelled setting—are excluded because they cannot display a quartet tree on four distinct taxa.

The coefficients $1/2$ and $1/4$ are purely combinatorial. Each unordered adjacent pair has two orientations, which gives $1/2$ for a correction in one tree and $1/4$ when both tree coordinates are corrected.

3.4 Reduction to the E_2-core

The local argument depends only on quartet-relevant edges and their adjacent side-pairs. The following reduction removes the remaining parts of the trees without changing these data.

Lemma 3.2 (E_2-core contraction)**.** *Let T be an X-tree and suppose that $E_2(T) \neq \varnothing$. Let $K_2(T)$ be the subgraph whose edges are those in $E_2(T)$ and whose vertices are their endpoints. Then $K_2(T)$ is a connected subtree. Contract every component of $T - V(K_2(T))$, together with its unique incident edge to $K_2(T)$, to its attachment vertex, and transfer all taxa in that component to the attachment vertex. Denote the resulting X-tree by T°. Then:*

(i) *under the natural identification, $E(T^\circ) = E_2(T)$, and the adjacency relations among these edges are unchanged;*

(ii) *every split induced by an edge $e \in E_2(T)$ is unchanged as a split of X;*

(iii) *every adjacent side-pair $C_{e,e'}$, with $(e, e') \in \overrightarrow{A}_2(T)$, is unchanged;*

(iv) *replacing either argument of f_1 by its E_2-core contraction does not change f_1; in particular,*

$$f_1(T_1, T_2) = f_1(T_1^\circ, T_2^\circ)$$

whenever both contractions are defined.

In the chart notation formalized in Subsection 4.1, the endpoint classes a_e and b_e index the adjacent E_2-directions. The atom belonging to one such direction contains every taxon lying beyond that adjacent core edge, including taxa transferred to more distant core vertices in the same branch. Only taxa located at an endpoint of e after contraction (whether already there or transferred there) and lying beyond no adjacent E_2-direction enter the residual vertex classes c_e and d_e.

Proof. Let $e, g \in E_2(T)$, and let h be an edge on the path from e to g. One component of $T - h$ contains the side of e directed away from h and therefore contains at least two taxa. The other contains the corresponding side of g and also contains at least two taxa. Hence $h \in E_2(T)$, so the edges in $E_2(T)$ form the connected subtree $K_2(T)$.

Every component of $T - V(K_2(T))$ has a unique attachment vertex in $K_2(T)$; otherwise the path between two attachment vertices would contain an edge outside $E_2(T)$. Contracting such a component moves taxa only within a single component of $T - e$ for each $e \in E_2(T)$. Thus the two taxon sets of every E_2-split are unchanged, proving (ii). By construction every edge outside $K_2(T)$ is contracted, while no edge of $K_2(T)$ and no two distinct core vertices are identified. Hence $E(T^\circ) = E_2(T)$, and two surviving edges are adjacent in T° exactly when they were adjacent in T, proving (i).

The contracted tree is again reduced. Indeed, let v be a vertex whose degree in T° is at most two. If no edge incident with v was contracted, then its degree in T was already at most two, so v carried a taxon by reducedness of T. Otherwise at least one component attached to v was contracted onto v. Every such component contains a taxon: an edge joining it to the core separates a nonempty taxon set from the rest of X, as every edge of an X-tree does. Those taxa are transferred to v. Hence every vertex of degree at most two in T° carries a nonempty label set, as required.

For adjacent core edges e, e', each contracted component was already contained in one of the three taxon blocks obtained by deleting e and e'. The side-pair $C_{e,e'}$ is therefore unchanged, proving (iii). Since the definition of f_1 uses only the E_2-splits and these ordered adjacent side-pairs, (iv) follows. The final statement follows from the construction of the contracted chart. □

If $E_2(T_1) = \varnothing$ or $E_2(T_2) = \varnothing$, every term in the definition of $f_1(T_1, T_2)$ is absent and hence $f_1(T_1, T_2) = 0$. The nonnegativity statement is then immediate. In the remainder of the local proof we therefore assume that both E_2-cores are nonempty and work with T_1° and T_2°.

4 Symmetric decomposition and local nonnegativity

The goal of this section is to reduce $f_1 \geq 0$ to a finite coefficient inequality. Throughout this section and Appendix A, a *relevant edge* means an edge in the quartet-relevant set E_2 of the contracted core under discussion. A pair of relevant edges (e, f), one from each tree, determines a *principal chart*. Each edge is surrounded by endpoint atoms, which index adjacent corrections, and possible residual atoms, which do not. We use the following convention throughout the local calculation:

symbol	coordinate	meaning
e, f	tree edges	principal edges of the chart
i, j	row copy	atoms around e
k, l, u	column copy	atoms around f
a, b	either copy	endpoint atoms indexing adjacent edges
c, d	either copy	residual atoms at the two endpoints

The construction below first defines these objects, then proves the one coefficient inequality needed in Appendix A.

4.1 Principal charts

We use the E_2-core contractions from Lemma 3.2. Let T° be such a contraction and let $e = \{v_-, v_+\} \in E_2(T^\circ)$. Orient e from v_- to v_+ only to name its two sides, and let X_e^- and X_e^+ be the taxon sets in the two components of $T^\circ - e$. If $g \in E_2(T^\circ) \setminus \{e\}$ is incident with v_-, let $A_e(g)$ be the taxa in the component beyond g away from v_-. Define $A_e(g)$ analogously for edges incident with v_+. The residual vertex blocks are

$$V_e^- := X_e^- \setminus \bigsqcup_{\substack{g \in E_2(T^\circ)\setminus\{e\},\ g\sim e \\ g \ni v_-}} A_e(g), \qquad V_e^+ := X_e^+ \setminus \bigsqcup_{\substack{g \in E_2(T^\circ)\setminus\{e\},\ g\sim e \\ g \ni v_+}} A_e(g).$$

After empty residual blocks are omitted, these sets form the atom partition

$$\Pi_e = \{A_e(g) : g \in E_2(T^\circ),\ g \sim e\} \sqcup \{V_e^-, V_e^+\}_{\neq \varnothing}.$$

The atoms $A_e(g)$ at v_- and v_+ form the endpoint classes a_e and b_e, respectively; the possible blocks V_e^- and V_e^+ form c_e and d_e. Thus endpoint atoms are indexed by ordered adjacent pairs (e, g), whereas residual atoms do not index adjacent corrections.

Let Ω_e index the nonempty atoms of Π_e; thus $a_e, b_e, c_e, d_e \subseteq \Omega_e$ are the four atom-index classes just described. When this partition belongs to the first tree, write Ω_e^{r} for its row-labelled copy, and when it belongs to the second tree, write Ω_e^{c} for its column-labelled copy. The superscripts distinguish chart copies and do not alter the underlying atom partition.

For two contracted trees, a principal chart is simply a pair

$$\chi = (e, f) \in E_2(T_1^\circ) \times E_2(T_2^\circ).$$

Write

$$\Omega_{\mathrm{r}}^\chi := \Omega_e^{\mathrm{r}}, \qquad \Omega_{\mathrm{c}}^\chi := \Omega_f^{\mathrm{c}},$$

and write R_i^χ and C_k^χ for the corresponding row and column taxon atoms. For $P \subseteq \Omega_{\mathrm{r}}^\chi$ and $Q \subseteq \Omega_{\mathrm{c}}^\chi$, put

$$R_P^\chi := \bigsqcup_{i \in P} R_i^\chi, \qquad C_Q^\chi := \bigsqcup_{k \in Q} C_k^\chi.$$

For the column copy, set $a^\chi := a_f$, $b^\chi := b_f$, $c^\chi := c_f$, and $d^\chi := d_f$. For every $k \in \Omega_{\mathrm{c}}^\chi$, define its same-side block, same-side aggregate, and opposite-side block by

$$S_\chi(k) := \begin{cases} a^\chi \cup c^\chi, & k \in a^\chi \cup c^\chi, \\ b^\chi \cup d^\chi, & k \in b^\chi \cup d^\chi, \end{cases} \qquad k^{+,\chi} := S_\chi(k), \qquad \bar{k}_\chi := \Omega_{\mathrm{c}}^\chi \setminus S_\chi(k).$$

When the chart is fixed, we suppress the chart decoration. If k is an endpoint atom corresponding to an edge g adjacent to f, the ordered bridge $\alpha = (f, g)$ has three taxon blocks

$$K_\alpha = C_k^\chi, \qquad H_\alpha = C_{k^{+,\chi}\setminus\{k\}}^\chi, \qquad L_\alpha = C_{\bar{k}_\chi}^\chi.$$

Its reversal $\rho\alpha = (g, f)$ belongs to the different principal chart $\rho\chi = (e, g)$ and has selected, middle, and opposite blocks $(L_\alpha, H_\alpha, K_\alpha)$. Thus reversal is a fixed-point-free involution of the complete ordered endpoint-chart range. In particular, it does not mean replacing k by the aggregate $\bar{k}$ inside a single chart.

Every sum over $a_e \cup b_e$ therefore ranges over atom indices associated with adjacent directions in $E_2(T^\circ)$. An endpoint atom may contain taxa transferred to any more distant core vertex in its branch. The residual blocks $c_e \cup d_e$ contain only the taxa located at the two endpoints of e that lie beyond no such adjacent direction. These residual classes may be nonempty even when the original tree is binary and leaf-labelled.

4.2 Local notation

Fix a principal chart $\chi = (e, f)$ and suppress χ from the notation. Let the local clusters induced by the split of e be indexed by i, j, and those induced by the split of f be indexed by k, l. Write

$$r_i = \frac{|R_i|}{N}, \qquad c_k = \frac{|C_k|}{N}, \qquad a_{i,k} = \frac{|R_i \cap C_k|}{N}.$$

Let

$$x_{i,k} = a_{i,k} - r_i c_k.$$

Then

$$\sum_i r_i = 1, \qquad \sum_k c_k = 1,$$

$$\sum_i a_{i,k} = c_k, \qquad \sum_k a_{i,k} = r_i,$$

and hence

$$\sum_i x_{i,k} = 0, \qquad \sum_k x_{i,k} = 0.$$

For a local column index k, the blocks $k^+ = S_\chi(k)$ and $\bar{k} = \Omega_c^\chi \setminus S_\chi(k)$ are the same-side and opposite-side blocks defined in the preceding subsection. More generally, for any block U of local column indices, use the additive notation

$$c_U = \sum_{l \in U} c_l, \qquad a_{i,U} = \sum_{l \in U} a_{i,l}, \qquad x_{i,U} = \sum_{l \in U} x_{i,l}.$$

These are respectively the mass, mixed mass, and centered mass of the union of the column clusters indexed by U; in particular this convention applies to $U = \bar{k}$. We assume $c_k > 0$ and $c_{\bar{k}} > 0$; zero-mass indices contribute nothing and may be omitted.

4.3 The side-reversal symmetric decomposition

For each k, set

$$r_i^k = \frac{1}{2}\left(\frac{a_{i,k}}{c_k} + \frac{a_{i,\bar{k}}}{c_{\bar{k}}}\right).$$

This is the equal algebraic average associated with the taxon side-pair $(C_k, C_{\bar{k}})$; it is not a mass-weighted average over the two blocks. When k is an endpoint atom, exchanging the two taxon blocks is realized by the global bridge-reversal reindexing to the reversed principal chart. For a residual atom the formula is only a same-side/opposite-side algebraic average; no reversed endpoint chart is asserted. For each fixed k,

$$\sum_i r_i^k = \frac{1}{2}\left(\frac{\sum_i a_{i,k}}{c_k} + \frac{\sum_i a_{i,\bar{k}}}{c_{\bar{k}}}\right) = \frac{1}{2}\left(\frac{c_k}{c_k} + \frac{c_{\bar{k}}}{c_{\bar{k}}}\right) = 1.$$

Lemma 4.1 (Domination generated by the decomposition)**.** *For any i, j, k, l,*

$$4c_k c_l r_i^k r_j^l = a_{i,k}a_{j,l} + \frac{c_l}{c_{\bar{l}}}a_{i,k}a_{j,\bar{l}} + \frac{c_k}{c_{\bar{k}}}a_{i,\bar{k}}a_{j,l} + \frac{c_k c_l}{c_{\bar{k}} c_{\bar{l}}}a_{i,\bar{k}}a_{j,\bar{l}}.$$

In particular,

$$4c_k c_l r_i^k r_j^l \geq \frac{c_k c_l}{c_{\bar{k}} c_{\bar{l}}}a_{i,\bar{k}}a_{j,\bar{l}}.$$

Proof. Expanding the definition gives

$$\begin{aligned} 4c_k c_l r_i^k r_j^l &= c_k c_l \left(\frac{a_{i,k}}{c_k} + \frac{a_{i,\bar{k}}}{c_{\bar{k}}} \right) \left(\frac{a_{j,l}}{c_l} + \frac{a_{j,\bar{l}}}{c_{\bar{l}}} \right) \\ &= a_{i,k}a_{j,l} + \frac{c_l}{c_{\bar{l}}} a_{i,k}a_{j,\bar{l}} + \frac{c_k}{c_{\bar{k}}} a_{i,\bar{k}}a_{j,l} + \frac{c_k c_l}{c_{\bar{k}} c_{\bar{l}}} a_{i,\bar{k}}a_{j,\bar{l}}. \end{aligned}$$

All terms on the right are nonnegative. □

4.4 Index sets for the local configuration

We now fix the local four-sector data used in the coefficient calculation. The proof uses only the following finite combinatorial structure. There is a finite index set Ω together with a disjoint decomposition

$$\Omega = a \sqcup b \sqcup c \sqcup d.$$

Here a, b, c, d are four *sets of atom indices*; they are not the scalar entries denoted by the same letters in the earlier 2×2 contingency table. In a principal chart $\chi = (e, f)$ this generic partition is instantiated separately in the two coordinates:

$$\begin{aligned} (a_{\mathrm{r}}^{\chi}, b_{\mathrm{r}}^{\chi}, c_{\mathrm{r}}^{\chi}, d_{\mathrm{r}}^{\chi}) &= (a_e, b_e, c_e, d_e), \\ (a_{\mathrm{c}}^{\chi}, b_{\mathrm{c}}^{\chi}, c_{\mathrm{c}}^{\chi}, d_{\mathrm{c}}^{\chi}) &= (a_f, b_f, c_f, d_f). \end{aligned}$$

The subscripts r and c are mandatory whenever both copies occur in one calculation; the undecorated letters are used only for one fixed copy. This decomposition determines two same-side blocks,

$$S_L := a \cup c, \qquad S_R := b \cup d.$$

In the local X-tree interpretation, the classes a and b index ordinary incident branch classes at the two endpoints of the central local edge, while c and d index taxa blocks carried by the two endpoint vertices themselves. No particular cardinalities are assumed; the argument below uses only the partition of Ω and the same-side relation determined by S_L and S_R.

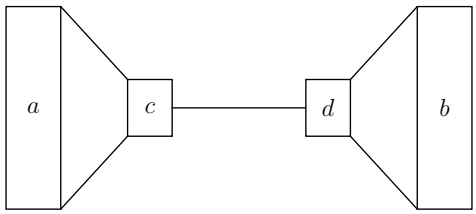


Figure 1: The local four-sector configuration used to define the index sets. From left to right the four displayed blocks are a, c, d, b; hence the two same-side blocks are $a \cup c$ and $b \cup d$.

With the two same-side blocks shown in Figure 1, let $S(k)$ denote the same-side block containing $k \in \Omega$:

$$S(k) = \begin{cases} S_L, & k \in S_L, \\ S_R, & k \in S_R. \end{cases}$$

We set

$$k^+ := S(k), \qquad \bar{k} := \Omega \setminus S(k).$$

Thus k^+ is the whole same-side block containing k, whereas $\bar{k}$ is the opposite-side block. In particular, $\bar{k}$ is a block of indices, not a second singleton index.

4.5 The exact four-class decomposition

For a principal chart $\chi = (e, f)$, write

$$U_\chi = T[\sigma_e, \tau_f], \qquad R_\chi = T[\sigma_e^+, \tau_f], \qquad C_\chi = T[\sigma_e, \tau_f^+], \qquad B_\chi = T[\sigma_e^+, \tau_f^+].$$

Before declaring the four square families, define their row relation sets. Put

$$\mathcal{E}_{\mathrm{r}}^\chi := a_{\mathrm{r}}^\chi \cup b_{\mathrm{r}}^\chi, \qquad \Delta_{\mathcal{E}_{\mathrm{r}}^\chi} := \{(i,i) : i \in \mathcal{E}_{\mathrm{r}}^\chi\},$$

$$D_\chi := \{(i,j) \in (\Omega_{\mathrm{r}}^\chi)^2 : i \in \mathcal{E}_{\mathrm{r}}^\chi,\ j \in \bar{i}\}, \qquad I_\chi := (\Omega_{\mathrm{r}}^\chi)^2 \setminus \big(D_\chi \sqcup \Delta_{\mathcal{E}_{\mathrm{r}}^\chi}\big).$$

Here $\bar{i}$ is the opposite-side row block defined by the row copy of the four-sector partition above. Appendix A performs the coefficient calculation for the global sum of $4U_\chi - 2R_\chi - 2C_\chi + B_\chi$. Its output consists of four quadratic forms $\mathcal{S}_{1,\chi}, \dots, \mathcal{S}_{4,\chi}$. For a linear reading of the main proof, we declare those forms here rather than leaving their notation implicit. The symbols $\mathcal{P}_{ij,k}^\chi$ and $\Psi_{ij}^\chi(u)$ mean, by definition, the explicit chart-local completed-square polynomial and the explicit six-row coefficient displayed in (24) and (A.30a), respectively. With every mass and aggregate computed in the displayed chart, put

$$\begin{aligned}
\mathcal{S}_{2,\chi} &:= \sum_{(i,j)\in I_\chi} \sum_{k \in a^\chi \cup b^\chi} \mathcal{P}_{ij,k}^\chi(x_{\bar{i},\bar{k}}, x_{\bar{i},k}),\\
\mathcal{S}_{1,\chi} &:= \sum_{(i,j)\in D_\chi} \sum_{k \in a^\chi \cup b^\chi} \mathcal{P}_{ij,k}^\chi(x_{\bar{i},\bar{k}} + x_{i,\bar{k}}, x_{\bar{i},k} + x_{i,k}),\\
\mathcal{S}_{4,\chi} &:= \frac{1}{4} \sum_{(i,j)\in I_\chi} \sum_{u \in \Omega_{\mathrm{c}}^\chi} \Psi_{ij}^\chi(u) x_{\bar{i},\bar{u}}^2,\\
\mathcal{S}_{3,\chi} &:= \frac{1}{4} \sum_{(i,j)\in D_\chi} \sum_{u \in \Omega_{\mathrm{c}}^\chi} \Psi_{ij}^\chi(u)(x_{\bar{i},\bar{u}} + x_{i,\bar{u}})^2.
\end{aligned}$$

Thus $\mathcal{S}_1, \mathcal{S}_2$ are completed-square families and $\mathcal{S}_3, \mathcal{S}_4$ are coefficient-weighted square families. The same definitions are repeated beside their derivation in (25)–(28). No fixed-chart identity is asserted below: the exact equality is obtained only after the complete principal-chart sum.

Square-face convention. Whenever a weighted completed square is written with its variables in the order X, Y, we use

$$\gamma(\lambda X + \mu Y)^2 = \underbrace{\gamma\lambda^2 X^2}_{\text{first square face}} + \underbrace{2\gamma\lambda\mu XY}_{\text{mixed face}} + \underbrace{\gamma\mu^2 Y^2}_{\text{second square face}}.$$

Each "face" includes its displayed scalar coefficient and monomial, not merely the bare monomial. The abbreviated terms *first face* and *second face* always mean these X^2- and Y^2-monomial contributions. They do not refer to the first and second trees. If the square summand is indexed by an endpoint atom k, its three displayed contributions are called its *endpoint-indexed first square face*, *endpoint-indexed mixed face*, and *endpoint-indexed second square face*. The generic phrase *endpoint-indexed monomial face* may refer to any of these three. We reserve *endpoint square face* for the first or second square face; it never includes the mixed face. Here "endpoint-indexed" refers to the source index k, not to a claim that every displayed block variable is a singleton endpoint block in the current orientation. Under bridge reversal the second endpoint square face in one chart

is reindexed as the first endpoint square face in the reversed chart, up to an irrelevant common sign in the underlying linear form.

The identity below is deliberately stated after summing over all principal charts. The second square face of a completed square in one ordered bridge chart is identified with the corresponding first square face only in the reversed chart, so there is generally no corresponding identity for a single fixed chart.

Lemma 4.2 (Exact four-class decomposition)**.** *For any two contracted E_2-core trees,*

$$f_1(T_1, T_2) = \sum_{\chi} \left(\frac{1}{4}\mathcal{S}_{1,\chi} + \frac{1}{2}\mathcal{S}_{2,\chi} + \frac{1}{2}\mathcal{S}_{3,\chi} + \mathcal{S}_{4,\chi} \right). \tag{1}$$

Proof. The formal proof in Appendix A is the occurrence-wise settlement of Lemmas A.4 and A.5. The earlier residual calculation through (A.15a), its row–column transpose, and (A.15b) independently derives the same seven-source numerator and records its algebraic provenance; it is not a second global source-settlement argument. In the formal settlement, the terms are first regrouped as $2(2U - R) - (2C - B)$ and the row coordinate is decomposed with each column occurrence fixed. Next, (A.22)–(A.25) complete the mixed terms, while bridge reversal reindexes each second endpoint square face as the corresponding first endpoint square face in the reversed chart inside the global chart sum. Finally, the block identity (A.30b) computes the remaining coefficient. Lemma A.5 gives

$$\sum_{\chi}(4U_\chi - 2R_\chi - 2C_\chi + B_\chi) = \sum_{\chi}(\mathcal{S}_{1,\chi} + 2\mathcal{S}_{2,\chi} + 2\mathcal{S}_{3,\chi} + 4\mathcal{S}_{4,\chi}).$$

Dividing by four gives (1). □

4.6 Domination and local nonnegativity

Proposition 4.3 (Global nonnegativity of the auxiliary functional)**.** *For any two binary leaf-labelled phylogenetic trees on the same taxon set,*

$$f_1(T_1, T_2) \geq 0.$$

Proof. If either $E_2(T_1)$ or $E_2(T_2)$ is empty, the defining sum for f_1 is empty and the conclusion is immediate. Otherwise replace each tree by its contracted E_2-core. Lemma 3.2 shows that this does not change f_1, and Lemma 4.2 now applies to the two nonempty cores. It writes f_1 as the positive linear combination in (1). The forms $\mathcal{S}_{1,\chi}$ and $\mathcal{S}_{2,\chi}$ are sums of squares. In the other two forms, the full coefficient $\Psi_{ij}^{\chi}(u)$ is nonnegative by the explicit endpoint/residual calculation (A.32): for an endpoint target its $l = u$ positive term cancels the endpoint-diagonal subtraction, whereas for a residual target that subtraction is absent. The remaining same-side subtraction is controlled term by term by Lemma 4.1, and all other terms are nonnegative. Thus each summand on the right-hand side of (1) is nonnegative. □

5 Counting identity and quartet consequence

The preceding sections prove $f_1(T_1, T_2) \geq 0$. We now identify what this functional counts. The argument has a short chain of implications: side-pair indicators multiply along adjacent edges; the four levels of f_1 therefore combine into one signed edge-path weight; this weight defines a kernel on

X^4; and the kernel is then evaluated on the 24 orderings of each four-subset. The same pointwise kernel bound also controls tuples with repeated labels.

Let

$$\mathfrak{P} := \{12 \mid 34, 13 \mid 24, 14 \mid 23\}$$

be the three pairings of the four positions. For an ordered side-pair $o = (A, B)$, a tuple $\boldsymbol{q} = (q_1, q_2, q_3, q_4) \in X^4$, and $\pi = ij \mid k\ell \in \mathfrak{P}$, put

$$\begin{aligned} I_o^{ij|k\ell}(\boldsymbol{q}) := {} & \mathbf{1}_A(q_i)\mathbf{1}_A(q_j)\mathbf{1}_B(q_k)\mathbf{1}_B(q_\ell) \\ & + \mathbf{1}_B(q_i)\mathbf{1}_B(q_j)\mathbf{1}_A(q_k)\mathbf{1}_A(q_\ell). \end{aligned} \tag{2}$$

Because $A \cap B = \varnothing$, this is a $\{0,1\}$-valued indicator and is unchanged when (A, B) is reversed. Formula (2) is defined on all of X^4; no distinctness assumption is made. If one label occurs in both positional groups, the indicator is zero.

For an edge e, write $I_e^\pi := I_{o_e}^\pi$, where o_e is either ordering of its split. This is independent of that ordering.

Lemma 5.1 (Adjacent correction on the full positional domain)**.** *Let T be a binary leaf-labelled phylogenetic tree. For every $(e, e') \in \overrightarrow{A}_2(T)$, every $\boldsymbol{q} \in X^4$, and every $\pi \in \mathfrak{P}$,*

$$I_{C_{e,e'}}^\pi(\boldsymbol{q}) = I_e^\pi(\boldsymbol{q}) I_{e'}^\pi(\boldsymbol{q}). \tag{3}$$

Proof. Write $C_{e,e'} = (K, L)$ and $H = X \setminus (K \sqcup L)$. With the convention of Section 3.3,

$$\sigma_e = L \mid (K \sqcup H), \qquad \sigma_{e'} = K \mid (H \sqcup L).$$

Relative to the displayed sides L and K, respectively, the three blocks have signatures

	e	e'
K	0	1
H	0	0
L	1	0.

If both edge indicators on the right of (3) are one, the two positional groups have complementary signatures. The only complementary signatures in the table are those of K and L. Thus one group lies wholly in K and the other wholly in L, which says that the left side is one. The converse is immediate. If a label occurs in both positional groups, every indicator in (3) is zero because the relevant sides are disjoint. Thus the argument also covers every repeated-label tuple. □

For each $e \in E_2(T)$, choose an orientation $o_e = (A_e, A_e^c)$ of its split. Define the occurrence-indexed signed multiset

$$\mathfrak{O}_T := \bigsqcup_{e \in E_2(T)} \{(e; o_e, 1)\} \sqcup \bigsqcup_{(e,e') \in \overrightarrow{A}_2(T)} \{((e, e'); C_{e,e'}, -\tfrac{1}{2})\}. \tag{4}$$

The edge or adjacent-pair index is part of each term, so two terms remain distinct even when they determine the same underlying side-pair. For $\eta \in \mathfrak{O}_T$, write o_η for its ordered side-pair and ω_η for its weight. Expanding the four levels in the definition of f_1 gives

$$f_1(T_1, T_2) = \sum_{\eta \in \mathfrak{O}_{T_1}} \sum_{\theta \in \mathfrak{O}_{T_2}} \omega_\eta \omega_\theta T[o_\eta, o_\theta]. \tag{5}$$

The four products of weights are $1, -1/2, -1/2, 1/4$, exactly the four coefficients in the definition of f_1.

For an ordered side-pair $o = (A, B)$ define

$$\phi_o(x, y) := \mathbf{1}_A(x)\mathbf{1}_B(y) - \mathbf{1}_B(x)\mathbf{1}_A(y), \tag{6}$$

and put

$$K_T(q_1, q_2; q_3, q_4) := \sum_{\eta \in \mathfrak{O}_T} \omega_\eta \phi_{o_\eta}(q_1, q_2)\phi_{o_\eta}(q_3, q_4). \tag{7}$$

Reversing o_η changes both ϕ factors by a sign, so the kernel is independent of every arbitrarily chosen split orientation. In particular, $C_{e',e}$ is the reversal of $C_{e,e'}$, and both ordered correction occurrences are handled consistently.

Lemma 5.2 (Ordered-quadruple kernel identity)**.** *For all binary leaf-labelled phylogenetic trees T_1, T_2 on X,*

$$N^4 f_1(T_1, T_2) = \frac{1}{4} \sum_{\boldsymbol{q} \in X^4} K_{T_1}(q_1, q_2; q_3, q_4) K_{T_2}(q_1, q_2; q_3, q_4). \tag{8}$$

Proof. For ordered side-pairs $o = (A, B)$ and $p = (C, D)$, set

$$\Delta(o, p) := |A \cap C||B \cap D| - |A \cap D||B \cap C|.$$

Direct expansion gives

$$\sum_{x,y \in X} \phi_o(x, y)\phi_p(x, y) = 2\Delta(o, p), \qquad T[o, p] = \frac{\Delta(o, p)^2}{N^4}.$$

Using finite expansion and (5),

$$\begin{aligned}
&\frac{1}{4} \sum_{\boldsymbol{q} \in X^4} K_{T_1}(q_1, q_2; q_3, q_4) K_{T_2}(q_1, q_2; q_3, q_4) \\
&\quad = \frac{1}{4} \sum_{\eta \in \mathfrak{O}_{T_1}} \sum_{\theta \in \mathfrak{O}_{T_2}} \omega_\eta \omega_\theta \left(\sum_{x,y \in X} \phi_{o_\eta}(x, y)\phi_{o_\theta}(x, y) \right)^2 \\
&\quad = \sum_{\eta \in \mathfrak{O}_{T_1}} \sum_{\theta \in \mathfrak{O}_{T_2}} \omega_\eta \omega_\theta \Delta(o_\eta, o_\theta)^2 \\
&\quad = N^4 f_1(T_1, T_2).
\end{aligned}$$

The factor $1/4$ is exactly the inverse of the factor 4 in $(2\Delta)^2$. □

We next identify the kernel pointwise. Writing $A_r = \mathbf{1}_A(q_r)$ and $B_r = \mathbf{1}_B(q_r)$, one has on the full domain X^4

$$\begin{aligned}
\phi_o(q_1, q_2)\phi_o(q_3, q_4) &= (A_1B_2 - B_1A_2)(A_3B_4 - B_3A_4) \\
&= A_1A_3B_2B_4 + B_1B_3A_2A_4 \\
&\quad - A_1A_4B_2B_3 - B_1B_4A_2A_3 \\
&= I_o^{13|24}(\boldsymbol{q}) - I_o^{14|23}(\boldsymbol{q}).
\end{aligned} \tag{9}$$

For $\boldsymbol{q} \in X^4$ and $\pi \in \mathfrak{P}$, let

$$E_{T,2}(\boldsymbol{q}, \pi) := \{e \in E_2(T) : I_e^\pi(\boldsymbol{q}) = 1\}$$

and let

$$\overrightarrow{A}_{T,2}(\boldsymbol{q},\pi) := \{(e,e') \in \overrightarrow{A}_2(T) : e,e' \in E_{T,2}(\boldsymbol{q},\pi)\}.$$

Define

$$\lambda_T(\boldsymbol{q},\pi) := |E_{T,2}(\boldsymbol{q},\pi)| - \frac{1}{2}|\overrightarrow{A}_{T,2}(\boldsymbol{q},\pi)|. \tag{10}$$

Lemma 5.3 (Kernel-to-path bridge and uniform bound)**.** *For every binary leaf-labelled phylogenetic tree T, every $\boldsymbol{q} \in X^4$, and every $\pi \in \mathfrak{P}$,*

$$\lambda_T(\boldsymbol{q},\pi) \in \{0,1\}. \tag{11}$$

Moreover,

$$K_T(q_1,q_2;q_3,q_4) = \lambda_T(\boldsymbol{q}, 13 \mid 24) - \lambda_T(\boldsymbol{q}, 14 \mid 23), \tag{12}$$

and consequently

$$K_T(q_1,q_2;q_3,q_4) \in \{-1,0,1\}. \tag{13}$$

Proof. If one label occurs in both positional groups of π, then no edge can separate the groups and $E_{T,2}(\boldsymbol{q},\pi) = \varnothing$. Otherwise let U, V be the two nonempty disjoint sets of labels occurring in the two groups. The edges separating U from V form the bridge path between the minimal subtrees spanned by U and V, or form the empty set if those minimal subtrees meet.

In a binary leaf-labelled phylogenetic tree, $E_2(T)$ is exactly the set of non-pendant edges. Indeed, a pendant edge has a singleton side, while each side of an internal edge contains a leaf in each of the two branches leaving the corresponding internal endpoint. Intersecting a path with the non-pendant edges removes only possible end edges, so $E_{T,2}(\boldsymbol{q},\pi)$ is empty or a path. If it has $m \geq 1$ edges, it contains $2(m-1)$ ordered adjacent pairs. Therefore

$$\lambda_T(\boldsymbol{q},\pi) = m - \frac{1}{2}\,2(m-1) = 1;$$

the empty path gives zero. This proves (11).

By Lemma 5.1,

$$\begin{aligned}\sum_{\eta \in \mathfrak{O}_T} \omega_\eta I^\pi_{o_\eta}(\boldsymbol{q}) &= \sum_{e \in E_2(T)} I^\pi_e(\boldsymbol{q}) - \frac{1}{2} \sum_{(e,e') \in \overrightarrow{A}_2(T)} I^\pi_e(\boldsymbol{q}) I^\pi_{e'}(\boldsymbol{q}) \\ &= \lambda_T(\boldsymbol{q},\pi).\end{aligned}$$

Summing (9) over the signed occurrence multiset now gives (12). Equations (11) and (12) imply (13). □

Proposition 5.4 (Counting identity with explicit remainder)**.** *For two binary leaf-labelled phylogenetic trees T_1, T_2 on the same taxon set X, $|X| = N$, one has*

$$N^4 f_1(T_1,T_2) = 4\,\mathrm{Same}(T_1,T_2) - 2\,\mathrm{Diff}(T_1,T_2) + R_N, \tag{14}$$

where

$$R_N := \frac{1}{4} \sum_{\substack{\boldsymbol{q}=(q_1,q_2,q_3,q_4) \in X^4 \\ |\{q_1,q_2,q_3,q_4\}|<4}} K_{T_1}(q_1,q_2;q_3,q_4) K_{T_2}(q_1,q_2;q_3,q_4), \tag{15}$$

and

$$|R_N| \leq \frac{1}{4}(6N^3 - 11N^2 + 6N). \tag{16}$$

In particular,

$$N^4 f_1(T_1, T_2) = 4\,\mathrm{Same}(T_1, T_2) - 2\,\mathrm{Diff}(T_1, T_2) + O(N^3),$$

with an absolute constant independent of the two tree shapes.

Proof. We separate the kernel sum in (8) into tuples with four distinct labels and tuples with a repeated label.

First fix a four-subset $Y \subseteq X$ and let $\boldsymbol{q}$ range over its 24 orderings. For such a tuple, Lemma 5.3 gives

$$K_T(\boldsymbol{q}) = \begin{cases} 0, & T|Y = q_1q_2 \mid q_3q_4, \\ 1, & T|Y = q_1q_3 \mid q_2q_4, \\ -1, & T|Y = q_1q_4 \mid q_2q_3. \end{cases}$$

Write $\mathrm{pos}_{\boldsymbol{q}}(\rho)$ for the positional pairing occupied by a label pairing ρ under the ordering $\boldsymbol{q}$.

Suppose first that $T_1|Y = T_2|Y = \rho$. Exactly eight orderings place the fixed pairing ρ in each one of the three positional pairing classes. Thus eight orderings give kernel product zero and sixteen give kernel product one. Their contribution to (8) is

$$\frac{1}{4}(16) = 4.$$

Now suppose that $T_1|Y = \rho_1$ and $T_2|Y = \rho_2$ are different. The two displayed pairings occupy distinct positional classes. Each of the six ordered pairs of distinct positional classes occurs for four orderings: this is the action of S_4 on the three pairings, whose kernel has order four. The kernel product is nonzero only for

$$(\mathrm{pos}_{\boldsymbol{q}}(\rho_1), \mathrm{pos}_{\boldsymbol{q}}(\rho_2)) = (13 \mid 24, 14 \mid 23) \quad \text{or} \quad (14 \mid 23, 13 \mid 24).$$

These two cases comprise eight orderings, each with product -1; the other sixteen products vanish. Their contribution is

$$\frac{1}{4}(-8) = -2.$$

Summing over all four-subsets gives the distinct-label contribution $4\,\mathrm{Same}(T_1, T_2) - 2\,\mathrm{Diff}(T_1, T_2)$.

For a tuple with a repeated label, (13) bounds the absolute value of its summand in (8) by $1/4$. The exact number of such ordered tuples is

$$N^4 - N(N-1)(N-2)(N-3) = 6N^3 - 11N^2 + 6N.$$

Their total R_N therefore satisfies (16), proving the proposition. □

Theorem 5.5 (Asymptotic quartet bound)**.** *For two binary leaf-labelled phylogenetic trees T_1, T_2 on the same taxon set X, with $|X| = N \geq 4$,*

$$\frac{\mathrm{Same}(T_1, T_2)}{\binom{N}{4}} \geq \frac{1}{3} - \varepsilon_N, \qquad \frac{\mathrm{Diff}(T_1, T_2)}{\binom{N}{4}} \leq \frac{2}{3} + \varepsilon_N,$$

where

$$\varepsilon_N := \frac{6N^3 - 11N^2 + 6N}{24\binom{N}{4}} = \frac{6N^2 - 11N + 6}{(N-1)(N-2)(N-3)}.$$

In particular, the two bounds are respectively $1/3 - O(1/N)$ and $2/3 + O(1/N)$.

Proof. By Propositions 4.3 and 5.4,

$$0 \le N^4 f_1(T_1, T_2) = 4\,\mathrm{Same}(T_1, T_2) - 2\,\mathrm{Diff}(T_1, T_2) + R_N,$$

where

$$|R_N| \le \frac{1}{4}(6N^3 - 11N^2 + 6N).$$

Since $\mathrm{Same}(T_1, T_2) + \mathrm{Diff}(T_1, T_2) = \binom{N}{4}$, it follows that

$$\mathrm{Same}(T_1, T_2) \ge \frac{1}{3}\binom{N}{4} - \frac{1}{24}(6N^3 - 11N^2 + 6N).$$

Equivalently,

$$\mathrm{Diff}(T_1, T_2) \le \frac{2}{3}\binom{N}{4} + \frac{1}{24}(6N^3 - 11N^2 + 6N).$$

Dividing by $\binom{N}{4} = \Theta(N^4)$ gives the two stated estimates. □

Corollary 5.6 (Sharp asymptotic value)**.** *The maximum quartet distance between two binary leaf-labelled phylogenetic trees on N taxa is*

$$\left(\frac{2}{3} + o(1)\right)\binom{N}{4}.$$

Proof. The upper bound is Theorem 5.5. For the lower bound, fix two unlabelled binary tree shapes with N leaf positions and choose independent uniformly random bijections from their leaf positions to X. For any fixed four-subset $Q \subset X$, each tree induces one of the three quartet trees on Q. By symmetry of the two independent random bijections, conditional on the quartet tree induced by the first labelled tree, the quartet tree induced by the second labelled tree is uniformly distributed over the three possible quartet trees on Q. Hence the two induced quartet trees agree with probability $1/3$, and the expected quartet distance of the two randomly labelled trees is

$$\frac{2}{3}\binom{N}{4}.$$

Therefore there exists a labelled pair with quartet distance at least $(2/3)\binom{N}{4}$. Combining this lower bound with Theorem 5.5 gives the stated asymptotic value. □

6 Remarks on X-trees

The local algebra and the quartet-counting argument have different ranges of validity. The local algebra uses only cluster masses and therefore applies to the vertex-labelled X-trees of Section 2 whenever $c_k, c_{\bar{k}} > 0$. The counting argument in Section 5 uses the stronger fact that every four-subset of a binary leaf-labelled tree displays one resolved quartet. If taxa may occur at internal vertices, a four-subset can be degenerate or unresolved, so Corollary 5.6 does not follow without an additional definition and a separate estimate for such subsets. We therefore state the sharp distance result only for binary leaf-labelled phylogenetic trees.

Appendix

A Local coefficient calculations and the exact four-class identity

This appendix proves the nonnegative decomposition used in Lemma 4.2. The calculation is long because one must keep the row and column atom indices distinct until the final global sum. Its logical structure is nevertheless simple:

formulas	purpose
(A.14a), (A.14b)	establish the weighted-centered and two-sided determinant identities used by the row corrections
(A.15a), (A.15b)	decompose one coordinate into retained squares and a removed-square term
(A.17), (A.18)	write the retained ordinary and adjacent-correction squares explicitly
(A.21)–(A.29)	complete every mixed term and compute the available square coefficient
(A.29c)–(A.29g)	sum over charts, cancel the row-boundary polynomial, and pair the column endpoint square faces by reversal
(A.30), (A.32)	identify the four square families and prove their coefficients are nonnegative

For traceability to the source calculation underlying this appendix, the displayed tags (A.15a)–(A.32) are retained as stable identifiers. Consequently, auxiliary tags such as (A.30a) and (A.30b) are introduced before the final identity (A.30), and the historical tag (A.32) is retained although (A.31) is not used in this streamlined presentation. The roadmap above records the logical order of the proof. Thus the argument is local in the row coordinate, blockwise in the column coordinate, and global only at the two bridge-reversal steps. No equality of complete fixed-principal-chart subtotals is asserted. The only cross-chart termwise identities are the explicitly indexed square-face coefficient identities transported by the bridge-reversal bijection; cancellation is claimed only after summation over the full ordered bridge orbit.

A.1 Coordinates, blocks, and index relations

We begin by fixing all notation. As in Section 4, r_i^k is the symmetric coefficient, c_k is the mass of a column block, $a_{i,k}$ is the mixed mass, and $x_{i,k}$ is the centered mixed mass. The four local index classes are denoted by a, b, c, d, and

$$\Omega = a \sqcup b \sqcup c \sqcup d.$$

As shown in Figure 1, their left-to-right order is a, c, d, b; hence $a \cup c$ and $b \cup d$ are respectively the left and right sides of the central local edge. These four letters denote index classes throughout this appendix, not scalar entries of a 2×2 table. We also retain the square-face convention stated before Lemma 4.2: in the expansion

$$\gamma(\lambda X + \mu Y)^2 = \gamma\lambda^2 X^2 + 2\gamma\lambda\mu XY + \gamma\mu^2 Y^2,$$

the three displayed monomials are, in order, the first square face, mixed face, and second square face. Whenever the abbreviated phrases "first face" or "second face" are used, they mean the first square face or second square face relative to this displayed order of X, Y. For a summand indexed by an endpoint atom, the three terms are called its endpoint-indexed first square face, endpoint-indexed mixed face, and endpoint-indexed second square face; collectively they are its endpoint-indexed monomial faces. The phrase "endpoint square face" means only the first or second square face and

excludes the mixed face. The two blocks $a \cup c$ and $b \cup d$ are called the *central sides* of the fixed coordinate; they are the same-side blocks defined in Subsection 4.4. For any $U \subseteq \Omega$ write

$$\mathrm{Off}(U) = \{(p,q) \in U \times U : p \neq q\}, \qquad \Delta_U = \{(p,p) : p \in U\}.$$

For every $u \in \Omega$ define the same-side block

$$S(u) = \begin{cases} a \cup c, & u \in a \cup c, \\ b \cup d, & u \in b \cup d, \end{cases}$$

and define

$$u^+ = S(u), \qquad S^0(u) = S(u) \setminus \{u\}, \qquad \bar{u} = \Omega \setminus S(u).$$

Thus

$$c_{u^+} = \sum_{l \in S(u)} c_l, \qquad c_{\bar{u}} = \sum_{l \in \bar{u}} c_l, \qquad c_{u^+} + c_{\bar{u}} = 1.$$

Throughout this appendix, principal charts, endpoint atoms and residual blocks are those of Section 4.1. Thus sums over $a \cup b$ index adjacent E_2-directions, whereas the possible residual blocks $c \cup d$ remain part of the full atom set Ω. Whenever $u \in a \cup b$, one has $c_{u^+} \geq c_u > 0$, so the quotients below are well-defined. Empty sums are interpreted as zero. For row and column blocks P, Q, respectively, we use the additive notation

$$R_P = \bigsqcup_{i \in P} R_i, \qquad C_Q = \bigsqcup_{k \in Q} C_k,$$

$$r_P = \sum_{i \in P} r_i, \qquad c_Q = \sum_{k \in Q} c_k, \qquad a_{P,Q} = \sum_{i \in P} \sum_{k \in Q} a_{i,k}, \qquad x_{P,Q} = a_{P,Q} - r_P c_Q.$$

In particular, $\bar{i}$ and $\bar{k}$ denote aggregate opposite-side blocks in their respective charts, never singleton indices.

The following ordered-pair relations separate diagonal endpoint terms, same-side terms, and opposite-side terms:

$$I = \mathrm{Off}(a) \cup (a \times c) \cup \mathrm{Off}(b) \cup (b \times d) \cup ((c \cup d) \times \Omega),$$

$$D = a \times (b \cup d) \cup b \times (a \cup c),$$

$$M_S = \mathrm{Off}(a) \cup (a \times c) \cup \mathrm{Off}(b) \cup (b \times d),$$

$$N_S = c \times (a \cup c) \cup d \times (b \cup d), \qquad M_D = c \times (b \cup d) \cup d \times (a \cup c).$$

For a fixed first coordinate u write

$$M_S(u) = \{l \in \Omega : (u,l) \in M_S\}, \qquad M_D(u) = \{l \in \Omega : (u,l) \in M_D\}.$$

The six sets

$$\Omega^2 = \Delta_a \sqcup \Delta_b \sqcup M_S \sqcup D \sqcup N_S \sqcup M_D$$

form a disjoint partition, and $I = M_S \sqcup N_S \sqcup M_D$. The same relations are used in the row and column copies; whenever both copies occur in one formula we write the subscripts r and c explicitly.

The principal chart χ fixes the central edge in each tree before an adjacent direction is chosen. An endpoint $u \in a^\chi \cup b^\chi$ then determines one ordered bridge. By contrast, an index in $c^\chi \cup d^\chi$ is residual data of the principal chart and does not determine an additional bridge. Hence

every residual target is counted once. If every nonprincipal edge incident with the endpoints of a principal edge belongs to E_2, the corresponding residual classes may be empty. After E_2-core contraction, however, even a binary leaf-labelled tree may have nonempty vertex blocks c^χ, d^χ; their contributions occur in the nonendpoint residual terms.

For a principal chart $\chi = (e, f)$, the row atom set is $\Omega_{\mathrm{r}}^\chi = \Omega_e^{\mathrm{r}}$ and the column atom set is $\Omega_{\mathrm{c}}^\chi = \Omega_f^{\mathrm{c}}$. Variables i, j always belong to the row copy, whereas u, l, k belong to the column copy. Thus I_χ, D_χ denote row-copy relations when indexed by (i, j), whereas $a^\chi, b^\chi, c^\chi, d^\chi, S_\chi, M_{S,\chi}, M_{D,\chi}$ denote column-copy objects when indexed by (u, l). Explicit subscripts r, c are restored whenever both copies occur in one display. For that purpose, set

$$a_{\mathrm{r}}^\chi := a_e, \quad b_{\mathrm{r}}^\chi := b_e, \quad c_{\mathrm{r}}^\chi := c_e, \quad d_{\mathrm{r}}^\chi := d_e, \qquad a_{\mathrm{c}}^\chi := a_f, \quad b_{\mathrm{c}}^\chi := b_f, \quad c_{\mathrm{c}}^\chi := c_f, \quad d_{\mathrm{c}}^\chi := d_f,$$

and write $I_{\mathrm{r}}^\chi := I_\chi$, $D_{\mathrm{r}}^\chi := D_\chi$ and $I_{\mathrm{c}}^\chi, D_{\mathrm{c}}^\chi$ for the column-copy versions of the displayed definitions of I, D above. In (A.14a) and (A.14b), abbreviate

$$a_{\mathrm{row}} := a_{\mathrm{r}}^\chi, \qquad b_{\mathrm{row}} := b_{\mathrm{r}}^\chi, \qquad a_{\mathrm{col}} := a_{\mathrm{c}}^\chi, \qquad b_{\mathrm{col}} := b_{\mathrm{c}}^\chi.$$

Within a fixed chart, a *row relation* means an ordered pair (i, j) in I_{r}^χ or D_{r}^χ. A *target* means a column atom $u \in \Omega_{\mathrm{c}}^\chi$ at which a square coefficient is recorded; it introduces no additional index set. The correction notation σ^+ or τ^+ always has the occurrence-sum meaning defined in Subsection 3.3. By contrast, the phrase *plus form* below refers only to an algebraic linear form such as $x_{\bar{i},A} + x_{i,A}$ in a D-row; it does not denote another correction operator. For $u \in \Omega_{\mathrm{c}}^\chi$, in particular,

$$u^{+,\chi} := S_\chi(u), \qquad \bar{u}_\chi := \Omega_{\mathrm{c}}^\chi \setminus S_\chi(u),$$

$$M_{S,\chi}(u) := \{l \in \Omega_{\mathrm{c}}^\chi : (u, l) \in M_{S,\chi}\}, \qquad M_{D,\chi}(u) := \{l \in \Omega_{\mathrm{c}}^\chi : (u, l) \in M_{D,\chi}\},$$

and

$$r_i^{u,\chi} := \frac{1}{2}\left(\frac{a_{i,u}}{c_u} + \frac{a_{i,\bar{u}_\chi}}{c_{\bar{u}_\chi}}\right).$$

Every aggregate subscript denotes the additive sum in that chart; for example,

$$c_{\bar{u}_\chi} = \sum_{l \in \bar{u}_\chi} c_l, \qquad a_{i,\bar{u}_\chi} = \sum_{l \in \bar{u}_\chi} a_{i,l}.$$

Row aggregates $\bar{i}_\chi$ are defined analogously; explicitly,

$$r_{\bar{i}_\chi}^{u,\chi} := \sum_{j \in \bar{i}_\chi} r_j^{u,\chi}, \qquad a_{\bar{i}_\chi,u} := \sum_{j \in \bar{i}_\chi} a_{j,u}, \qquad x_{\bar{i}_\chi,A} := \sum_{j \in \bar{i}_\chi} x_{j,A}$$

for every column atom u and every column block A used below. Inside a fixed chart we suppress chart decorations when no ambiguity can result.

A.2 The two algebraic identities from which the calculation starts

We now derive (A.17) and (A.18) from the local terms of f_1. The first identity concerns a correction in the row tree; the second concerns corrections in both trees. They are written out in full so that the meanings of $i, \bar{i}, k, \bar{k}$ do not change during the later completion of squares.

For any atom k in a fixed chart, use the taxon side-pair $(C_k, C_{\bar{k}})$ and define the side-reversal symmetric coefficient

$$r_i^k = \frac{1}{2}\left(\frac{a_{i,k}}{c_k} + \frac{a_{i,\bar{k}}}{c_{\bar{k}}}\right).$$

This expression is algebraically symmetric under exchanging the two taxon blocks C_k and $C_{\bar{k}}$. When k is an endpoint atom, bridge reversal realizes this symmetry by passing to the reversed endpoint chart, where the old opposite block is the selected endpoint block. It is not a substitution $k \mapsto \bar{k}$ within one chart. For a residual atom the formula is only the same-side/opposite-side algebraic average; no endpoint chart is asserted. For each fixed side-pair it satisfies

$$\sum_{i\in\Omega_{\mathrm{r}}} r_i^k = \frac{1}{2}\left(\frac{\sum_{i\in\Omega_{\mathrm{r}}} a_{i,k}}{c_k} + \frac{\sum_{i\in\Omega_{\mathrm{r}}} a_{i,\bar{k}}}{c_{\bar{k}}}\right) = \frac{1}{2}\left(\frac{c_k}{c_k} + \frac{c_{\bar{k}}}{c_{\bar{k}}}\right) = 1.$$

Hence the global normalization inserted below is

$$\sum_{p\in\Omega_{\mathrm{c}}}\sum_{i\in\Omega_{\mathrm{r}}} r_i^p c_p = 1.$$

Explicitly,

$$\begin{aligned}\sum_{p\in\Omega_{\mathrm{c}}}\sum_{i\in\Omega_{\mathrm{r}}} r_i^p c_p &= \sum_{p\in\Omega_{\mathrm{c}}}\sum_{i\in\Omega_{\mathrm{r}}} \frac{1}{2}\left(\frac{a_{i,p}}{c_p} + \frac{a_{i,\bar{p}}}{c_{\bar{p}}}\right) c_p \\ &= \frac{1}{2}\sum_{p\in\Omega_{\mathrm{c}}}\sum_{i\in\Omega_{\mathrm{r}}} a_{i,p} + \frac{1}{2}\sum_{p\in\Omega_{\mathrm{c}}}\frac{c_p}{c_{\bar{p}}}\sum_{i\in\Omega_{\mathrm{r}}} a_{i,\bar{p}} \\ &= \frac{1}{2}\sum_{p\in\Omega_{\mathrm{c}}} c_p + \frac{1}{2}\sum_{p\in\Omega_{\mathrm{c}}}\frac{c_p}{c_{\bar{p}}} c_{\bar{p}} = \frac{1}{2} + \frac{1}{2}\sum_{p\in\Omega_{\mathrm{c}}} c_p = 1.\end{aligned}$$

Only this global normalization is used; r_i^k is not an ordinary row marginal. Indeed,

$$\sum_{k\in\Omega_{\mathrm{c}}} r_i^k c_k = \frac{1}{2} r_i + \frac{1}{2}\sum_{k\in\Omega_{\mathrm{c}}}\frac{c_k}{c_{\bar{k}}} a_{i,\bar{k}},$$

which need not equal r_i. The coefficients r_i^k are side-reversal indexing weights, not ordinary row marginals.

The next proposition is the key cancellation behind the row correction. It says that replacing the original row mass r_R by the averaged quantity A_R adds the same multiple of the centered column mass to both rows of a 2×2 determinant; that common addition cancels from the determinant.

Proposition A.1 (Weighted centered cancellation)**.** *Fix a local column chart and let*

$$\Omega_{\mathrm{c}} = S \sqcup B$$

be its two central sides, in either order. Thus, for every atom $p \in S$ one has $\bar{p} = B$, while for every atom $p \in B$ one has $\bar{p} = S$. Both sides are nonempty, so their masses satisfy $c_S > 0$ and $c_B > 0$. For every row block R, define

$$A_R := \sum_{p\in\Omega_{\mathrm{c}}} c_p r_R^p, \qquad r_R^p := \frac{1}{2}\left(\frac{a_{R,p}}{c_p} + \frac{a_{R,\bar{p}}}{c_{\bar{p}}}\right).$$

Then

$$A_R = r_R + \theta_S x_{R,B}, \qquad \theta_S := \frac{1}{2}\left(\frac{c_S}{c_B} - \frac{c_B}{c_S}\right).$$

Consequently, for arbitrary row-index blocks R and R',

$$A_R x_{R',B} - A_{R'} x_{R,B} = r_R x_{R',B} - r_{R'} x_{R,B}.$$

In particular, if an endpoint index k is chosen and $(S,B)=(k^+,\bar{k})$, specialize to a row endpoint index i and its aggregate opposite-side block $\bar{i}$ to obtain

$$\sum_{i\in a_{\mathrm{row}}\cup b_{\mathrm{row}}}\left(r_i x_{\bar{i},\bar{k}}-r_{\bar{i}}x_{i,\bar{k}}\right)^2=\sum_{i\in a_{\mathrm{row}}\cup b_{\mathrm{row}}}\left(\sum_{p\in\Omega_{\mathrm{c}}}c_p r_i^p x_{\bar{i},\bar{k}}-\sum_{l\in\Omega_{\mathrm{c}}}c_l r_{\bar{i}}^l x_{i,\bar{k}}\right)^2. \tag{A.14a}$$

Proof. Since $p\in S$ implies $\bar{p}=B$, and $p\in B$ implies $\bar{p}=S$, we have

$$\begin{aligned}A_R&=\frac{1}{2}\sum_{p\in\Omega_{\mathrm{c}}}a_{R,p}+\frac{1}{2}\sum_{p\in\Omega_{\mathrm{c}}}\frac{c_p}{c_{\bar{p}}}a_{R,\bar{p}}\\&=\frac{1}{2}r_R+\frac{1}{2}\left(\frac{c_S}{c_B}a_{R,B}+\frac{c_B}{c_S}a_{R,S}\right).\end{aligned}$$

Using

$$a_{R,S}=r_R c_S+x_{R,S},\qquad a_{R,B}=r_R c_B+x_{R,B},$$

we obtain

$$A_R=r_R+\frac{1}{2}\left(\frac{c_S}{c_B}x_{R,B}+\frac{c_B}{c_S}x_{R,S}\right).$$

The blocks S and B form a partition of the column side, so centeredness gives

$$x_{R,S}+x_{R,B}=x_{R,\Omega_{\mathrm{c}}}=0.$$

Hence $x_{R,S}=-x_{R,B}$, and therefore

$$A_R=r_R+\theta_S x_{R,B}.$$

Applying this identity to arbitrary R,R' gives

$$\begin{aligned}A_R x_{R',B}-A_{R'}x_{R,B}&=(r_R+\theta_S x_{R,B})x_{R',B}-(r_{R'}+\theta_S x_{R',B})x_{R,B}\\&=r_R x_{R',B}-r_{R'}x_{R,B},\end{aligned}$$

because the two terms containing θ_S cancel identically. Squaring and then specializing to $R=i$, $R'=\bar{i}$ and summing over the endpoint indices proves the final identity. The algebraic cancellation itself does not use complementarity of R and R'. □

For the corresponding calculation when both coordinates are corrected, the relevant adjacent endpoint directions in the two contracted E_2-cores are $a_{\mathrm{row}}\cup b_{\mathrm{row}}$ and $a_{\mathrm{col}}\cup b_{\mathrm{col}}$. The bars in the next display are computed in the corresponding row or column chart; vertex blocks in $c_{\mathrm{row}}\cup d_{\mathrm{row}}$ and $c_{\mathrm{col}}\cup d_{\mathrm{col}}$ do not index an adjacent correction.

$$\sum_{i\in a_{\mathrm{row}}\cup b_{\mathrm{row}}}\sum_{k\in a_{\mathrm{col}}\cup b_{\mathrm{col}}}\left(a_{i,k}\cdot a_{\bar{i},\bar{k}}-a_{\bar{i},k}\cdot a_{i,\bar{k}}\right)^2=\sum_{i\in a_{\mathrm{row}}\cup b_{\mathrm{row}}}\sum_{k\in a_{\mathrm{col}}\cup b_{\mathrm{col}}}\left[r_i^k\cdot\left(c_k\cdot x_{\bar{i},\bar{k}}-c_{\bar{k}}\cdot x_{\bar{i},k}\right)-r_{\bar{i}}^k\cdot\left(c_k\cdot x_{i,\bar{k}}-c_{\bar{k}}\cdot x_{i,k}\right)\right]^2. \tag{A.14b}$$

To prove (A.14b), fix an admissible pair (i,k). Using $x_{R,S}=a_{R,S}-r_R c_S$ and substituting the definitions of r_i^k and $r_{\bar{i}}^k$ gives

$$\begin{aligned}&r_i^k\cdot\left(c_k\cdot x_{\bar{i},\bar{k}}-c_{\bar{k}}\cdot x_{\bar{i},k}\right)-r_{\bar{i}}^k\cdot\left(c_k\cdot x_{i,\bar{k}}-c_{\bar{k}}\cdot x_{i,k}\right)\\&=\frac{1}{2}\left(a_{i,k}+\frac{a_{i,\bar{k}}}{c_{\bar{k}}}\cdot c_k\right)\cdot\left(a_{\bar{i},\bar{k}}-r_{\bar{i}}\cdot c_{\bar{k}}\right)-\frac{1}{2}\left(a_{i,\bar{k}}+\frac{a_{i,k}}{c_k}\cdot c_{\bar{k}}\right)\cdot\left(a_{\bar{i},k}-r_{\bar{i}}\cdot c_k\right)\\&\quad-\frac{1}{2}\left(a_{\bar{i},k}+\frac{a_{\bar{i},\bar{k}}}{c_{\bar{k}}}\cdot c_k\right)\cdot\left(a_{i,\bar{k}}-r_i\cdot c_{\bar{k}}\right)+\frac{1}{2}\left(a_{\bar{i},\bar{k}}+\frac{a_{\bar{i},k}}{c_k}\cdot c_{\bar{k}}\right)\cdot\left(a_{i,k}-r_i\cdot c_k\right)\\&=a_{i,k}\cdot a_{\bar{i},\bar{k}}-a_{i,\bar{k}}\cdot a_{\bar{i},k}.\end{aligned}$$

Squaring this pointwise identity and summing over $i \in a_{\mathrm{row}} \cup b_{\mathrm{row}}$ and $k \in a_{\mathrm{col}} \cup b_{\mathrm{col}}$ proves (A.14b).

Fix a principal chart $\chi = (e, f)$, write $\sigma := \sigma_e$ and $\tau := \tau_f$, and suppress the superscript χ. The normalization above gives

$$\sum_{p,l \in \Omega_{\mathrm{c}}} \sum_{i,j \in \Omega_{\mathrm{r}}} r_i^p c_p \, r_j^l c_l = 1.$$

In the calculation below, p and l retain their positions as the two ordered column indices.

Identities (A.14a) and (A.14b) are the algebraic support for the square completions below. Proposition A.1 records the first identity in the row-transposed form needed for the first-tree adjacent correction; its proof uses only the column-side complement $k^+ \sqcup \bar{k} = \Omega$ and does not require the row atom i and the aggregate row block $\bar{i}$ to be complementary blocks.

The two-sided adjacent correction is governed by the determinant of the two adjacent side-pairs. The left-hand side of (A.14b) is precisely

$$T[\sigma^+, \tau^+] = \sum_{i \in a_{\mathrm{row}} \cup b_{\mathrm{row}}} \sum_{k \in a_{\mathrm{col}} \cup b_{\mathrm{col}}} \left(a_{i,k} a_{\bar{i},\bar{k}} - a_{\bar{i},k} a_{i,\bar{k}} \right)^2.$$

For later use, denote the bracket on the right-hand side of (A.14b) by

$$Q_i(k) = r_i^k (c_k x_{\bar{i},\bar{k}} - c_{\bar{k}} x_{\bar{i},k}) - r_{\bar{i}}^k (c_k x_{i,\bar{k}} - c_{\bar{k}} x_{i,k}).$$

The centered parts remove the same row mean:

$$c_k x_{\bar{i},\bar{k}} - c_{\bar{k}} x_{\bar{i},k} = c_k a_{\bar{i},\bar{k}} - c_{\bar{k}} a_{\bar{i},k},$$

and similarly

$$c_k x_{i,\bar{k}} - c_{\bar{k}} x_{i,k} = c_k a_{i,\bar{k}} - c_{\bar{k}} a_{i,k}.$$

Consequently, the defining expression for $Q_i(k)$ is exactly the left-hand side of the pointwise calculation above, and hence

$$Q_i(k) = a_{i,k} a_{\bar{i},\bar{k}} - a_{\bar{i},k} a_{i,\bar{k}}.$$

Thus the neighboring contrast $Q_i(k)$ used in (A.18) is not an auxiliary replacement: it is exactly the two-sided adjacent determinant from the original calculation. These two identities are the computational basis for separating the retained residuals and then completing the mixed terms in (A.22)–(A.25).

A.3 The four determinant levels and a one-coordinate residual identity

Fix a principal chart $\chi = (e, f)$, write $\sigma := \sigma_e$ and $\tau := \tau_f$, and abbreviate

$$U_\chi = T[\sigma_e, \tau_f], \quad R_\chi = T[\sigma_e^+, \tau_f], \quad C_\chi = T[\sigma_e, \tau_f^+], \quad B_\chi = T[\sigma_e^+, \tau_f^+].$$

Thus U is uncorrected in both coordinates, R is corrected only in the row tree, C only in the column tree, and B in both trees:

	row coordinate	column coordinate
U	ordinary split	ordinary split
R	adjacent correction	ordinary split
C	ordinary split	adjacent correction
B	adjacent correction	adjacent correction

We work with the exact global numerator

$$4f_1 = \sum_{\chi} (4U_\chi - 2R_\chi - 2C_\chi + B_\chi). \tag{17}$$

Every determinant in (17) is expanded with its own row and column indices. Equal block squares are combined only after the relevant finite sum has been written. Every later reindexing is an explicit bijection of that complete sum.

Lemma A.2 (One-coordinate residual identity)**.** *Let Ω be partitioned into two central sides, and let $E \subseteq \Omega$ be the set of endpoint indices. For every $i \in \Omega$, write $\bar{i}$ for the whole central side opposite to i, and put*

$$D := \{(i,j) : i \in E,\ j \in \bar{i}\}, \qquad \Delta_E := \{(i,i) : i \in E\}, \qquad I := \Omega^2 \setminus (D \sqcup \Delta_E).$$

Let $w_i \geq 0$ satisfy $\sum_{i\in\Omega} w_i = 1$, and define

$$w_{\bar{i}} := \sum_{j\in\bar{i}} w_j \qquad (i \in E).$$

Suppose that $X_i, Y_i \in \mathbb{R}$ and that $X_i^2 = X_{i'}^2$ for all $i, i' \in \Omega$. Then

$$\begin{aligned} &X^2 - \sum_{i\in E} (w_i X_i - w_{\bar{i}} Y_i)^2 \\ &\quad = \sum_{(i,j)\in I} w_i w_j X_i^2 + \sum_{(i,j)\in D} w_i w_j (X_i + Y_i)^2 - \sum_{i\in E} w_{\bar{i}}(w_i + w_{\bar{i}}) Y_i^2, \end{aligned} \tag{18}$$

where X^2 denotes the common value of the X_i^2. The final sum is an ordinary nonnegative square sum; its minus sign records that it is the part removed by the adjacent correction.

Proof. Because $\sum_i w_i = 1$ and all X_i^2 have the same value,

$$X^2 = \sum_{(i,j)\in\Omega^2} w_i w_j X_i^2.$$

For every family F_i depending only on the first coordinate, the three finite sums produced by the partition have the explicit forms

$$\sum_{(i,j)\in D} w_i w_j F_i = \sum_{i\in E} w_i \left(\sum_{j\in\bar{i}} w_j \right) F_i = \sum_{i\in E} w_i w_{\bar{i}} F_i,$$

$$\sum_{(i,j)\in\Delta_E} w_i w_j F_i = \sum_{i\in E} w_i^2 F_i, \qquad \sum_{(i,j)\in\Omega^2} w_i w_j X_i^2 = X^2 \left(\sum_i w_i \right) \left(\sum_j w_j \right) = X^2.$$

Thus the passage from the ordered-pair partition to the following coefficient calculation is termwise. Using the disjoint partition $\Omega^2 = I \sqcup D \sqcup \Delta_E$ and $\sum_{j\in\bar{i}} w_j = w_{\bar{i}}$, we obtain

$$\begin{aligned} &\sum_{(i,j)\in I} w_i w_j X_i^2 + \sum_{(i,j)\in D} w_i w_j (X_i + Y_i)^2 \\ &= X^2 - \sum_{i\in E} w_i^2 X_i^2 - \sum_{i\in E} w_i w_{\bar{i}} X_i^2 + \sum_{i\in E} w_i w_{\bar{i}} (X_i + Y_i)^2 \\ &= X^2 - \sum_{i\in E} w_i^2 X_i^2 + 2\sum_{i\in E} w_i w_{\bar{i}} X_i Y_i + \sum_{i\in E} w_i w_{\bar{i}} Y_i^2. \end{aligned}$$

Subtracting $\sum_{i\in E} w_{\bar{i}}(w_i + w_{\bar{i}})Y_i^2$ gives

$$X^2 - \sum_{i\in E}\big(w_i^2 X_i^2 - 2w_i w_{\bar{i}} X_i Y_i + w_{\bar{i}}^2 Y_i^2\big),$$

which is the left-hand side of (18). □

Row correction with an ordinary column split: proof of (A.15a). Fix an oriented side B_{c} of the central column split. Define

$$\omega_i := \sum_{p\in\Omega_{\mathrm{c}}} c_p r_i^p, \qquad \omega_{\bar{i}} := \sum_{j\in\bar{i}} \omega_j = \sum_{p\in\Omega_{\mathrm{c}}} c_p r_{\bar{i}}^p.$$

The global normalization gives $\sum_{i\in\Omega_{\mathrm{r}}} \omega_i = 1$. Put

$$X_i := x_{\bar{i},B_{\mathrm{c}}}, \qquad Y_i := x_{i,B_{\mathrm{c}}}.$$

Let $S_{\mathrm{r}} \sqcup B_{\mathrm{r}} = \Omega_{\mathrm{r}}$ be the two central row sides. Centeredness gives

$$x_{S_{\mathrm{r}},B_{\mathrm{c}}} + x_{B_{\mathrm{r}},B_{\mathrm{c}}} = x_{\Omega_{\mathrm{r}},B_{\mathrm{c}}} = 0.$$

For every row atom i, the aggregate $\bar{i}$ is one of these two sides. Consequently, changing the row representative changes X_i at most by an overall sign, and

$$X_i^2 = x_{\bar{i},B_{\mathrm{c}}}^2 = T[\sigma,\tau] \qquad (i\in\Omega_{\mathrm{r}}).$$

For an endpoint $i \in a_{\mathrm{row}} \cup b_{\mathrm{row}}$, the original ordered adjacent row occurrence has determinant $(r_i X_i - r_{\bar{i}} Y_i)^2$. By Proposition A.1, it is equivalently represented as $(\omega_i X_i - \omega_{\bar{i}} Y_i)^2$. Therefore

$$T[\sigma^+,\tau] = \sum_{i\in a_{\mathrm{row}}\cup b_{\mathrm{row}}} (\omega_i X_i - \omega_{\bar{i}} Y_i)^2.$$

The two expressions are exactly the two sides required by Lemma A.2; no coefficient has been identified by an additional convention. Lemma A.2, with $E = a_{\mathrm{row}} \cup b_{\mathrm{row}}$, gives the exact identity

$$T[\sigma,\tau] - T[\sigma^+,\tau] = \mathcal{R}_1 - \mathcal{Q}_1, \tag{A.15a}$$

where

$$\mathcal{R}_1 = \sum_{(i,j)\in I_{\mathrm{r}}} \omega_i\omega_j x_{\bar{i},B_{\mathrm{c}}}^2 + \sum_{(i,j)\in D_{\mathrm{r}}} \omega_i\omega_j (x_{\bar{i},B_{\mathrm{c}}} + x_{i,B_{\mathrm{c}}})^2, \tag{19}$$

$$\mathcal{Q}_1 = \sum_{i\in a_{\mathrm{row}}\cup b_{\mathrm{row}}} \omega_{\bar{i}}(\omega_i + \omega_{\bar{i}}) x_{\bar{i},B_{\mathrm{c}}}^2 \geq 0. \tag{20}$$

To keep every coefficient visible, expand

$$\omega_i\omega_j = \sum_{p,l\in\Omega_{\mathrm{c}}} c_p r_i^p\, c_l r_j^l.$$

Since the sums are finite, direct substitution in (19) gives

$$\begin{aligned}\mathcal{R}_1 = &\sum_{(i,j)\in I_{\mathrm{r}}} \sum_{(p,l)\in\Omega_{\mathrm{c}}^2} c_p r_i^p\, c_l r_j^l\, x_{\bar{i},B_{\mathrm{c}}}^2 \\ &+ \sum_{(i,j)\in D_{\mathrm{r}}} \sum_{(p,l)\in\Omega_{\mathrm{c}}^2} c_p r_i^p\, c_l r_j^l\, (x_{\bar{i},B_{\mathrm{c}}} + x_{i,B_{\mathrm{c}}})^2.\end{aligned}$$

Let $S_{\mathrm{c}} \sqcup B_{\mathrm{c}} = \Omega_{\mathrm{c}}$ be the central column split. For each $p \in \Omega_{\mathrm{c}}$ put

$$\varepsilon_p = \begin{cases} 1, & \bar{p} = B_{\mathrm{c}}, \\ -1, & \bar{p} = S_{\mathrm{c}}. \end{cases}$$

The centered identity $x_{R,S_{\mathrm{c}}} + x_{R,B_{\mathrm{c}}} = 0$ gives

$$x_{R,B_{\mathrm{c}}} = \varepsilon_p x_{R,\bar{p}}$$

for every row block R. The same sign occurs in both summands of the plus form, and hence

$$x^2_{\bar{i},B_{\mathrm{c}}} = x^2_{\bar{i},\bar{p}}, \qquad (x_{\bar{i},B_{\mathrm{c}}} + x_{i,B_{\mathrm{c}}})^2 = (x_{\bar{i},\bar{p}} + x_{i,\bar{p}})^2.$$

Renaming p as k gives the first equality in (A.17); expanding the plus square gives the second equality. More explicitly, if $\omega_{i,p} := c_p r_i^p$, then

$$\begin{aligned} \mathcal{Q}_1 = &\sum_{i \in a_{\mathrm{row}} \cup b_{\mathrm{row}}} \sum_{p,l \in \Omega_{\mathrm{c}}} \sum_{j \in \bar{i}} \omega_{i,p}\omega_{j,l}\, x^2_{i,B_{\mathrm{c}}} \\ &+ \sum_{i \in a_{\mathrm{row}} \cup b_{\mathrm{row}}} \sum_{p,l \in \Omega_{\mathrm{c}}} \sum_{j,j' \in \bar{i}} \omega_{j,p}\omega_{j',l}\, x^2_{i,B_{\mathrm{c}}}. \end{aligned}$$

Thus every removed square and its coefficient is explicit. For the term indexed by p, replacing the representative side B_{c} by $\bar{p}$ changes both linear forms only by the common sign ε_p displayed above. This completes the source-by-source derivation of (A.17).

Row correction with a corrected column term: proof of (A.15b). For every relevant column endpoint $k \in a_{\mathrm{col}} \cup b_{\mathrm{col}}$ and every row index i, define

$$\begin{aligned} U_i(k) &:= c_k x_{\bar{i},\bar{k}} - c_{\bar{k}} x_{\bar{i},k}, \\ V_i(k) &:= c_k x_{i,\bar{k}} - c_{\bar{k}} x_{i,k}. \end{aligned}$$

Let $S_{\mathrm{r}} \sqcup B_{\mathrm{r}} = \Omega_{\mathrm{r}}$ be the two central row sides. For every column block Q, centeredness gives

$$x_{S_{\mathrm{r}},Q} + x_{B_{\mathrm{r}},Q} = 0.$$

Thus changing the row representative changes both $x_{\bar{i},\bar{k}}$ and $x_{\bar{i},k}$ by the same sign. It follows that $U_i(k)$ changes only by an overall sign, so $U_i(k)^2$ has a common value, denoted by $U(k)^2$, independent of i. Consequently

$$T[\sigma, \tau^+] = \sum_{k \in a_{\mathrm{col}} \cup b_{\mathrm{col}}} U(k)^2.$$

Moreover, (A.14b) says that the (i,k)-indexed summand of $T[\sigma^+, \tau^+]$ is

$$\big(r_i^k U_i(k) - r_{\bar{i}}^k V_i(k)\big)^2,$$

where

$$r_{\bar{i}}^k = \sum_{j \in \bar{i}} r_j^k, \qquad \sum_{i \in \Omega_{\mathrm{r}}} r_i^k = 1.$$

Therefore

$$T[\sigma^+, \tau^+] = \sum_{k \in a_{\mathrm{col}} \cup b_{\mathrm{col}}} \sum_{i \in a_{\mathrm{row}} \cup b_{\mathrm{row}}} \big(r_i^k U_i(k) - r_{\bar{i}}^k V_i(k)\big)^2.$$

Apply Lemma A.2 for each fixed k with

$$w_i = r_i^k, \qquad X_i = U_i(k), \qquad Y_i = V_i(k),$$

and then sum over all relevant column endpoints k. This gives the exact identity

$$T[\sigma, \tau^+] - T[\sigma^+, \tau^+] = \mathcal{R}_2 - \mathcal{Q}_2, \tag{A.15b}$$

where

$$\mathcal{R}_2 = \sum_{k \in a_{\mathrm{col}} \cup b_{\mathrm{col}}} \Bigg[\sum_{(i,j) \in I_{\mathrm{r}}} r_i^k r_j^k \, U_i(k)^2 + \sum_{(i,j) \in D_{\mathrm{r}}} r_i^k r_j^k \, (U_i(k) + V_i(k))^2 \Bigg], \tag{21}$$

$$\mathcal{Q}_2 = \sum_{k \in a_{\mathrm{col}} \cup b_{\mathrm{col}}} \sum_{i \in a_{\mathrm{row}} \cup b_{\mathrm{row}}} r_{\bar{i}}^k (r_i^k + r_{\bar{i}}^k) V_i(k)^2 \geq 0. \tag{22}$$

Equivalently, the coefficient of the removed square in (22) expands as

$$r_{\bar{i}}^k (r_i^k + r_{\bar{i}}^k) = \sum_{j \in \bar{i}} r_i^k r_j^k + \sum_{j,j' \in \bar{i}} r_j^k r_{j'}^k.$$

Thus the complete expansion by the original row indices is

$$\mathcal{Q}_2 = \sum_{k \in a_{\mathrm{col}} \cup b_{\mathrm{col}}} \sum_{i \in a_{\mathrm{row}} \cup b_{\mathrm{row}}} \left[\sum_{j \in \bar{i}} r_i^k r_j^k + \sum_{j,j' \in \bar{i}} r_j^k r_{j'}^k \right] V_i(k)^2.$$

Thus (A.15b) is also an exact finite polynomial identity. Its retained D-row plus-form square is

$$U_i(k) + V_i(k) = c_k (x_{\bar{i},\bar{k}} + x_{i,\bar{k}}) - c_{\bar{k}} (x_{\bar{i},k} + x_{i,k}). \tag{23}$$

Formal row–column transpose of (A.15a). Let $\mathfrak{t}_{\mathrm{rc}}$ act on the data of a fixed principal chart by

$$\begin{aligned} &\mathfrak{t}_{\mathrm{rc}}\big(\Omega_{\mathrm{r}}, \Omega_{\mathrm{c}}; i, j, u, l;\ r_i, c_u;\ a_{i,u}, x_{i,u};\ \sigma, \tau\big) \\ &\qquad := \big(\Omega_{\mathrm{c}}, \Omega_{\mathrm{r}}; u, l, i, j;\ c_u, r_i;\ a_{u,i}^{\mathsf{T}}, x_{u,i}^{\mathsf{T}};\ \tau, \sigma\big), \end{aligned}$$

where

$$a_{u,i}^{\mathsf{T}} := a_{i,u}, \qquad x_{u,i}^{\mathsf{T}} := x_{i,u}.$$

Here B_{c} is the oriented column side fixed in (A.15a); write B_{r} for its row-side counterpart after transposition. Extend $\mathfrak{t}_{\mathrm{rc}}$ to every finite polynomial in these data by substitution. The same exchange transports the aggregate opposite blocks and ordered-pair relations:

$$\bar{i} \longleftrightarrow \bar{u}, \qquad I_{\mathrm{r}} \longleftrightarrow I_{\mathrm{c}}, \qquad D_{\mathrm{r}} \longleftrightarrow D_{\mathrm{c}}, \qquad B_{\mathrm{c}} \longleftrightarrow B_{\mathrm{r}}, \qquad a_{\mathrm{r}} \cup b_{\mathrm{r}} \longleftrightarrow a_{\mathrm{c}} \cup b_{\mathrm{c}}.$$

On determinant occurrences this means

$$\mathfrak{t}_{\mathrm{rc}}(T[\xi, \eta]) = T[\eta, \xi], \qquad \mathfrak{t}_{\mathrm{rc}}\big(T[\xi^+, \eta]\big) = T[\eta^+, \xi].$$

Thus $\mathfrak{t}_{\mathrm{rc}}^2 = \mathrm{id}$; after the two arguments are returned to the original row–column order, determinant symmetry gives $T[\eta^+, \xi] = T[\xi, \eta^+]$.

The resulting column-coordinate residual can be written explicitly. Fix an oriented side B_{r} of the central row split and put

$$\check{r}^i_u := \frac{1}{2}\left(\frac{a_{i,u}}{r_i} + \frac{a^-_{\bar{i},u}}{r^-_{\bar{i}}}\right), \qquad \nu_u := \sum_{i\in\Omega_{\mathrm{r}}} r_i \check{r}^i_u, \qquad \nu_{\bar{u}} := \sum_{l\in\bar{u}} \nu_l.$$

For a fixed row atom i, the family $\check{r}^i_u$, $u \in \Omega_{\mathrm{c}}$, gives the column-coordinate one-sided neighboring-correction weights. The quantity ν_u is their row-mass average, and is therefore the effective weight of the column atom u after the row-side correction has been transposed. Explicitly,

$$\nu_u = \frac{1}{2}\sum_{i\in\Omega_{\mathrm{r}}}\left(a_{i,u} + \frac{r_i}{r^-_{\bar{i}}}a^-_{\bar{i},u}\right) = \frac{1}{2}c_u + \frac{1}{2}\sum_{i\in\Omega_{\mathrm{r}}}\frac{r_i}{r^-_{\bar{i}}}a^-_{\bar{i},u}.$$

Thus ν_u need not equal the ordinary column marginal c_u. The notation $\nu_{\bar{u}}$ does not denote the weight of a new atom: $\bar{u}$ is the whole column block opposite to u, and $\nu_{\bar{u}}$ is the sum of the effective weights of all atoms in that block. For every fixed i,

$$\sum_{u\in\Omega_{\mathrm{c}}} \check{r}^i_u = \frac{1}{2}\left(\frac{\sum_{u\in\Omega_{\mathrm{c}}} a_{i,u}}{r_i} + \frac{\sum_{u\in\Omega_{\mathrm{c}}} a^-_{\bar{i},u}}{r^-_{\bar{i}}}\right) = 1, \qquad \sum_{u\in\Omega_{\mathrm{c}}} \nu_u = 1.$$

In the notation of (A.15a), the same substitution sends

$$r^p_i \longmapsto \check{r}^i_u, \qquad \omega_i \longmapsto \nu_u, \qquad x^-_{\bar{i},B_{\mathrm{c}}} \longmapsto x_{B_{\mathrm{r}},\bar{u}}, \qquad x_{i,B_{\mathrm{c}}} \longmapsto x_{B_{\mathrm{r}},u}.$$

In particular, the definition of ν_u is forced by the transpose of $\omega_i = \sum_p c_p r^p_i$: the column mass c_p becomes the row mass r_i, and the corrected row weight r^p_i becomes $\check{r}^i_u$. The normalization above is exactly what is required in order to apply Lemma A.2 in the transposed column coordinate. Define

$$\begin{aligned}
\widetilde{\mathcal{R}}_1 &:= \sum_{(u,l)\in I_{\mathrm{c}}} \nu_u\nu_l\, x^2_{B_{\mathrm{r}},\bar{u}} \\
&\quad + \sum_{(u,l)\in D_{\mathrm{c}}} \nu_u\nu_l\big(x_{B_{\mathrm{r}},\bar{u}} + x_{B_{\mathrm{r}},u}\big)^2, \\
\widetilde{\mathcal{Q}}_1 &:= \sum_{u\in a_{\mathrm{c}}\cup b_{\mathrm{c}}} \nu_{\bar{u}}\big(\nu_u + \nu_{\bar{u}}\big)x^2_{B_{\mathrm{r}},u}.
\end{aligned}$$

Equivalently,

$$\widetilde{\mathcal{R}}_1 := \mathfrak{t}_{\mathrm{rc}}(\mathcal{R}_1), \qquad \widetilde{\mathcal{Q}}_1 := \mathfrak{t}_{\mathrm{rc}}(\mathcal{Q}_1).$$

Hence the factors $\nu_u\nu_l$ are the effective ordered-pair coefficients in the transposed retained terms, while $\nu_{\bar{u}}(\nu_u + \nu_{\bar{u}})$ is the corresponding removed endpoint-square coefficient. The quantities ν_u and $\nu_{\bar{u}}$ therefore enter the proof through the actual coefficients of $\widetilde{\mathcal{R}}_1$ and $\widetilde{\mathcal{Q}}_1$, rather than as detached bookkeeping notation. Applying $\mathfrak{t}_{\mathrm{rc}}$ to (A.15a) and using determinant symmetry gives

$$\begin{aligned}
U - C &= T[\sigma,\tau] - T[\sigma,\tau^+] \\
&= T[\tau,\sigma] - T[\tau^+,\sigma] = \widetilde{\mathcal{R}}_1 - \widetilde{\mathcal{Q}}_1.
\end{aligned}$$

Interface with bridge reversal. The row–column transpose $\mathfrak{t}_{\mathrm{rc}}$ and bridge reversal ρ are distinct operations. The former exchanges the two coordinates in the data of a fixed principal chart and produces the effective column weights ν_u used in the exact source identity for $U - C$. The latter reverses an ordered adjacent bridge, passes to a different principal chart, and is used only after the sum over the complete ordered-bridge range has been written explicitly. In particular, no termwise cross-chart identity between two ν-weights is asserted. The relevant dependency chain is

$$\begin{aligned}(\check{r}^{i}_{u}, \nu_u, \nu_{\bar{u}}) \implies (\widetilde{\mathcal{R}}_1, \widetilde{\mathcal{Q}}_1) \implies U - C,\\ U - C \implies 4U - 2R - 2C + B \implies \text{global square-face pairing under } \rho.\end{aligned}$$

Before the last implication, the seven source terms are regrouped by (A.29b). The subsequent bridge reversal therefore acts on the block-additive occurrence weights and their exposed square faces over a complete reversal orbit; it does not identify ν_u with a value from another chart. This ν-calculation is a consistency derivation of the transposed residual and of the seven-source numerator. The formal global proof below uses the occurrence-wise settlement in Lemma A.4; no separate cross-chart identification is inferred from the ν-weights.

How the residual identities enter $4U-2R-2C+B$. Here and below, the *seven source terms* mean exactly the seven polynomial entries $U, 2\mathcal{R}_1, \widetilde{\mathcal{R}}_1, -\mathcal{R}_2, -2\mathcal{Q}_1, -\widetilde{\mathcal{Q}}_1, \mathcal{Q}_2$ displayed in (A.29a); the word "source" records their determinant-level origin and introduces no additional operator. Equation (A.15a), its formal row–column transpose, and equation (A.15b) are substituted into the complete expression (17). These three substitutions first give

$$R = U - \mathcal{R}_1 + \mathcal{Q}_1, \qquad C = U - \widetilde{\mathcal{R}}_1 + \widetilde{\mathcal{Q}}_1, \qquad B = C - \mathcal{R}_2 + \mathcal{Q}_2.$$

Substitution, without any regrouping, gives

$$\begin{aligned}4U - 2R - 2C + B &= 4U - 2(U - \mathcal{R}_1 + \mathcal{Q}_1) - C - \mathcal{R}_2 + \mathcal{Q}_2\\ &= U + 2\mathcal{R}_1 + \widetilde{\mathcal{R}}_1 - \mathcal{R}_2 - 2\mathcal{Q}_1 - \widetilde{\mathcal{Q}}_1 + \mathcal{Q}_2.\end{aligned}$$

This calculation is recorded later as the seven-source identity (A.29a). The family $\mathcal{Q}_2$ enters with positive sign. The negative diagonal terms from $\mathcal{Q}_1$ and from the endpoint diagonal of the neighboring correction become the fifth row of (A.30a). No term is assigned by an extra convention: all signs come directly from the four coefficients $4, -2, -2, 1$.

We shall also use the following side-relabeling identity. Every full row sum and every full column sum of the centered table is zero. Reversing either local side therefore changes the relevant linear form only by an overall sign, and hence

$$x^2_{\bar{i},\bar{k}} = x^2_{\bar{i},k^+}, \qquad (x_{\bar{i},\bar{k}} + x_{i,\bar{k}})^2 = (x_{\bar{i},k^+} + x_{i,k^+})^2. \tag{A.16}$$

Substituting the finite expansion of $\omega_i\omega_j$ into (19), applying the common-sign relabelling above, and then expanding the D-row plus form gives

$$\begin{aligned}\mathcal{R}_1 &= \sum_{(i,j)\in I_{\mathrm{r}}} \sum_{(k,l)\in\Omega^2_{\mathrm{c}}} r^k_i c_k r^l_j c_l x^2_{\bar{i},\bar{k}}\\ &\quad + \sum_{(i,j)\in D_{\mathrm{r}}} \sum_{(k,l)\in\Omega^2_{\mathrm{c}}} r^k_i c_k r^l_j c_l (x_{\bar{i},\bar{k}} + x_{i,\bar{k}})^2\\ &= \sum_{(i,j)\in I_{\mathrm{r}}} \sum_{(k,l)\in\Omega^2_{\mathrm{c}}} r^k_i c_k r^l_j c_l x^2_{\bar{i},\bar{k}}\\ &\quad + \sum_{(i,j)\in D_{\mathrm{r}}} \sum_{(k,l)\in\Omega^2_{\mathrm{c}}} r^k_i c_k r^l_j c_l \left(x^2_{\bar{i},\bar{k}} + 2x_{\bar{i},\bar{k}}x_{i,\bar{k}} + x^2_{i,\bar{k}}\right)\end{aligned} \tag{A.17}$$

By (A.16), the relabelling records the same block square, not a new one. Thus (A.17) is exactly the retained part of the one-sided row decomposition.

Now apply the same row-index partition to the neighboring correction in the second tree. The row-index classes I_{r} and D_{r} are unchanged, because the correction is now in the column tree. The correction in the column tree is supported only on endpoint directions $k \in a_{\mathrm{c}} \cup b_{\mathrm{c}}$. By (21),

$$U_i(k) + V_i(k) = c_k(x_{\bar{i},\bar{k}} + x_{i,\bar{k}}) - c_{\bar{k}}(x_{\bar{i},k} + x_{i,k}).$$

Direct substitution of $U_i(k)$ and this displayed plus form into (21), followed only by interchanging finite sums, gives

$$\begin{aligned}\mathcal{R}_2 = &\sum_{(i,j)\in I_{\mathrm{r}}} \sum_{k\in a_{\mathrm{c}}\cup b_{\mathrm{c}}} r_i^k r_j^k (c_k x_{\bar{i},\bar{k}} - c_{\bar{k}} x_{\bar{i},k})^2 \\ &+ \sum_{(i,j)\in D_{\mathrm{r}}} \sum_{k\in a_{\mathrm{c}}\cup b_{\mathrm{c}}} r_i^k r_j^k \left[c_k(x_{\bar{i},\bar{k}} + x_{i,\bar{k}}) - c_{\bar{k}}(x_{\bar{i},k} + x_{i,k})\right]^2.\end{aligned} \tag{A.18}$$

This proves (A.18) before any monomial is collected. We now expand its two square integrands in order to read their endpoint-indexed first square, mixed, and second square faces:

$$\begin{aligned}(c_k x_{\bar{i},\bar{k}} - c_{\bar{k}} x_{\bar{i},k})^2 &= c_k^2 x_{\bar{i},\bar{k}}^2 - 2c_k c_{\bar{k}} x_{\bar{i},\bar{k}} x_{\bar{i},k} + c_{\bar{k}}^2 x_{\bar{i},k}^2, \\ \left[c_k(x_{\bar{i},\bar{k}} + x_{i,\bar{k}}) - c_{\bar{k}}(x_{\bar{i},k} + x_{i,k})\right]^2 &= c_k^2 (x_{\bar{i},\bar{k}} + x_{i,\bar{k}})^2 \\ &\quad - 2c_k c_{\bar{k}} (x_{\bar{i},\bar{k}} + x_{i,\bar{k}})(x_{\bar{i},k} + x_{i,k}) \\ &\quad + c_{\bar{k}}^2 (x_{\bar{i},k} + x_{i,k})^2.\end{aligned}$$

Because $\mathcal{R}_2$ has coefficient -1 in the seven-source numerator, these three endpoint-indexed monomial faces enter there with signs $-$, $+$, and $-$, respectively. In the I-row, the first square face produces (A.19) and the mixed face produces (A.20). The D-row has the same three numerical coefficients with the ordinary variables replaced by the plus-form variables. In both rows the second endpoint square face remains present until the explicit bridge-reversal cancellation (A.29e).

Role of the calligraphic residual calculation in the later occurrence proof. The variables $\mathcal{R}_1, \mathcal{Q}_1, \widetilde{\mathcal{R}}_1, \widetilde{\mathcal{Q}}_1, \mathcal{R}_2, \mathcal{Q}_2$ are not a second family of target quantities. They resolve the four determinant levels at one fixed principal chart and make the algebraic provenance of every retained and omitted square visible. Their first role is to prove

$$U - R = \mathcal{R}_1 - \mathcal{Q}_1, \qquad U - C = \widetilde{\mathcal{R}}_1 - \widetilde{\mathcal{Q}}_1, \qquad C - B = \mathcal{R}_2 - \mathcal{Q}_2,$$

and hence the seven-source identity (A.29a) and its direct regrouping (A.29b). After that regrouping has been established, the formal global proof works with fixed column occurrences and no longer needs to carry the calligraphic symbols as separate notation.

The tilded pair is the formally transposed provenance of $U - C$. It does not create an additional occurrence family: it is used in the second bracket of (A.29b), where

$$\begin{aligned}U + \mathcal{R}_1 - \mathcal{Q}_1 &= 2U - R, \\ U - \widetilde{\mathcal{R}}_1 + \widetilde{\mathcal{Q}}_1 + \mathcal{R}_2 - \mathcal{Q}_2 &= C + (C - B) = 2C - B.\end{aligned}$$

This explains why the calligraphic calculation disappears from the notation after (A.29b) while its algebraic content remains in the two occurrence polynomials $2U - R$ and $2C - B$. Their coefficients come directly from

$$4U - 2R - 2C + B = 2(2U - R) - (2C - B).$$

No occurrence symbol is used at this stage. The exact crosswalk from $\mathcal{R}_1, \mathcal{Q}_1, \mathcal{R}_2, \mathcal{Q}_2$ to the occurrence quantities is stated immediately after Lemma A.4, once all occurrence quantities and the boundary polynomial have been formally defined. That later crosswalk also records the full ordered-adjacent summation domain and shows explicitly where the omitted square and the boundary polynomial enter.
The geometry in Figure 2 gives

$$K^+ = K \sqcup H, \qquad L^+ = L \sqcup H, \qquad 1 - c_K = c_{L^+}, \qquad 1 - c_L = c_{K^+}.$$

Here and in the bridge-reversal lemma below, if $A \subseteq X$ is a taxon block then

$$c_A := \frac{|A|}{N}, \qquad a_{i,A} := \frac{|R_i \cap A|}{N}.$$

Thus subscripts such as K, L, K^+, L^+ name taxon blocks, whereas lower-case subscripts such as $k, \bar{k}, k^+$ name the corresponding atom index or aggregate index block in a fixed chart. These four block identities are the source of the coefficient correspondences $(A.22) \leftrightarrow (A.23)$ and $(A.24) \leftrightarrow (A.25)$.

The next lemma gives the coefficient-transfer mechanism used at every bridge-reversal step. It applies to the complete ordered-bridge sum, not to a fixed chart in isolation.

Lemma A.3 (Bridge-reversal correspondence of the endpoint square faces)**.** *Let $\alpha = (e, e')$ be an ordered pair of adjacent relevant edges. Removing e, e' gives three blocks $(K_\alpha, H_\alpha, L_\alpha)$, where K_α and L_α are the endpoint blocks and H_α is the middle block. For the reversed chart $\rho\alpha = (e', e)$,*

$$(K_{\rho\alpha}, H_{\rho\alpha}, L_{\rho\alpha}) = (L_\alpha, H_\alpha, K_\alpha).$$

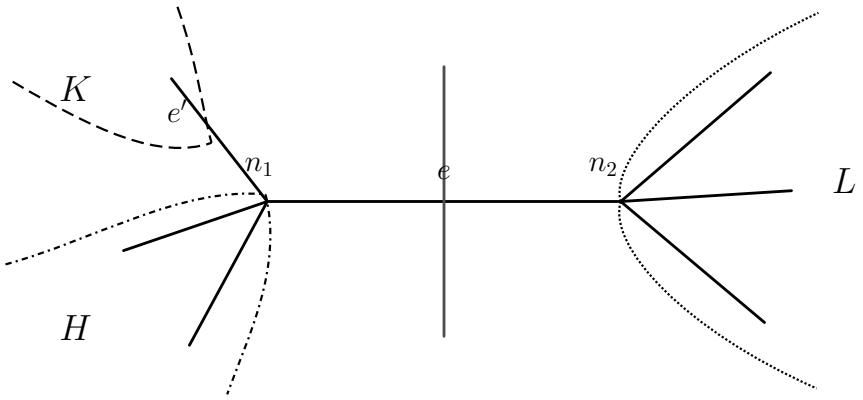


Figure 2: The three-block geometry of an ordered bridge chart $\alpha = (e, e')$. The principal edge e and its adjacent selected edge e' meet at n_1; n_2 is the other endpoint of e. If k is the endpoint atom selected by e', then $K = C_k$, $H = C_{k^+ \setminus \{k\}}$, and $L = C_{\bar{k}}$. The long-dashed, dash-dotted, and dotted curves delimit K, H, L, respectively. The solid transversal bar marks the split induced by e; neither it nor the block boundaries are additional tree edges. Reversal of the ordered pair reindexes (K, H, L) as (L, H, K).

For every test function $Z : X \to \mathbb{R}$ satisfying $\sum_{t\in X} Z(t) = 0$, define, for every taxon block $A \subseteq X$,

$$x_{Z,A} := \frac{1}{N} \sum_{t\in A} Z(t),$$

and put

$$X_\alpha(Z) := x_{Z,L_\alpha}, \qquad Y_\alpha(Z) := x_{Z,K_\alpha}.$$

Then

$$\Lambda_\alpha(Z) = c_{K_\alpha} X_\alpha(Z) - c_{L_\alpha} Y_\alpha(Z)$$

satisfies $\Lambda_{\rho\alpha}(Z) = -\Lambda_\alpha(Z)$. When Z is fixed, write $X_\alpha := X_\alpha(Z)$ and $Y_\alpha := Y_\alpha(Z)$. For the plus form, also fix all data in the other coordinate: a row principal chart and a row relation (i, j), with $\bar{i}$ and $\bar{j}$ denoting the corresponding opposite row blocks. These spectator data are held fixed when α is replaced by $\rho\alpha$. Put

$$X^D_\alpha(i) := x_{\bar{i},L_\alpha} + x_{i,L_\alpha}, \qquad Y^D_\alpha(i) := x_{\bar{i},K_\alpha} + x_{i,K_\alpha},$$

so that

$$\Lambda^D_\alpha(i) = c_{K_\alpha} X^D_\alpha(i) - c_{L_\alpha} Y^D_\alpha(i)$$

satisfies the same reversal identity. Moreover, define the two symmetric coefficients

$$r^\alpha_s = \frac{1}{2}\left(\frac{a_{s,K_\alpha}}{c_{K_\alpha}} + \frac{a_{s,L_\alpha}}{c_{L_\alpha}}\right), \qquad s \in \{i, j\}.$$

Both are invariant under reversal: $r^{\rho\alpha}_i = r^\alpha_i$ and $r^{\rho\alpha}_j = r^\alpha_j$.

Consequently, for either the ordinary pair (X_α, Y_α) or the D-row plus-form pair $(X^D_\alpha(i), Y^D_\alpha(i))$, reversal interchanges the first and second endpoint square faces while preserving their symmetric weights. Since ρ is a fixed-point-free involution of the full ordered-adjacent summation range, the X^2 coefficients in one orientation therefore give the Y^2 coefficients after global reindexing. At the ordinary-polynomial level, the map sends the second endpoint square face $c^2_{L_\alpha} r^\alpha_i r^\alpha_j Y^2_\alpha$ to $c^2_{K_{\rho\alpha}} r^{\rho\alpha}_i r^{\rho\alpha}_j X^2_{\rho\alpha}$. This lemma asserts only that abstract transport statement. Its concrete application is carried out after the four completion formulas have been introduced and expanded.

Proof. Reversing the ordered pair interchanges K_α and L_α and leaves H_α fixed, so both displayed contrasts change only by an overall sign. The formula for each r^α_s, $s \in \{i, j\}$, is symmetric in K_α, L_α, so both coefficients are reversal-invariant. Also $c_{K_{\rho\alpha}} = c_{L_\alpha}$ and $X_{\rho\alpha}(Z)^2 = Y_\alpha(Z)^2$, with the analogous identity for the plus pair. These identities prove the endpoint-square-face correspondence directly. No completion formula is used at this point. The later completion-coefficient table verifies the four concrete substitutions. Finally $\rho^2 = \mathrm{id}$ and $\rho\alpha \neq \alpha$, proving that reversal is a bijective reindexing of the full ordered-chart range. □

In the forward chart, the notation of Figure 2 means

$$(K, H, L) = \big(C_k, C_{k^+\setminus\{k\}}, C_{\bar{k}}\big).$$

In the reversed chart the selected, middle, and opposite taxon blocks are (L, H, K). Hence

$$1 - c_K = c_H + c_L = c_{L^+}, \qquad 1 - c_L = c_H + c_K = c_{K^+}.$$

If k is the selected atom index in the forward chart and k^ρ is the selected atom index in the reversed chart, these equalities read $1 - c_k = c_{(k^\rho)^+}$ and $1 - c_{k^\rho} = c_{k^+}$, with each aggregate taken in its own

chart. They are cross-chart block identities, not a replacement of an atom by an aggregate in one chart. This is the block symmetry behind the coefficient pairs

$$(A.22) \longleftrightarrow (A.23), \qquad (A.24) \longleftrightarrow (A.25).$$

Thus bridge reversal pairs the four square completions.

A.4 Why bridge reversal is a global reindexing

Every use of the column bridge reversal in this subsection means the bijective change of summation index $\alpha = (f, g) \mapsto \rho\alpha = (g, f)$ on the complete ordered-adjacent range. If a row principal edge e and row indices i, j are present, they are held fixed, so the principal chart changes from (e, f) to (e, g). The taxon-block triple changes from $(K_\alpha, H_\alpha, L_\alpha)$ to $(L_\alpha, H_\alpha, K_\alpha)$. Lemma A.3 therefore identifies the second-square coefficient in the chart (e, f) with the first-square coefficient in the reversed chart (e, g) and cancels the two ordinary endpoint square faces only after summation over the full ordered range. No same-chart coefficient identity is asserted. After this reindexing, the remaining coefficients are calculated directly from the seven terms in (A.29a); the row and column refinements are never identified term by term.

A.5 The available first-square coefficient: derivation of (A.21)

Fix a principal chart χ and a row relation $(i, j) \in I_{\mathrm{r}}^{\chi}$. For every $u \in \Omega_{\mathrm{c}}^{\chi}$ put

$$X_u := x_{\bar{i},\bar{u}},$$

and, only for an endpoint $k \in a_{\mathrm{c}}^{\chi} \cup b_{\mathrm{c}}^{\chi}$, put

$$Y_k := x_{\bar{i},k}.$$

The retained I-row component of the fixed-chart seven-source numerator uses $2\mathcal{R}_1 - \mathcal{R}_2$, not a single copy of (A.17). Using

$$(\Omega_{\mathrm{c}}^{\chi})^2 = (I_{\mathrm{c}}^{\chi} \cup D_{\mathrm{c}}^{\chi}) \sqcup \Delta_{a_{\mathrm{c}}^{\chi}} \sqcup \Delta_{b_{\mathrm{c}}^{\chi}},$$

direct expansion gives the complete identity

$$\begin{aligned} 2 \sum_{(u,l)\in(\Omega_{\mathrm{c}}^{\chi})^2} r_i^u c_u r_j^l c_l X_u^2 - \sum_{k\in a_{\mathrm{c}}^{\chi}\cup b_{\mathrm{c}}^{\chi}} r_i^k r_j^k (c_k X_k - c_{\bar{k}} Y_k)^2 \\ = 2 \sum_{(u,l)\in I_{\mathrm{c}}^{\chi}\cup D_{\mathrm{c}}^{\chi}} r_i^u c_u r_j^l c_l X_u^2 \\ + \sum_{k\in a_{\mathrm{c}}^{\chi}\cup b_{\mathrm{c}}^{\chi}} r_i^k r_j^k \big(c_k^2 X_k^2 + 2 c_k c_{\bar{k}} X_k Y_k - c_{\bar{k}}^2 Y_k^2\big). \end{aligned}$$

Thus the negative neighboring square removes one of the two positive endpoint diagonals, changes the sign of its mixed term, and leaves a negative second endpoint square face. The formal chart-indexed notation and the complete table of the three endpoint-indexed monomial faces are given in (A.29d) and the table following it. The reversal-paired first and negative second endpoint square faces have not disappeared chartwise; their cancellation over the full bridge orbit is precisely (A.29e).

The magnitude of the first endpoint diagonal removed by $-\mathcal{R}_2$ is

$$\sum_{(i,j)\in I_{\mathrm{r}}} \sum_{k\in a_{\mathrm{c}}\cup b_{\mathrm{c}}} r_i^k r_j^k c_k^2. \tag{A.19}$$

The corresponding mixed face enters with the positive sign

$$2\sum_{(i,j)\in I_{\mathrm{r}}}\sum_{k\in a_{\mathrm{c}}\cup b_{\mathrm{c}}} r_i^k r_j^k c_k c_{\bar{k}} X_k Y_k. \tag{A.20}$$

After the endpoint square faces are paired over the complete bridge orbit, the remaining polynomial has two copies of the nonendpoint first-square pool and one copy of the mixed face. One first-square copy together with the mixed face forms the following first-square completion ledger: Here and below, the superscript rep means *representative-square ledger*.

$$\begin{aligned}\mathcal{C}_I^{\mathrm{rep}} := \sum_{(i,j)\in I_{\mathrm{r}}^{\chi}}\Bigg[&\sum_{(u,l)\in I_{\mathrm{c}}^{\chi}\cup D_{\mathrm{c}}^{\chi}} r_i^u c_u r_j^l c_l X_u^2 \\ &+2\sum_{k\in a_{\mathrm{c}}^{\chi}\cup b_{\mathrm{c}}^{\chi}} r_i^k r_j^k c_k c_{\bar{k}} X_k Y_k\Bigg].\end{aligned} \tag{A.21}$$

Equation (A.21) is therefore a first-square completion ledger: it records both the available X^2 coefficient and the mixed XY term that must be absorbed. It is neither a single square face nor the complete fixed-chart polynomial $2\mathcal{R}_1 - \mathcal{R}_2$. Lemma A.3 supplies the matching second-square coefficient by reindexing the complete global sum through the reversed chart. It does not replace k by $\bar{k}$ within a fixed chart.

The D-row calculation is parallel and is recorded explicitly. For $(i,j)\in D_{\mathrm{r}}^{\chi}$, put

$$X_u^D := x_{\bar{i},\bar{u}} + x_{i,\bar{u}}, \qquad Y_k^D := x_{\bar{i},k} + x_{i,k}.$$

Its parallel first-square completion ledger is

$$\begin{aligned}\mathcal{C}_D^{\mathrm{rep}} := \sum_{(i,j)\in D_{\mathrm{r}}^{\chi}}\Bigg[&\sum_{(u,l)\in I_{\mathrm{c}}^{\chi}\cup D_{\mathrm{c}}^{\chi}} r_i^u c_u r_j^l c_l (X_u^D)^2 \\ &+2\sum_{k\in a_{\mathrm{c}}^{\chi}\cup b_{\mathrm{c}}^{\chi}} r_i^k r_j^k c_k c_{\bar{k}} X_k^D Y_k^D\Bigg].\end{aligned}$$

The complete table of its three endpoint-indexed monomial faces is obtained from the displayed I-row table by the substitution $(X_k, Y_k)\mapsto(X_k^D, Y_k^D)$. Hence the ordinary and D-row plus-form rows use the same numerical coefficients but remain separate polynomial families.

A.6 Completion of the cross term: equations (A.22)–(A.25)

We now split the mixed coefficient in (A.21) among four completed squares. For $k\in a_{\mathrm{c}}^{\chi}\cup b_{\mathrm{c}}^{\chi}$,

$$r_i^k r_j^k c_k c_{\bar{k}} = \frac{1}{4}\left(\frac{c_{\bar{k}}}{c_k}a_{i,k}a_{j,k} + a_{i,k}a_{j,\bar{k}} + a_{i,\bar{k}}a_{j,k} + \frac{c_k}{c_{\bar{k}}}a_{i,\bar{k}}a_{j,\bar{k}}\right). \tag{A.21a}$$

The selected endpoint block, its opposite block, and its same-side aggregate are nonempty. Hence

$$c_k > 0, \qquad c_{\bar{k}} > 0, \qquad c_{k^+} > 0, \qquad 1 - c_k > 0,$$

so every denominator and square root in the next four displays is well defined. Thus the cross term in (A.21) is absorbed by the following four square sums:

$$\frac{1}{4}\sum_{(i,j)\in I_{\mathrm{r}}^{\chi}}\sum_{k\in a_{\mathrm{c}}^{\chi}\cup b_{\mathrm{c}}^{\chi}} c_k\frac{a_{i,\bar{k}}a_{j,\bar{k}}}{c_{\bar{k}}}\left(\sqrt{\frac{1-c_k}{c_{\bar{k}}}}x_{\bar{i},\bar{k}} + \sqrt{\frac{c_{\bar{k}}}{1-c_k}}x_{\bar{i},k}\right)^2, \tag{A.22}$$

$$\frac{1}{4}\sum_{(i,j)\in I_{\mathrm{r}}^{\chi}}\sum_{k\in a_{\mathrm{c}}^{\chi}\cup b_{\mathrm{c}}^{\chi}} c_{\bar{k}}\frac{a_{i,k}a_{j,k}}{c_k}\left(\sqrt{\frac{c_k}{c_{k^+}}}x_{\bar{i},\bar{k}}+\sqrt{\frac{c_{k^+}}{c_k}}x_{\bar{i},k}\right)^2, \tag{A.23}$$

$$\frac{1}{4}\sum_{(i,j)\in I_{\mathrm{r}}^{\chi}}\sum_{k\in a_{\mathrm{c}}^{\chi}\cup b_{\mathrm{c}}^{\chi}} a_{i,k}a_{j,\bar{k}}(x_{\bar{i},\bar{k}}+x_{\bar{i},k})^2, \tag{A.24}$$

$$\frac{1}{4}\sum_{(i,j)\in I_{\mathrm{r}}^{\chi}}\sum_{k\in a_{\mathrm{c}}^{\chi}\cup b_{\mathrm{c}}^{\chi}} a_{i,\bar{k}}a_{j,k}(x_{\bar{i},\bar{k}}+x_{\bar{i},k})^2. \tag{A.25}$$

The numerical coefficients are justified by the following elementary identities. For any scalar γ and formal variables X, Y,

$$\frac{1}{4}\gamma(X+Y)^2=\frac{1}{4}\gamma X^2+\frac{1}{2}\gamma XY+\frac{1}{4}\gamma Y^2.$$

More generally, for any positive scalar s,

$$\frac{1}{4}\gamma(sX+s^{-1}Y)^2=\frac{1}{4}\gamma s^2X^2+\frac{1}{2}\gamma XY+\frac{1}{4}\gamma s^{-2}Y^2.$$

Fix the chart χ and one triple (i,j,k), and in the corresponding summand of (A.22)–(A.25) write $X := x_{\bar{i},\bar{k}}$ and $Y := x_{\bar{i},k}$. For any polynomial $F(X,Y)$ and monomial M, write $[M]F := \mathrm{coeff}_M(F)$. For

$$t\in\{22,23,24,25\},\qquad M\in\{X^2,XY,Y^2\},$$

define

$$[M]_t := \mathrm{coeff}_M\left(\text{the fixed } (i,j,k)\text{-summand of equation (A.}t\text{)}\right).$$

Thus the subscript t names the source equation, while the expression inside the brackets names the monomial whose coefficient is extracted. The dependence on χ, i, j, k is suppressed only to keep the table readable. For example, the outside coefficient and the square of the X-multiplier in (A.22) give

$$[X^2]_{22}=\left(\frac{1}{4}c_k\frac{a_{i,\bar{k}}a_{j,\bar{k}}}{c_{\bar{k}}}\right)\left(\frac{1-c_k}{c_{\bar{k}}}\right)=\frac{1}{4}\frac{c_k(1-c_k)}{c_{\bar{k}}^2}a_{i,\bar{k}}a_{j,\bar{k}}.$$

Direct expansion gives all twelve coefficients:

$$\begin{aligned}
&[X^2]_{22}=\frac{1}{4}\frac{c_k(1-c_k)}{c_{\bar{k}}^2}a_{i,\bar{k}}a_{j,\bar{k}}, && [XY]_{22}=\frac{1}{2}\frac{c_k}{c_{\bar{k}}}a_{i,\bar{k}}a_{j,\bar{k}},\\
&[Y^2]_{22}=\frac{1}{4}\frac{c_k}{1-c_k}a_{i,\bar{k}}a_{j,\bar{k}},\\
&[X^2]_{23}=\frac{1}{4}\frac{c_{\bar{k}}}{c_{k^+}}a_{i,k}a_{j,k}, && [XY]_{23}=\frac{1}{2}\frac{c_{\bar{k}}}{c_k}a_{i,k}a_{j,k},\\
&[Y^2]_{23}=\frac{1}{4}\frac{c_{\bar{k}}c_{k^+}}{c_k^2}a_{i,k}a_{j,k},\\
&[X^2]_{24}=\frac{1}{4}a_{i,k}a_{j,\bar{k}}, && [XY]_{24}=\frac{1}{2}a_{i,k}a_{j,\bar{k}}, && [Y^2]_{24}=\frac{1}{4}a_{i,k}a_{j,\bar{k}},\\
&[X^2]_{25}=\frac{1}{4}a_{i,\bar{k}}a_{j,k}, && [XY]_{25}=\frac{1}{2}a_{i,\bar{k}}a_{j,k}, && [Y^2]_{25}=\frac{1}{4}a_{i,\bar{k}}a_{j,k}.
\end{aligned}$$

Define the chart-indexed first- and second-square coefficients by

$$p_{ij}^{\chi}(k):=\sum_{t=22}^{25}[X^2]_t,\qquad q_{ij}^{\chi}(k):=\sum_{t=22}^{25}[Y^2]_t.$$

When the chart is fixed, we suppress the superscript χ and write $p_{ij}(k), q_{ij}(k)$. Adding the four mixed coefficients and using (A.21a) gives

$$\sum_{t=22}^{25}[XY]_t = 2c_k c_{\bar{k}} r_i^k r_j^k,$$

which is exactly the coefficient in (A.20).

The total first- and second-square coefficients consumed by the completed package are therefore

$$\begin{aligned}
4p_{ij}(k) &= \frac{c_k(1-c_k)}{c_{\bar{k}}^2} a_{i,\bar{k}} a_{j,\bar{k}} + \frac{c_{\bar{k}}}{c_{k^+}} a_{i,k} a_{j,k} \\
&\quad + a_{i,k} a_{j,\bar{k}} + a_{i,\bar{k}} a_{j,k}, \\
4q_{ij}(k) &= \frac{c_k}{1-c_k} a_{i,\bar{k}} a_{j,\bar{k}} + \frac{c_{\bar{k}} c_{k^+}}{c_k^2} a_{i,k} a_{j,k} \\
&\quad + a_{i,k} a_{j,\bar{k}} + a_{i,\bar{k}} a_{j,k}.
\end{aligned}$$

Finally, let α have three-block data (K, H, L) and let $\rho\alpha$ have (L, H, K). Since $1 - c_K = c_{L^+}$ and $1 - c_L = c_{K^+}$, the preceding twelve coefficients give the four coefficient-preserving identities below. In this display, $[M](\mathrm{A}.t)_\alpha$ means the coefficient of M in the fixed (i, j, k)-summand of equation $(\mathrm{A}.t)$ belonging to the ordered bridge α:

$$\begin{aligned}
&[Y^2](A.22)_\alpha = [X^2](A.23)_{\rho\alpha}, \quad [Y^2](A.23)_\alpha = [X^2](A.22)_{\rho\alpha}, \\
&[Y^2](A.24)_\alpha = [X^2](A.25)_{\rho\alpha}, \quad [Y^2](A.25)_\alpha = [X^2](A.24)_{\rho\alpha}.
\end{aligned}$$

This is the explicit coefficient calculation behind $(A.22) \leftrightarrow (A.23)$ and $(A.24) \leftrightarrow (A.25)$; no same-chart substitution is used.

A.7 Expansion of the available square coefficient: equations (A.26)–(A.29)

It remains to express the available first-square coefficient in the common a-monomial language used in the final collection. The two factors in the square coefficient satisfy

$$2c_k r_i^k = a_{i,k} + \frac{c_k}{c_{\bar{k}}} a_{i,\bar{k}}, \qquad 2c_l r_j^l = a_{j,l} + \frac{c_l}{c_{\bar{l}}} a_{j,\bar{l}}.$$

Multiplying these two binomials, without suppressing any term, gives

$$\begin{aligned}
4r_i^k c_k r_j^l c_l &= \left(a_{i,k} + \frac{c_k}{c_{\bar{k}}} a_{i,\bar{k}}\right)\left(a_{j,l} + \frac{c_l}{c_{\bar{l}}} a_{j,\bar{l}}\right) \\
&= a_{i,k} a_{j,l} + \frac{c_l}{c_{\bar{l}}} a_{i,k} a_{j,\bar{l}} + \frac{c_k}{c_{\bar{k}}} a_{i,\bar{k}} a_{j,l} + \frac{c_k c_l}{c_{\bar{k}} c_{\bar{l}}} a_{i,\bar{k}} a_{j,\bar{l}}.
\end{aligned} \tag{A.26–A.29}$$

The four summands reproduce, in order, the original equations (26), (27), (28), and (29). In particular, the denominator $c_{\bar{l}}$ belongs to the term containing $a_{j,\bar{l}}$, whereas $c_{\bar{k}}$ belongs to the term containing $a_{i,\bar{k}}$.

A.8 From the four determinant levels to four square families

We now assemble the preceding local identities. All equalities are equalities of quadratic polynomials. The order is important: first expand the four determinant levels into seven terms, then decompose the row coordinate for each fixed column term, next pair the endpoint square faces by bridge reversal, and finally sum the remaining coefficient over each same-side column block. The row and column refinements remain distinct throughout.

Dependency map for Subsections A.4–A.7. The preliminary expansion below obtains the seven-source numerator from the determinant-level identities of Subsection A.3. Step 1 in the proof of Lemma A.5 then decomposes the row coordinate while keeping each column term fixed. Once the retained row polynomial has been isolated, Subsections A.4–A.7 enter in the following precise way:

input	object supplied to A.8	exact use inside A.8
A.4	the global bridge-reversal reindexing, with first and second square faces interchanged only in the reversed chart	equation (A.29e) cancels the ordinary endpoint square faces, and the q-to-p reindexing in (A.29f) transfers the second square face of a completed package to the first square face in the reversed chart
A.5	the available first-square pool and the three endpoint-indexed monomial faces: first square, mixed, and negative second square	these terms are named and collected in (A.29d); they become the available coefficient and the contributions $d_k Z^2$, $m_k ZW$, and $-e_k W^2$ in Step 2
A.6	the four completed squares (A.22)–(A.25), together with their first, mixed, and second monomial coefficients	their sum is the completed-square package used in (A.29f); its mixed coefficient absorbs the retained cross term, while its two square coefficients record the amounts consumed on the first and second square faces
A.7	the four-term expansion (A.26)–(A.29) of the available square coefficient in the common a-monomial language	Step 3 compares that expansion with the completed-package coefficient and the six rows of the remaining coefficient, producing the block identity (A.30b) and then (A.29g)

Thus A.8 is not a second independent local calculation. It first uses A.5 to supply the uncompleted coefficient ledger, uses A.4 twice to settle the reversal-paired endpoint square faces globally, uses A.6 to complete the mixed term, and uses A.7 to identify the coefficient left after that completion. In terms of the stable equation labels, the three remaining assembly steps are

$$\begin{aligned}
&\text{Step 2:} \quad \text{(A.21)} \longrightarrow \text{(A.29d)} \longrightarrow \text{(A.29e)} \longrightarrow \text{(A.29f)},\\
&\text{Step 3:} \quad \text{(A.29f)} \longrightarrow \text{(A.30b)} \longrightarrow \text{(A.29g)},\\
&\text{Step 4:} \quad \text{(A.29g)} \longrightarrow \text{(A.29h)}.
\end{aligned}$$

Step 1 performs the fixed-column row decomposition and cancels its boundary over the complete row-bridge range. Step 2 computes the retained row polynomial, cancels its ordinary endpoint square faces, and inserts the completed-square package. Step 3 identifies the coefficient left after that completion by summing one same-side block. Step 4 restores the two row multiplicities and identifies the four resulting terms with the four nonnegative square families.

We first record the seven-source numerator used in Step 1. Using the formally defined row–column transpose of (A.15a), the exact seven-source identity is

$$\begin{aligned}
4U - 2R - 2C + B = U + 2\mathcal{R}_1 + \widetilde{\mathcal{R}}_1 - \mathcal{R}_2 \\
- 2\mathcal{Q}_1 - \widetilde{\mathcal{Q}}_1 + \mathcal{Q}_2.
\end{aligned} \tag{A.29a}$$

Indeed,

$$U - R = \mathcal{R}_1 - \mathcal{Q}_1, \qquad U - C = \widetilde{\mathcal{R}}_1 - \widetilde{\mathcal{Q}}_1, \qquad C - B = \mathcal{R}_2 - \mathcal{Q}_2.$$

Equivalently,

$$R = U - \mathcal{R}_1 + \mathcal{Q}_1, \qquad C = U - \widetilde{\mathcal{R}}_1 + \widetilde{\mathcal{Q}}_1, \qquad B = C - \mathcal{R}_2 + \mathcal{Q}_2.$$

Therefore

$$\begin{aligned} 4U - 2R - 2C + B &= 4U - 2(U - \mathcal{R}_1 + \mathcal{Q}_1) - C - \mathcal{R}_2 + \mathcal{Q}_2 \\ &= 2U + 2\mathcal{R}_1 - 2\mathcal{Q}_1 - (U - \widetilde{\mathcal{R}}_1 + \widetilde{\mathcal{Q}}_1) - \mathcal{R}_2 + \mathcal{Q}_2 \\ &= U + 2\mathcal{R}_1 + \widetilde{\mathcal{R}}_1 - \mathcal{R}_2 - 2\mathcal{Q}_1 - \widetilde{\mathcal{Q}}_1 + \mathcal{Q}_2, \end{aligned}$$

which proves every sign and multiplicity in (A.29a).

Why the row and column refinements remain distinct. Fix a principal chart χ. For a row atom i and a column atom u, put

$$\alpha^{\chi}_{i,u} := c_u r^u_i = \frac{1}{2}\left(a_{i,u} + \frac{c_u}{c_{\bar{u}}} a_{i,\bar{u}}\right).$$

For the transposed first passage, the preceding definition gives

$$\beta^{\chi}_{i,u} := r_i \check{r}^i_u = \frac{1}{2}\left(a_{i,u} + \frac{r_i}{r_{\bar{i}}} a_{\bar{i},u}\right).$$

The two refinements need not agree. Their difference is

$$\alpha^{\chi}_{i,u} - \beta^{\chi}_{i,u} = \frac{1}{2}\left(\frac{c_u}{c_{\bar{u}}} a_{i,\bar{u}} - \frac{r_i}{r_{\bar{i}}} a_{\bar{i},u}\right).$$

This difference is not identically zero as a function of the chart data, although it may vanish for particular choices of χ, i, u. Since $\nu_u = \sum_{i \in \Omega^{\chi}_{\mathrm{r}}} \beta^{\chi}_{i,u}$, the two products

$$\omega_i \omega_j = \sum_{u,l \in \Omega^{\chi}_{\mathrm{c}}} \alpha^{\chi}_{i,u} \alpha^{\chi}_{j,l}, \qquad \nu_u \nu_l = \sum_{i,j \in \Omega^{\chi}_{\mathrm{r}}} \beta^{\chi}_{i,u} \beta^{\chi}_{j,l}$$

retain the different ordered index pairs $(i,u),(j,l)$ and are not equated term by term. Both $\alpha^{\chi}_{i,u}$ and $\beta^{\chi}_{i,u}$ are chart-local quantities. Their superscript is suppressed only inside one fixed-chart calculation. Whenever they occur inside a sum over χ, their masses, opposite blocks, and aggregates are evaluated in that summand's chart χ. These subscripted scalar weights are unrelated to the bare symbol α used for an ordered bridge.

We therefore regroup the seven terms before applying either refinement:

$$\begin{aligned} 4U - 2R - 2C + B &= 2[U + \mathcal{R}_1 - \mathcal{Q}_1] \\ &\quad - [U - \widetilde{\mathcal{R}}_1 + \widetilde{\mathcal{Q}}_1 + \mathcal{R}_2 - \mathcal{Q}_2] \\ &= 2(2U - R) - (2C - B). \end{aligned} \tag{A.29b}$$

The following table verifies the coefficient of each of the seven terms:

	U	$\mathcal{R}_1$	$\widetilde{\mathcal{R}}_1$	$\mathcal{R}_2$	$\mathcal{Q}_1$	$\widetilde{\mathcal{Q}}_1$	$\mathcal{Q}_2$
$2[U + \mathcal{R}_1 - \mathcal{Q}_1]$	2	2	0	0	−2	0	0
$-[U - \widetilde{\mathcal{R}}_1 + \widetilde{\mathcal{Q}}_1 + \mathcal{R}_2 - \mathcal{Q}_2]$	−1	0	1	−1	0	−1	1
total	1	2	1	−1	−2	−1	1

The coefficient table shows directly that the transposed bracket reconstructs C before the row residual identity is applied. This is the reason no termwise equality between $\alpha^{\chi}_{i,u}$ and $\beta^{\chi}_{i,u}$ is needed.

For $k \in a^{\chi} \cup b^{\chi}$, both k and its opposite block $\bar{k}$ are nonempty. Hence

$$c_k > 0, \qquad c_{\bar{k}} > 0, \qquad c_{k^+} \geq c_k > 0, \qquad 1 - c_k = c_{\bar{k}} + c_{k^+ \setminus \{k\}} > 0.$$

All denominators and square roots below are therefore well defined. For independent formal variables X, Y, define the completed-square polynomial

$$\begin{aligned} \mathcal{P}_{ij,k}(X,Y) := {} & \frac{1}{4} c_k \frac{a_{i,\bar{k}} a_{j,\bar{k}}}{c_{\bar{k}}} \left(\sqrt{\frac{1-c_k}{c_{\bar{k}}}} X + \sqrt{\frac{c_{\bar{k}}}{1-c_k}} Y \right)^2 \\ & + \frac{1}{4} c_{\bar{k}} \frac{a_{i,k} a_{j,k}}{c_k} \left(\sqrt{\frac{c_k}{c_{k^+}}} X + \sqrt{\frac{c_{k^+}}{c_k}} Y \right)^2 \\ & + \frac{1}{4} a_{i,k} a_{j,\bar{k}} (X+Y)^2 + \frac{1}{4} a_{i,\bar{k}} a_{j,k} (X+Y)^2. \end{aligned} \tag{24}$$

When more than one principal chart is present, we write $\mathcal{P}^{\chi}_{ij,k}$ for the polynomial whose masses and aggregates are computed in chart χ; within one fixed chart the superscript is suppressed. Also define

$$\begin{aligned} \Psi^{\chi}_{ij}(u) := {} & \sum_{l \in M_{S,\chi}(u)} \frac{c_{\bar{u}}}{c_{u^+}} a_{i,u} a_{j,l} + \sum_{l \in S_{\chi}(u)} 4 c_u c_l r^{u,\chi}_i r^{l,\chi}_j \\ & + \mathbf{1}_{u \in c^{\chi} \cup d^{\chi}} \left(a_{i,u} a_{j,\bar{u}} + a_{i,\bar{u}} a_{j,u} \right) \\ & - \sum_{l \in M_{S,\chi}(u)} \frac{c_u c_l}{c_{\bar{u}} c_{\bar{l}}} a_{i,\bar{u}} a_{j,\bar{l}} - \mathbf{1}_{u \in a^{\chi} \cup b^{\chi}} 4 c_u^2 r^{u,\chi}_i r^{u,\chi}_j \\ & + \sum_{l \in M_{D,\chi}(u)} \left(\frac{c_u}{c_{\bar{u}}} a_{i,\bar{u}} a_{j,l} + \frac{c_l}{c_{\bar{l}}} a_{i,u} a_{j,\bar{l}} \right). \end{aligned} \tag{A.30a}$$

For reference, the six rows of (A.30a) are organized as follows; the terms themselves are exactly those displayed in that equation.

row	support or index set	role in the coefficient
1	$l \in M_{S,\chi}(u)$	same-side endpoint contribution
2	$l \in S_{\chi}(u)$	same-side symmetric contribution
3	$u \in c^{\chi} \cup d^{\chi}$	nonendpoint residual contribution
4	$l \in M_{S,\chi}(u)$	same-side subtraction
5	$u \in a^{\chi} \cup b^{\chi}$	endpoint diagonal subtraction
6	$l \in M_{D,\chi}(u)$	opposite-side residual contribution

Table 1: The six terms of $\Psi^{\chi}_{ij}(u)$ in (A.30a).

Lemma A.4 (Row decomposition with one column term fixed)**.** *Fix a column principal edge $f \in E_2(T_2^{\circ})$. The column term η is either the ordinary split $\eta = \tau_f$ or one adjacent side-pair $\eta = C_{f,h}$ for one ordered adjacent pair $(f,h) \in \overrightarrow{A}_2(T_2^{\circ})$. In the second case let $k = k(f,h) \in a_f \cup b_f$ be the endpoint atom of Π_f indexed by h.*

Now fix a row principal edge $e \in E_2(T_1^\circ)$ and work in the principal chart $\chi = (e, f)$. Choose either side of the split τ_f and let $B_f \subseteq \Omega_f^{\mathrm{c}}$ be the set of column atom indices whose taxon union is that side. Put

$$S_f := \Omega_f^{\mathrm{c}} \setminus B_f.$$

Thus C_{B_f} and C_{S_f} are the two taxon sides of τ_f. For an actual row taxon block $A \subseteq X$, define

$$a_{A,p} := \frac{|A \cap C_p|}{N}, \qquad r_A^p := \frac{1}{2}\left(\frac{a_{A,p}}{c_p} + \frac{a_{A,\bar{p}}}{c_{\bar{p}}}\right),$$

and define the row-chart-independent additive weight of A for the fixed column term η by

$$w^\eta(A) := \begin{cases} \displaystyle\sum_{p \in \Omega_f^{\mathrm{c}}} c_p r_A^p, & \eta = \tau_f, \\ r_A^k, & \eta = C_{f,h}. \end{cases}$$

The masses and opposite blocks in this formula belong to the fixed column chart of f. Thus $w^\eta(A)$ depends on A and the fixed column term η, but not on the row principal edge used to partition A; moreover it is additive on disjoint taxon blocks. For $i \in \Omega_e^{\mathrm{r}}$ set $w_i^\eta := w^\eta(R_i)$ and define the two row forms by

η	w_i^η	X_i^η	Y_i^η
τ_f	$\displaystyle\sum_{p \in \Omega_f^{\mathrm{c}}} c_p r_i^p$	$x_{\bar{i},B_f}$	x_{i,B_f}
$C_{f,h}$	r_i^k	$c_k x_{\bar{i},\bar{k}} - c_{\bar{k}} x_{\bar{i},k}$	$c_k x_{i,\bar{k}} - c_{\bar{k}} x_{i,k}$

Put

$$w_{\bar{i}}^\eta := w^\eta(R_{\bar{i}}) = \sum_{j \in \bar{i}} w_j^\eta, \qquad \mathcal{E}_e := a_e \cup b_e,$$

and define the row-copy relation sets explicitly by

$$D_e := \{(i,j) \in (\Omega_e^{\mathrm{r}})^2 : i \in \mathcal{E}_e,\ j \in \bar{i}\}, \qquad \Delta_{\mathcal{E}_e} := \{(i,i) : i \in \mathcal{E}_e\},$$

$$I_e := (\Omega_e^{\mathrm{r}})^2 \setminus \big(D_e \sqcup \Delta_{\mathcal{E}_e}\big).$$

All quantities below depend on this fixed choice of η:

$$\begin{aligned} E_e^\eta &:= T[\sigma_e, \eta] = (X_i^\eta)^2 \quad \text{for any } i \in \Omega_e^{\mathrm{r}}, \\ N_e^\eta &:= \sum_{g:\,(e,g) \in \overrightarrow{A}_2(T_1^\circ)} T[C_{e,g}, \eta] = \sum_{s \in \mathcal{E}_e} \big(w_s^\eta X_s^\eta - w_{\bar{s}}^\eta Y_s^\eta\big)^2, \\ I_e^\eta &:= \sum_{(i,j) \in I_e} w_i^\eta w_j^\eta (X_i^\eta)^2, \\ D_e^\eta &:= \sum_{(i,j) \in D_e} w_i^\eta w_j^\eta (X_i^\eta + Y_i^\eta)^2, \\ \mathcal{B}_e^\eta &:= \sum_{s \in \mathcal{E}_e} (w_s^\eta + w_{\bar{s}}^\eta)\left(w_s^\eta (X_s^\eta)^2 - w_{\bar{s}}^\eta (Y_s^\eta)^2\right). \end{aligned}$$

We call the explicitly displayed polynomial $\mathcal{B}_e^\eta$ the row-boundary polynomial. *Every later reference to the row boundary means this polynomial, with the indicated e and fixed column term η. Then, for every e and every fixed η,*

$$2E_e^\eta - N_e^\eta = 2I_e^\eta + D_e^\eta + \mathcal{B}_e^\eta. \tag{A.29c$_0$}$$

Moreover, the row boundary vanishes only after summing over the full range of row principal edges:

$$\sum_{e\in E_2(T_1^\circ)} \mathcal{B}_e^\eta = 0 \qquad \textit{for every fixed } \eta. \tag{A.29c$_1$}$$

Finally, writing $U_{e,f}, R_{e,f}, C_{e,f}, B_{e,f}$ for the four polynomials of the principal chart (e,f), the identification is

η	E_e^η	N_e^η
τ_f	$U_{e,f}$	$R_{e,f}$
$C_{f,h}$	$T[\sigma_e, C_{f,h}]$	$\displaystyle\sum_{g:\,(e,g)\in\overrightarrow{A}_2(T_1^\circ)} T[C_{e,g}, C_{f,h}]$

and hence, for every fixed (e,f),

$$\begin{aligned} E_e^{\tau_f} &= U_{e,f}, & N_e^{\tau_f} &= R_{e,f}, \\ \sum_{h:\,(f,h)\in\overrightarrow{A}_2(T_2^\circ)} E_e^{C_{f,h}} &= C_{e,f}, & \sum_{h:\,(f,h)\in\overrightarrow{A}_2(T_2^\circ)} N_e^{C_{f,h}} &= B_{e,f}. \end{aligned} \tag{A.29c$_2$}$$

Thus every indexed term in U, R, C, B appears exactly once in these two ranges.

Proof. For $\eta = \tau_f$, apply Proposition A.1 to the central-side partition $\Omega_f^{\mathrm{c}} = S_f \sqcup B_f$. It identifies the summand belonging to the endpoint atom $s = s(e,g)$ with

$$T[C_{e,g}, \tau_f] = \big(w_s^{\tau_f} X_s^{\tau_f} - w_{\bar{s}}^{\tau_f} Y_s^{\tau_f}\big)^2.$$

For $\eta = C_{f,h}$ the corresponding identity is the fixed-k identity in the proof of (A.15b):

$$T[C_{e,g}, C_{f,h}] = \big(r_s^k X_s^{C_{f,h}} - r_{\bar{s}}^k Y_s^{C_{f,h}}\big)^2.$$

The bijection $g \leftrightarrow s(e,g)$ from the formal chart construction therefore proves the displayed formula for N_e^η in both cases. Changing the row orientation changes X_i^η only by sign, so $(X_i^\eta)^2 = E_e^\eta$ for every i.

Both weight systems are nonnegative and additive on actual row blocks. Their atomic normalizations are

$$\sum_{i\in\Omega_e^{\mathrm{r}}} w_i^{\tau_f} = \sum_{p\in\Omega_f^{\mathrm{c}}} c_p \sum_{i\in\Omega_e^{\mathrm{r}}} r_i^p = 1, \qquad \sum_{i\in\Omega_e^{\mathrm{r}}} w_i^{C_{f,h}} = \sum_{i\in\Omega_e^{\mathrm{r}}} r_i^k = 1.$$

Lemma A.2 applies and gives $E_e^\eta - N_e^\eta = I_e^\eta + D_e^\eta - Q_e^\eta$, where

$$Q_e^\eta = \sum_{s\in\mathcal{E}_e} w_{\bar{s}}^\eta (w_s^\eta + w_{\bar{s}}^\eta)(Y_s^\eta)^2.$$

The disjoint partition $(\Omega_e^{\mathrm{r}})^2 = I_e \sqcup D_e \sqcup \Delta_{\mathcal{E}_e}$ also gives

$$E_e^\eta - I_e^\eta = \sum_{s\in\mathcal{E}_e} (w_s^\eta + w_{\bar{s}}^\eta) w_s^\eta (X_s^\eta)^2.$$

The required identity is now obtained without suppressing the intermediate coefficient:

$$\begin{aligned} 2E_e^\eta - N_e^\eta &= (E_e^\eta - N_e^\eta) + E_e^\eta \\ &= (I_e^\eta + D_e^\eta - Q_e^\eta) + I_e^\eta + (E_e^\eta - I_e^\eta) \\ &= 2I_e^\eta + D_e^\eta + (E_e^\eta - I_e^\eta - Q_e^\eta) \\ &= 2I_e^\eta + D_e^\eta + \mathcal{B}_e^\eta. \end{aligned}$$

This proves (A.29c$_0$) and also shows exactly where the row-boundary polynomial enters.

The row reversal used in the boundary sum is the explicit map

$$\rho_{\mathrm{r}} : ((e,g),\eta) \longmapsto ((g,e),\eta);$$

in particular, the column term η remains fixed. Now fix one ordered row bridge $\alpha = (e,g)$ with taxon blocks $(K_\alpha, H_\alpha, L_\alpha)$. Let $s = s(e,g)$ be its endpoint atom in the chart of e, and define

$$X^\eta_\alpha := X^\eta_s, \qquad Y^\eta_\alpha := Y^\eta_s.$$

Using additivity on row blocks, set

$$w^\eta_{K_\alpha} := w^\eta_s, \qquad w^\eta_{L_\alpha} := w^\eta_{\bar{s}}.$$

Its endpoint summand in $\mathcal{B}^\eta_e$ is

$$(w^\eta_{K_\alpha} + w^\eta_{L_\alpha})\left(w^\eta_{K_\alpha}(X^\eta_\alpha)^2 - w^\eta_{L_\alpha}(Y^\eta_\alpha)^2\right).$$

The reversed bridge $\rho_{\mathrm{r}}\alpha = (g,e)$ has endpoint/opposite blocks (L_α, K_α). Additivity on row blocks and the fact that the fixed column term η is unchanged give

$$w^\eta_{K_{\rho_{\mathrm{r}}\alpha}} = w^\eta_{L_\alpha}, \qquad w^\eta_{L_{\rho_{\mathrm{r}}\alpha}} = w^\eta_{K_\alpha}, \qquad (X^\eta_{\rho_{\mathrm{r}}\alpha})^2 = (Y^\eta_\alpha)^2, \qquad (Y^\eta_{\rho_{\mathrm{r}}\alpha})^2 = (X^\eta_\alpha)^2.$$

The reversed summand is therefore the negative of the forward summand. Reversal is a fixed-point-free involution on $\overrightarrow{A}_2(T_1^\circ)$, proving (A.29c$_1$).

The first row of the identification table is the definition of $T[\sigma_e^+, \tau_f]$. After summing the second row over all $(f,h) \in \overrightarrow{A}_2(T_2^\circ)$, its E^η_e-entry and N^η_e-entry are, respectively, the definitions of $T[\sigma_e, \tau_f^+]$ and $T[\sigma_e^+, \tau_f^+]$. This proves (A.29c$_2$) and the claim that each indexed term appears once. □

Calligraphic-to-occurrence crosswalk. Fix the chart $\chi = (e,f)$ and put

$$H_f := \{h \in E_2(T_2^\circ) : (f,h) \in \overrightarrow{A}_2(T_2^\circ)\}.$$

Thus H_f consists of all relevant edges adjacent to f, with f first in the ordered pair. For $h \in H_f$, let $k(f,h) \in a_f \cup b_f$ index the endpoint atom $A_f(h)$ of Π_f. In the column copy of the chart,

$$C^\chi_{k(f,h)} = A_f(h).$$

This atom is not the whole adjacent side-pair correction $C_{f,h}$: if $k = k(f,h)$, then

$$C_{f,h} = (K_{f,h}, L_{f,h}) = \left(C^\chi_k, C^\chi_{\bar{k}_\chi}\right).$$

The map $h \mapsto k(f,h)$ from H_f to $a_f \cup b_f$ is bijective. For injectivity, let $h_1 \neq h_2$. If they meet the same endpoint of f, then $A_f(h_1)$ and $A_f(h_2)$ lie in two distinct branches at that endpoint; if they meet opposite endpoints, the two atoms lie in different components of $T_2^\circ - f$. In either case the atoms are nonempty and disjoint, so $k(f,h_1) \neq k(f,h_2)$. For surjectivity, recall that $a_f \cup b_f$ is defined to index precisely the atoms $A_f(h)$ arising from relevant edges h adjacent to f. Hence, for every endpoint-indexed family $F(k)$,

$$\sum_{h \in H_f} F\big(k(f,h)\big) = \sum_{k \in a_f \cup b_f} F(k).$$

Thus each ordered correction $C_{f,h}$ is in one-to-one correspondence with one endpoint atom of Π_f and is counted once. This "multiplicity one" does not concern its polynomial coefficient; residual indices in $c_f \cup d_f$ do not occur in this range. The resulting crosswalk is

column occurrence	retained square contribution	omitted square contribution
$\eta = \tau_f$	$\mathcal{R}_1 = I_e^{\tau_f} + D_e^{\tau_f}$	$\mathcal{Q}_1 = Q_e^{\tau_f}$
$\eta = C_{f,h},\ h \in H_f$	$\mathcal{R}_2 = \sum_{h\in H_f} \big(I_e^{C_{f,h}} + D_e^{C_{f,h}}\big)$	$\mathcal{Q}_2 = \sum_{h\in H_f} Q_e^{C_{f,h}}$

The first row is the residual partition behind (A.15a) and (A.17). The second row is the fixed-$k(f,h)$ partition behind (A.15b) and (A.18), summed over the complete ordered-adjacent column range H_f. In particular, the table records the two exact differences

$$\begin{aligned}
\mathcal{R}_1 - \mathcal{Q}_1 &= I_e^{\tau_f} + D_e^{\tau_f} - Q_e^{\tau_f} = E_e^{\tau_f} - N_e^{\tau_f} = U_{e,f} - R_{e,f},\\
\mathcal{R}_2 - \mathcal{Q}_2 &= \sum_{h\in H_f} \Big(I_e^{C_{f,h}} + D_e^{C_{f,h}} - Q_e^{C_{f,h}}\Big)\\
&= \sum_{h\in H_f} \Big(E_e^{C_{f,h}} - N_e^{C_{f,h}}\Big) = C_{e,f} - B_{e,f}.
\end{aligned}$$

Thus the first row translates one ordinary column-split occurrence, whereas the second row translates the complete family of adjacent corrected-column occurrences. The bijection above prevents both omission and double counting in the second row.

The omitted term is not discarded. For every fixed column occurrence η, the proof above gives

$$E_e^{\eta} - N_e^{\eta} = I_e^{\eta} + D_e^{\eta} - Q_e^{\eta},$$

and hence

$$\begin{aligned}
2E_e^{\eta} - N_e^{\eta} &= 2I_e^{\eta} + D_e^{\eta} + \mathcal{B}_e^{\eta},\\
\mathcal{B}_e^{\eta} &= (E_e^{\eta} - I_e^{\eta}) - Q_e^{\eta}.
\end{aligned}$$

The boundary polynomial cancels only after the complete row-bridge orbit is summed, by (A.29c$_1$). Thus the calligraphic calculation gives the local provenance of the occurrence ledger, while (A.29c$_0$)–(A.29c$_2$) give its formal global settlement. No equality of complete fixed-chart subtotals from two different principal charts is asserted.

Lemma A.5 (Global four-family coefficient identity)**.** *Let T_1, T_2 be two trees with nonempty contracted E_2-cores T_1°, T_2°, and let $\chi = (e,f)$ range over the full product $E_2(T_1^\circ) \times E_2(T_2^\circ)$. After summing over all these principal charts, the regrouping* (A.29b) *gives the following identity of quadratic polynomials:*

$$\begin{aligned}
\sum_{\chi}(4U_\chi - 2R_\chi - 2C_\chi + B_\chi) = \sum_{\chi}\Bigg[&\sum_{(i,j)\in D_\chi}\ \sum_{k\in a^\chi\cup b^\chi} \mathcal{P}_{ij,k}^{\chi}\big(x_{\bar{i},\bar{k}} + x_{i,\bar{k}}, x_{\bar{i},k} + x_{i,k}\big)\\
&+ 2\sum_{(i,j)\in I_\chi}\ \sum_{k\in a^\chi\cup b^\chi} \mathcal{P}_{ij,k}^{\chi}(x_{\bar{i},\bar{k}}, x_{\bar{i},k})\\
&+ \frac{1}{2}\sum_{(i,j)\in D_\chi}\ \sum_{u\in\Omega_c^\chi} \Psi_{ij}^{\chi}(u)(x_{\bar{i},\bar{u}} + x_{i,\bar{u}})^2\\
&+ \sum_{(i,j)\in I_\chi}\ \sum_{u\in\Omega_c^\chi} \Psi_{ij}^{\chi}(u)x_{\bar{i},\bar{u}}^2\Bigg].
\end{aligned} \tag{A.29h}$$

The row and column ordered refinements in this identity remain distinct.

Proof. Step 1: decompose the row coordinate with the column term fixed. Apply Lemma A.4 first to the ordinary term τ_f, and then separately to every ordered adjacent term $C_{f,h}$. The identification table gives

$$\sum_{\substack{e\in E_2(T_1^\circ)\\ f\in E_2(T_2^\circ)}} \big(2E_e^{\tau_f} - N_e^{\tau_f}\big) = \sum_{\chi=(e,f)} (2U_\chi - R_\chi),$$

$$\sum_{\substack{e\in E_2(T_1^\circ)\\ f\in E_2(T_2^\circ)}} \sum_{h:\,(f,h)\in \overrightarrow{A}_2(T_2^\circ)} \big(2E_e^{C_{f,h}} - N_e^{C_{f,h}}\big) = \sum_{\chi=(e,f)} (2C_\chi - B_\chi).$$

Thus (A.29b), followed by (A.29c$_0$) and the fixed-occurrence cancellation (A.29c$_1$), gives

$$\sum_\chi (4U_\chi - 2R_\chi - 2C_\chi + B_\chi) = 2 \sum_{\substack{e\in E_2(T_1^\circ)\\ f\in E_2(T_2^\circ)}} \big(2I_e^{\tau_f} + D_e^{\tau_f}\big) - \sum_{\substack{e\in E_2(T_1^\circ)\\ f\in E_2(T_2^\circ)}} \sum_{h:\,(f,h)\in \overrightarrow{A}_2(T_2^\circ)} \big(2I_e^{C_{f,h}} + D_e^{C_{f,h}}\big).$$

Set $\lambda_I := 2$ and $\lambda_D := 1$. For $\mathsf{R} \in \{I, D\}$, define R_e^η to mean I_e^η when $\mathsf{R} = I$ and D_e^η when $\mathsf{R} = D$; likewise, R_χ means I_χ or D_χ, respectively. For each fixed (e, f), this is exactly

$$2(2I_e^{\tau_f} + D_e^{\tau_f}) - \sum_{h:\,(f,h)\in \overrightarrow{A}_2(T_2^\circ)} (2I_e^{C_{f,h}} + D_e^{C_{f,h}}) = \sum_{\mathsf{R}\in\{I,D\}} \lambda_{\mathsf{R}} \left(2\mathsf{R}_e^{\tau_f} - \sum_{h:\,(f,h)\in \overrightarrow{A}_2(T_2^\circ)} \mathsf{R}_e^{C_{f,h}} \right),$$

This displayed equality is the complete source of the later multiplicities 2 and 1; no additional multiplicity is inserted afterwards. By (A.29c$_2$), every indexed term of U, R, C, B enters this display once, with its column term unchanged.

For a retained row relation (i, j) define

R	λ_{R}	$Z_{\mathsf{R}}(u)$	$W_{\mathsf{R}}(u)$
I	2	$x_{\bar{i},\bar{u}}$	$x_{\bar{i},u}$
D	1	$x_{\bar{i},\bar{u}} + x_{i,\bar{u}}$	$x_{\bar{i},u} + x_{i,u}$

When the chart must be displayed, write $Z_{\mathsf{R},\chi}(u), W_{\mathsf{R},\chi}(u)$ for these same expressions evaluated in chart χ; for a fixed row class we abbreviate them further as $Z_\chi(u), W_\chi(u)$. Inside one fixed-chart calculation we write simply $Z(u), W(u)$. The side-relabeling identity (A.16) shows that $Z(u)^2$ is the correct representative square for either orientation.

We now make explicit the substitutions that pass from the retained occurrence sums to the polynomial below. First, $\alpha_{i,u}^\chi = c_u r_i^u$ and the ordinary-column weight in Lemma A.4 satisfy

$$w_i^{\tau_f} = \sum_{u\in\Omega_{\mathrm{c}}^\chi} c_u r_i^u = \sum_{u\in\Omega_{\mathrm{c}}^\chi} \alpha_{i,u}^\chi.$$

Consequently, using (A.16) to replace either choice of the ordinary split side by the representative square $Z_{\mathsf{R},\chi}(u)^2$, one has, for each $\mathsf{R} \in \{I, D\}$,

$$\mathsf{R}_e^{\tau_f} = \sum_{(i,j)\in \mathsf{R}_\chi} \sum_{u,l\in\Omega_c^\chi} \alpha_{i,u}^\chi \alpha_{j,l}^\chi Z_{\mathsf{R},\chi}(u)^2.$$

Second, fix $h \in H_f$ and put $k = k(f,h)$. The fixed-k row forms in Lemma A.4 give

$$\mathsf{R}_e^{C_{f,h}} = \sum_{(i,j)\in \mathsf{R}_\chi} r_i^k r_j^k \big(c_k Z_{\mathsf{R},\chi}(k) - c_{\bar{k}} W_{\mathsf{R},\chi}(k)\big)^2.$$

Here it is essential that the map

$$h \longmapsto k(f,h)$$

proved in the calligraphic-to-occurrence crosswalk is a bijection from H_f onto $a^\chi \cup b^\chi$. It therefore converts the complete corrected-column occurrence sum, with neither omission nor repetition, into

$$\sum_{h\in H_f} \mathsf{R}_e^{C_{f,h}} = \sum_{(i,j)\in \mathsf{R}_\chi} \sum_{k\in a^\chi\cup b^\chi} r_i^k r_j^k \big(c_k Z_{\mathsf{R},\chi}(k) - c_{\bar{k}} W_{\mathsf{R},\chi}(k)\big)^2.$$

Thus, for every fixed (χ, R),

$$2\mathsf{R}_e^{\tau_f} - \sum_{h\in H_f} \mathsf{R}_e^{C_{f,h}} = \sum_{(i,j)\in \mathsf{R}_\chi} \Bigg[2 \sum_{u,l\in\Omega_c^\chi} \alpha_{i,u}^\chi \alpha_{j,l}^\chi Z_{\mathsf{R},\chi}(u)^2 - \sum_{k\in a^\chi\cup b^\chi} r_i^k r_j^k \big(c_k Z_{\mathsf{R},\chi}(k) - c_{\bar{k}} W_{\mathsf{R},\chi}(k)\big)^2 \Bigg].$$

The bracket is precisely $\mathcal{L}_{ij,\mathsf{R}}^\chi$ in (A.29d). The identities in this paragraph are fixed-chart substitutions. By contrast, the removal of the row-boundary polynomials is valid only after the complete sum over row principal edges, as proved in (A.29c$_1$). Combining the displayed fixed-chart identity with the globally summed form of (A.29b), after applying (A.29c$_0$) and cancelling the row-boundary polynomials, therefore gives

$$\sum_\chi (4U_\chi - 2R_\chi - 2C_\chi + B_\chi) = \sum_\chi \sum_{\mathsf{R}\in\{I,D\}} \sum_{(i,j)\in \mathsf{R}_\chi} \lambda_{\mathsf{R}}\, \mathcal{L}_{ij,\mathsf{R}}^\chi, \tag{A.29c}$$

where

$$\begin{aligned} \mathcal{L}_{ij,\mathsf{R}}^\chi := 2 &\sum_{u,l\in\Omega_c^\chi} \alpha_{i,u}^\chi \alpha_{j,l}^\chi Z_{\mathsf{R},\chi}(u)^2 \\ &- \sum_{k\in a^\chi\cup b^\chi} r_i^k r_j^k \big(c_k Z_{\mathsf{R},\chi}(k) - c_{\bar{k}} W_{\mathsf{R},\chi}(k)\big)^2. \end{aligned} \tag{A.29d}$$

The ordered index pairs in the first line remain (i,u) and (j,l). All undecorated masses and symmetric weights on the right-hand side of (A.29d) are evaluated in the displayed chart χ.

Step 2: compute the column coefficient and pair the endpoint square faces. For each chart χ, put

$$A_{ij}^\chi(u) := \sum_{l:(u,l)\in I_c^\chi \cup D_c^\chi} c_u c_l r_i^u r_j^l.$$

Within the fixed chart in the following calculation, write $A_{ij}(u) := A^{\chi}_{ij}(u)$. For fixed (i,j,u), the scalar $A_{ij}(u)$ is the coefficient of the representative square $Z(u)^2$ supplied by the non-endpoint-diagonal part of the positive ordered (u,l)-sum in (A.29d). It is the coefficient supplied before square completion, not a new quadratic form. Part of it will be used to complete the mixed term $Z(u)W(u)$. The resulting signed coefficient residue need not be nonnegative pointwise; only its same-side block sum is used below. Because

$$\Omega_{\mathrm{c}}^2 = (I_{\mathrm{c}} \cup D_{\mathrm{c}}) \sqcup \Delta_{a_{\mathrm{c}}} \sqcup \Delta_{b_{\mathrm{c}}},$$

the endpoint part of (A.29d) has the following monomial table. Put

$$d^{\chi}_{ij}(k) := c_k^2 r_i^k r_j^k, \qquad m^{\chi}_{ij}(k) := 2c_k c_{\bar{k}} r_i^k r_j^k, \qquad e^{\chi}_{ij}(k) := c_{\bar{k}}^2 r_i^k r_j^k.$$

Within the fixed summand (χ,i,j,k), abbreviate these three quantities as d_k, m_k, e_k. Splitting the ordered square pool by the displayed partition and expanding the negative neighboring square gives

$$\begin{aligned}\mathcal{L}^{\chi}_{ij,\mathsf{R}} = 2 &\sum_{(u,l)\in I_{\mathrm{c}}^{\chi}\cup D_{\mathrm{c}}^{\chi}} \alpha^{\chi}_{i,u}\alpha^{\chi}_{j,l} Z(u)^2 \\ &+ \sum_{k\in a^{\chi}\cup b^{\chi}} \left(d_k Z(k)^2 + m_k Z(k)W(k) - e_k W(k)^2\right).\end{aligned}$$

Thus every monomial of (A.29d) is represented in the following coefficient table, including the negative second square face $-e_k W(k)^2$:

source	$[Z^2]$	$[ZW]$	$[W^2]$
$2\times$ endpoint diagonal of the ordered (u,l) pool	$2d_k$	0	0
$-r_i^k r_j^k (c_k Z - c_{\bar{k}} W)^2$	$-d_k$	m_k	$-e_k$
sum at k	d_k	m_k	$-e_k$

Fix a row class R. On the complete sum over principal edges e,f, row relations $(i,j)\in\mathsf{R}_{\chi}$, and ordered adjacent pairs $(f,h)\in \overrightarrow{A}_2(T_2^{\circ})$, let $k = k(f,h)$ be the endpoint atom determined by (f,h) and define the column reversal by

$$\rho_{\mathrm{c}}(e,f,i,j,k(f,h)) := (e,h,i,j,k(h,f)).$$

This is a fixed-point-free involution on the just-displayed indexing range. It changes only the column chart, preserves the row principal edge and the ordered relation $(i,j)\in\mathsf{R}_{\chi}$, and sends the selected and opposite blocks into each other. Write $\rho\chi := (e,h)$ and $k^{\rho} := k(h,f)$. With every chart decoration exposed, this says

$$\begin{aligned} d^{\rho\chi}_{ij}(k^{\rho}) = e^{\chi}_{ij}(k), &\qquad e^{\rho\chi}_{ij}(k^{\rho}) = d^{\chi}_{ij}(k), \\ Z_{\rho\chi}(k^{\rho})^2 = W_{\chi}(k)^2, &\quad W_{\rho\chi}(k^{\rho})^2 = Z_{\chi}(k)^2.\end{aligned}$$

Hence

$$\begin{aligned}&\left(d^{\chi}_{ij}(k)Z_{\chi}(k)^2 - e^{\chi}_{ij}(k)W_{\chi}(k)^2\right) \\ &\quad + \left(d^{\rho\chi}_{ij}(k^{\rho})Z_{\rho\chi}(k^{\rho})^2 - e^{\rho\chi}_{ij}(k^{\rho})W_{\rho\chi}(k^{\rho})^2\right) \\ &\quad = \left(d_k Z(k)^2 - e_k W(k)^2\right) + \left(e_k W(k)^2 - d_k Z(k)^2\right) = 0.\end{aligned}$$

Summing this two-term identity over the fixed-point-free reversal orbits gives

$$\sum_{\chi}\sum_{(i,j)\in\mathsf{R}_\chi}\sum_{k\in a^\chi\cup b^\chi}\Big(d_{ij}^{\chi}(k)Z_\chi(k)^2-e_{ij}^{\chi}(k)W_\chi(k)^2\Big)=0. \tag{A.29e}$$

It follows, after summing over all column charts, that

$$\sum_{\chi}\sum_{(i,j)\in\mathsf{R}_\chi}\mathcal{L}_{ij,\mathsf{R}}^{\chi}=\sum_{\chi}\sum_{(i,j)\in\mathsf{R}_\chi}\left[2\sum_{u\in\Omega_c^\chi}A_{ij}^{\chi}(u)Z_{\mathsf{R},\chi}(u)^2+\sum_{k\in a^\chi\cup b^\chi}m_{ij}^{\chi}(k)Z_{\mathsf{R},\chi}(k)W_{\mathsf{R},\chi}(k)\right].$$

For an endpoint k, recall that

$$p_{ij}(k)=[X^2]\mathcal{P}_{ij,k}(X,Y),\qquad q_{ij}(k)=[Y^2]\mathcal{P}_{ij,k}(X,Y),$$

and set $p_{ij}(u)=0$ at a nonendpoint. Equations (A.21a) and (A.22)–(A.25) give

$$[XY]\mathcal{P}_{ij,k}(X,Y)=m_k.$$

$p_{ij}(k)$ and $q_{ij}(k)$ are therefore the first- and second-square coefficients of the completed-square polynomial $\mathcal{P}_{ij,k}$. Consequently $A_{ij}(u)-p_{ij}(u)$ is the signed first-square coefficient residue after completion. It need not be nonnegative pointwise; the block identity (A.30b) identifies its same-side block sum with the nonnegative Ψ coefficient. Lemma A.3 gives, under the same reindexing,

$$q_{ij}^{\chi}(k)=p_{ij}^{\rho\chi}(k^\rho),\qquad W_\chi(k)^2=Z_{\rho\chi}(k^\rho)^2.$$

Therefore

$$\sum_{\chi}\sum_{(i,j)\in\mathsf{R}_\chi}\sum_{k\in a^\chi\cup b^\chi}q_{ij}^{\chi}(k)W_\chi(k)^2=\sum_{\chi}\sum_{(i,j)\in\mathsf{R}_\chi}\sum_{k\in a^\chi\cup b^\chi}p_{ij}^{\chi}(k)Z_\chi(k)^2.$$

For each endpoint,

$$2A_{ij}(k)Z(k)^2+m_kZ(k)W(k)=\mathcal{P}_{ij,k}(Z(k),W(k))+(2A_{ij}(k)-p_{ij}(k))Z(k)^2-q_{ij}(k)W(k)^2.$$

Consequently, with every summation range retained,

$$\begin{aligned}
&\sum_{\chi}\sum_{(i,j)\in\mathsf{R}_\chi}\left[2\sum_{u\in\Omega_c^\chi}A_{ij}^{\chi}(u)Z_{\mathsf{R},\chi}(u)^2+\sum_{k\in a^\chi\cup b^\chi}m_{ij}^{\chi}(k)Z_{\mathsf{R},\chi}(k)W_{\mathsf{R},\chi}(k)\right]\\
&=\sum_{\chi}\sum_{(i,j)\in\mathsf{R}_\chi}\left[\sum_{k\in a^\chi\cup b^\chi}\mathcal{P}_{ij,k}^{\chi}(Z_{\mathsf{R},\chi}(k),W_{\mathsf{R},\chi}(k))+2\sum_{u\in\Omega_c^\chi}A_{ij}^{\chi}(u)Z_{\mathsf{R},\chi}(u)^2\right.\\
&\qquad\qquad\left.-\sum_{k\in a^\chi\cup b^\chi}p_{ij}^{\chi}(k)Z_{\mathsf{R},\chi}(k)^2-\sum_{k\in a^\chi\cup b^\chi}q_{ij}^{\chi}(k)W_{\mathsf{R},\chi}(k)^2\right]\\
&=\sum_{\chi}\sum_{(i,j)\in\mathsf{R}_\chi}\left[\sum_{k\in a^\chi\cup b^\chi}\mathcal{P}_{ij,k}^{\chi}(Z_{\mathsf{R},\chi}(k),W_{\mathsf{R},\chi}(k))+2\sum_{u\in\Omega_c^\chi}(A_{ij}^{\chi}(u)-p_{ij}^{\chi}(u))Z_{\mathsf{R},\chi}(u)^2\right].
\end{aligned}$$

The second equality is exactly the coefficient-preserving reversal $qW^2\leftrightarrow pZ^2$, and uses $p_{ij}^{\chi}(u)=0$ at nonendpoints. Hence

$$\sum_{\chi}\sum_{(i,j)\in\mathsf{R}_\chi}\mathcal{L}_{ij,\mathsf{R}}^{\chi}=\sum_{\chi}\sum_{(i,j)\in\mathsf{R}_\chi}\left[\sum_{k\in a^\chi\cup b^\chi}\mathcal{P}_{ij,k}^{\chi}(Z_{\mathsf{R},\chi}(k),W_{\mathsf{R},\chi}(k))+2\sum_{u\in\Omega_c^\chi}(A_{ij}^{\chi}(u)-p_{ij}^{\chi}(u))Z_{\mathsf{R},\chi}(u)^2\right]. \tag{A.29f}$$

Step 3: sum the remaining coefficient on one same-side block. Fix a chart, a row relation (i,j), and a same-side column block $S \in \{a^\chi \cup c^\chi, b^\chi \cup d^\chi\}$. Put $\bar{S} = \Omega_c^\chi \setminus S$. For an endpoint $u \in S$, expanding (24) gives

$$\begin{aligned} 4p_{ij}(u) = \frac{c_u(1-c_u)}{c_{\bar{S}}^2} a_{i,\bar{S}} a_{j,\bar{S}} + \frac{c_{\bar{S}}}{c_S} a_{i,u} a_{j,u} \\ + a_{i,u} a_{j,\bar{S}} + a_{i,\bar{S}} a_{j,u}. \end{aligned}$$

The four contributions to its first-square coefficient follow from

$$\begin{aligned} &\sum_{l \in \bar{u}} \frac{c_u}{c_{\bar{u}}} a_{i,\bar{u}} a_{j,l} + \sum_{l \in S \setminus \{u\}} \frac{c_u c_l}{c_{\bar{u}} c_{\bar{l}}} a_{i,\bar{u}} a_{j,\bar{l}} = \frac{c_u(1-c_u)}{c_{\bar{u}}^2} a_{i,\bar{u}} a_{j,\bar{u}}, \\ &\sum_{l \in \bar{u}} \frac{c_l}{c_{\bar{l}}} a_{i,u} a_{j,\bar{l}} = \frac{c_{\bar{u}}}{c_{u^+}} a_{i,u} a_{j,u} + \sum_{m \in M_S(u)} \frac{c_{\bar{u}}}{c_{u^+}} a_{i,u} a_{j,m}, \\ &\sum_{l \in \bar{u}} a_{i,u} a_{j,l} = a_{i,u} a_{j,\bar{u}}, \\ &\sum_{p \in S} \sum_{l \in \bar{S}} \frac{c_p c_l}{c_{\bar{S}} c_S} a_{i,\bar{S}} a_{j,S} = a_{i,\bar{S}} a_{j,S}. \end{aligned}$$

The first three identities hold pointwise at an endpoint; the fourth holds after summing the entire same-side block. We now perform the collection of (A.26)–(A.29) and the six rows of Table 1 explicitly. There are exactly two cases.

If $u \in a^\chi \cup b^\chi$ is an endpoint, then $u^+ = S$ and $\bar{u} = \bar{S}$. Term-by-term subtraction gives

$$\begin{aligned} 4A_{ij}(u) - \Psi_{ij}^\chi(u) &= \frac{c_u(1-c_u)}{c_{\bar{S}}^2} a_{i,\bar{S}} a_{j,\bar{S}} + \frac{c_{\bar{S}}}{c_S} a_{i,u} a_{j,u} \\ &\quad + a_{i,u} a_{j,\bar{S}} + \frac{c_u}{c_S} a_{i,\bar{S}} a_{j,S} \\ &= 4p_{ij}(u) + a_{i,\bar{S}} \left(\frac{c_u}{c_S} a_{j,S} - a_{j,u} \right). \end{aligned}$$

The first equality displays the four surviving monomials; the second uses the already expanded formula for $4p_{ij}(u)$.

If $u \in c^\chi \cup d^\chi$ is a residual index, then $p_{ij}(u) = 0$, the endpoint diagonal row of Ψ is absent, and direct cancellation gives

$$\begin{aligned} 4A_{ij}(u) - \Psi_{ij}^\chi(u) &= a_{i,u} a_{j,\bar{S}} + \frac{c_u}{c_S} a_{i,\bar{S}} a_{j,S} \\ &\quad - \left(a_{i,u} a_{j,\bar{S}} + a_{i,\bar{S}} a_{j,u} \right) \\ &= a_{i,\bar{S}} \left(\frac{c_u}{c_S} a_{j,S} - a_{j,u} \right). \end{aligned}$$

Thus both cases give, for every $u \in S$,

$$\begin{aligned} 4(A_{ij}(u) - p_{ij}(u)) &= \Psi_{ij}^\chi(u) + \Delta_{ij,S}(u), \\ \Delta_{ij,S}(u) &:= a_{i,\bar{S}} \left(\frac{c_u}{c_S} a_{j,S} - a_{j,u} \right). \end{aligned}$$

The discrepancy need not vanish pointwise, but

$$\sum_{u \in S} \Delta_{ij,S}(u) = a_{i,\bar{S}} \left(\frac{a_{j,S}}{c_S} \sum_{u \in S} c_u - \sum_{u \in S} a_{j,u} \right) = 0.$$

Therefore

$$4\left[\sum_{u\in S}A_{ij}(u)-\sum_{u\in S}p_{ij}(u)\right]=\sum_{u\in S}\Psi_{ij}^{\chi}(u). \tag{A.30b}$$

For this same-side block define $Z_{\mathsf{R}}(S):=Z_{\mathsf{R}}(u)$ for any $u\in S$. This is well-defined because $\bar{u}=\bar{S}$ throughout S. Hence all $u\in S$ multiply the same square $Z_{\mathsf{R}}(S)^2$. Multiplying (A.30b) by this common square gives the missing factor explicitly:

$$\begin{aligned}2\sum_{u\in S}(A_{ij}(u)-p_{ij}(u))Z_{\mathsf{R}}(u)^2&=2Z_{\mathsf{R}}(S)^2\sum_{u\in S}(A_{ij}(u)-p_{ij}(u))\\&=\frac{1}{2}Z_{\mathsf{R}}(S)^2\sum_{u\in S}\Psi_{ij}^{\chi}(u)\\&=\frac{1}{2}\sum_{u\in S}\Psi_{ij}^{\chi}(u)Z_{\mathsf{R}}(u)^2.\end{aligned}$$

The two same-side blocks $a^{\chi}\cup c^{\chi}$ and $b^{\chi}\cup d^{\chi}$ are disjoint and have union $\Omega_{\mathrm{c}}^{\chi}$. Summing the displayed identity over these two blocks is exactly the coefficient statement needed in (A.29f); it gives

$$\sum_{\chi}\sum_{(i,j)\in\mathsf{R}_{\chi}}\mathcal{L}_{ij,\mathsf{R}}^{\chi}=\sum_{\chi}\sum_{(i,j)\in\mathsf{R}_{\chi}}\left[\sum_{k\in a^{\chi}\cup b^{\chi}}\mathcal{P}_{ij,k}^{\chi}(Z_{\mathsf{R},\chi}(k),W_{\mathsf{R},\chi}(k))+\frac{1}{2}\sum_{u\in\Omega_{\mathrm{c}}^{\chi}}\Psi_{ij}^{\chi}(u)Z_{\mathsf{R},\chi}(u)^2\right]. \tag{A.30g}$$

Step 4: recover the four square families. Finally restore $\lambda_I=2$ and $\lambda_D=1$ from (A.29c). Substituting (A.29g) for the two row classes gives the following identity. Here $Z_{I,\chi},W_{I,\chi}$ and $Z_{D,\chi},W_{D,\chi}$ are the chart-indexed functions defined in Step 1; explicitly,

$$(Z_{I,\chi}(k),W_{I,\chi}(k))=(x_{\bar{i},\bar{k}},x_{\bar{i},k})$$

and

$$(Z_{D,\chi}(k),W_{D,\chi}(k))=(x_{\bar{i},\bar{k}}+x_{i,\bar{k}},x_{\bar{i},k}+x_{i,k}).$$

Consequently no argument of a completed-square polynomial is being suppressed:

$$\begin{aligned}&\sum_{\chi}\Bigg\{\lambda_I\left[\sum_{(i,j)\in I_{\chi}}\sum_{k\in a^{\chi}\cup b^{\chi}}\mathcal{P}_{ij,k}^{\chi}(Z_{I,\chi}(k),W_{I,\chi}(k))+\frac{1}{2}\sum_{(i,j)\in I_{\chi}}\sum_{u\in\Omega_{\mathrm{c}}^{\chi}}\Psi_{ij}^{\chi}(u)Z_{I,\chi}(u)^2\right]\\&\qquad+\lambda_D\left[\sum_{(i,j)\in D_{\chi}}\sum_{k\in a^{\chi}\cup b^{\chi}}\mathcal{P}_{ij,k}^{\chi}(Z_{D,\chi}(k),W_{D,\chi}(k))+\frac{1}{2}\sum_{(i,j)\in D_{\chi}}\sum_{u\in\Omega_{\mathrm{c}}^{\chi}}\Psi_{ij}^{\chi}(u)Z_{D,\chi}(u)^2\right]\Bigg\}\\&=\sum_{\chi}\Bigg[2\sum_{(i,j)\in I_{\chi}}\sum_{k\in a^{\chi}\cup b^{\chi}}\mathcal{P}_{ij,k}^{\chi}(Z_{I,\chi}(k),W_{I,\chi}(k))\\&\qquad+\sum_{(i,j)\in D_{\chi}}\sum_{k\in a^{\chi}\cup b^{\chi}}\mathcal{P}_{ij,k}^{\chi}(Z_{D,\chi}(k),W_{D,\chi}(k))\\&\qquad+\sum_{(i,j)\in I_{\chi}}\sum_{u\in\Omega_{\mathrm{c}}^{\chi}}\Psi_{ij}^{\chi}(u)Z_{I,\chi}(u)^2+\frac{1}{2}\sum_{(i,j)\in D_{\chi}}\sum_{u\in\Omega_{\mathrm{c}}^{\chi}}\Psi_{ij}^{\chi}(u)Z_{D,\chi}(u)^2\Bigg].\end{aligned}$$

All chart and index ranges are now visible. The four displayed terms are, respectively, the $2\mathcal{S}_2$, $\mathcal{S}_1$, $4\mathcal{S}_4$, and $2\mathcal{S}_3$ contributions. This is precisely (A.29h) and completes the proof. □

A.9 The four nonnegative square families

For a fixed principal chart χ, use the completed-square form $\mathcal{P}_{ij,k}$ defined in (24) and set

$$\mathcal{S}_{2,\chi} := \sum_{(i,j)\in I_\chi} \sum_{k\in a^\chi \cup b^\chi} \mathcal{P}_{ij,k}(x_{\bar{i},\bar{k}}, x_{\bar{i},k}), \tag{25}$$

$$\mathcal{S}_{1,\chi} := \sum_{(i,j)\in D_\chi} \sum_{k\in a^\chi \cup b^\chi} \mathcal{P}_{ij,k}(x_{\bar{i},\bar{k}} + x_{i,\bar{k}}, x_{\bar{i},k} + x_{i,k}), \tag{26}$$

$$\mathcal{S}_{4,\chi} := \frac{1}{4} \sum_{(i,j)\in I_\chi} \sum_{u\in\Omega_c^\chi} \Psi_{ij}^\chi(u) x_{\bar{i},\bar{u}}^2, \tag{27}$$

$$\mathcal{S}_{3,\chi} := \frac{1}{4} \sum_{(i,j)\in D_\chi} \sum_{u\in\Omega_c^\chi} \Psi_{ij}^\chi(u) (x_{\bar{i},\bar{u}} + x_{i,\bar{u}})^2. \tag{28}$$

The four families are distinguished by two independent choices: the row relation is I or D, and the column contribution is a completed square $\mathcal{P}$ or a remaining Ψ-square. After division by four, their coefficients in f_1 are

	$\mathcal{P}$-family	Ψ-family
D	$\frac{1}{4}\mathcal{S}_1$	$\frac{1}{2}\mathcal{S}_3$
I	$\frac{1}{2}\mathcal{S}_2$	$\mathcal{S}_4$.

Theorem A.6 (Exact four-class identity)**.** *Let T_1, T_2 be two trees on the same finite taxon set whose contracted E_2-cores T_1°, T_2° are nonempty, and let χ range over every principal chart $E_2(T_1^\circ)\times E_2(T_2^\circ)$. The auxiliary functional satisfies*

$$f_1(T_1, T_2) = \sum_\chi \left(\frac{1}{4}\mathcal{S}_{1,\chi} + \frac{1}{2}\mathcal{S}_{2,\chi} + \frac{1}{2}\mathcal{S}_{3,\chi} + \mathcal{S}_{4,\chi} \right). \tag{A.30}$$

Proof. By the four definitions immediately above,

$$\text{right-hand side of (A.29h)} = \sum_\chi \big(\mathcal{S}_{1,\chi} + 2\mathcal{S}_{2,\chi} + 2\mathcal{S}_{3,\chi} + 4\mathcal{S}_{4,\chi} \big).$$

Indeed, the unscaled D-row Ψ sum is $4\mathcal{S}_{3,\chi}$ and occurs with coefficient $1/2$, whereas the unscaled I-row Ψ sum is $4\mathcal{S}_{4,\chi}$ and occurs with coefficient 1. Combining this with (17) gives

$$4f_1 = \sum_\chi \big(\mathcal{S}_{1,\chi} + 2\mathcal{S}_{2,\chi} + 2\mathcal{S}_{3,\chi} + 4\mathcal{S}_{4,\chi} \big).$$

Dividing every displayed coefficient by four gives exactly (A.30). □

Appendix C evaluates equation (A.30) in exact arithmetic for a concrete pair of non-caterpillar binary trees with 25 labelled leaves and compares the result with a separate evaluation of the defining formula for f_1 and with a direct enumeration of all ordered quadruples in X^4.

A.10 Why the remaining coefficient Ψ is nonnegative

All terms in $\mathcal{P}$ are visibly nonnegative. It remains only to prove $\Psi_{ij}^{\chi_0}(u) \geq 0$. Fix a chart χ_0 and a target $u \in \Omega_c^{\chi_0}$. Throughout this paragraph, every undecorated mass, mixed mass, relation set, and aggregate is taken in the chart χ_0; we retain the subscript χ_0 where it prevents ambiguity. For

the coefficient $\Psi_{ij}^{\chi_0}(u)$, the first sum in (A.30a), the symmetric $u \in c^{\chi_0} \cup d^{\chi_0}$ residual, and the final M_D sum are termwise nonnegative. The only part requiring verification is the chart's core bracket

$$\sum_{l \in S_{\chi_0}(u)} 4c_u c_l r_i^{u,\chi_0} r_j^{l,\chi_0} - \sum_{l \in M_{S,\chi_0}(u)} \frac{c_u c_l a_{i,\bar{u}_{\chi_0}} a_{j,\bar{l}_{\chi_0}}}{c_{\bar{u}_{\chi_0}} c_{\bar{l}_{\chi_0}}} - \mathbf{1}_{u \in a^{\chi_0} \cup b^{\chi_0}} 4c_u^2 r_i^{u,\chi_0} r_j^{u,\chi_0}. \tag{A.32}$$

By the definitions of r_i^{u,χ_0} and r_j^{l,χ_0} in the same chart, the positive summand has the complete four-term expansion

$$\begin{aligned} 4c_u c_l r_i^{u,\chi_0} r_j^{l,\chi_0} = {} & a_{i,u} a_{j,l} + \frac{c_l}{c_{\bar{l}_{\chi_0}}} a_{i,u} a_{j,\bar{l}_{\chi_0}} \\ & + \frac{c_u}{c_{\bar{u}_{\chi_0}}} a_{i,\bar{u}_{\chi_0}} a_{j,l} + \frac{c_u c_l}{c_{\bar{u}_{\chi_0}} c_{\bar{l}_{\chi_0}}} a_{i,\bar{u}_{\chi_0}} a_{j,\bar{l}_{\chi_0}}. \end{aligned}$$

For every $l \in M_{S,\chi_0}(u)$, the fourth term is exactly the summand subtracted in the second sum of (A.32); the first three terms remain nonnegative. If $u \in a^{\chi_0} \cup b^{\chi_0}$, the diagonal term $l = u$ belongs to $S_{\chi_0}(u)$ and is disjoint from $M_{S,\chi_0}(u)$. Its whole value is $4c_u^2 r_i^{u,\chi_0} r_j^{u,\chi_0}$, so it exactly cancels the third term of (A.32). If $u \in c^{\chi_0} \cup d^{\chi_0}$, then $M_{S,\chi_0}(u) = \varnothing$ and the diagonal subtraction is absent. All terms left after these explicitly identified cancellations are nonnegative. Hence (A.32) is nonnegative for every chart target u.

Consequently

$$\Psi_{ij}^{\chi}(u) \geq 0$$

for every principal chart, row relation and target. Hence the coefficient-weighted square families $\mathcal{S}_{3,\chi}$ and $\mathcal{S}_{4,\chi}$ are nonnegative, while $\mathcal{S}_{1,\chi}$ and $\mathcal{S}_{2,\chi}$ are sums of squares. The exact identity in Theorem A.6 therefore gives $f_1(T_1, T_2) \geq 0$.

B A related identity for tree pseudometrics and distance covariance

This appendix is logically independent of the quartet-distance proof. It shows that the same split determinant square arises when two tree pseudometrics are double-centered in the empirical distance-covariance inner product [6]. The calculation gives a second, familiar reason for using determinant squares as the nonnegative building blocks of f_1.

Let $X = \{1, \ldots, n\}$, and let T_1, T_2 be two edge-weighted X-trees with nonnegative edge lengths. Each taxon of X is assigned to a vertex; the leaf-labelled case is the special case in which all taxa are assigned to leaves. Let d_1, d_2 be the induced tree pseudometrics, and put

$$a_{ij} = d_1(i, j), \qquad b_{ij} = d_2(i, j).$$

Define the double-centered matrices

$$A_{ij} = a_{ij} - a_{i\cdot} - a_{\cdot j} + a_{\cdot\cdot}, \qquad B_{ij} = b_{ij} - b_{i\cdot} - b_{\cdot j} + b_{\cdot\cdot},$$

where

$$a_{i\cdot} = \frac{1}{n} \sum_{j=1}^{n} a_{ij}, \qquad a_{\cdot j} = \frac{1}{n} \sum_{i=1}^{n} a_{ij}, \qquad a_{\cdot\cdot} = \frac{1}{n^2} \sum_{i,j=1}^{n} a_{ij},$$

and B is defined analogously. The empirical distance covariance inner product of the two pseudometrics is

$$V_n^2(T_1, T_2) = \frac{1}{n^2} \sum_{i,j=1}^{n} A_{ij} B_{ij}.$$

Proposition B.1. *Let $\Sigma(T_1)$ and $\Sigma(T_2)$ be the edge-induced split systems of T_1 and T_2. If $e \in E(T_1)$ has length w_e and induces the split $\sigma_e = P_e \mid P_e^c$, and if $f \in E(T_2)$ has length v_f and induces the split $\tau_f = Q_f \mid Q_f^c$, then*

$$V_n^2(T_1, T_2) = 4 \sum_{e \in E(T_1)} \sum_{f \in E(T_2)} w_e v_f D(\sigma_e, \tau_f),$$

where $D(\sigma_e, \tau_f)$ is the determinant square associated with the normalized 2×2 intersection table of the two splits. In particular,

$$V_n^2(T_1, T_2) \geq 0.$$

Proof. For an edge e of T_1, define

$$I_e : X \to \{0, 1\}, \qquad I_e(i) = \begin{cases} 1, & i \in P_e, \\ 0, & i \in P_e^c. \end{cases}$$

Let

$$p_e = \frac{|P_e|}{n}, \qquad z_e(i) = I_e(i) - p_e.$$

The edge e separates two taxa i, j if and only if $I_e(i) \neq I_e(j)$. Thus

$$d_1(i, j) = \sum_{e \in E(T_1)} w_e \big(I_e(i) - I_e(j)\big)^2.$$

First consider the split pseudometric

$$\delta_e(i, j) = \big(I_e(i) - I_e(j)\big)^2.$$

Since $I_e(i)^2 = I_e(i)$ and $I_e(j)^2 = I_e(j)$, we have

$$\delta_e(i, j) = I_e(i) + I_e(j) - 2 I_e(i) I_e(j).$$

Averaging over j with i fixed gives

$$\begin{aligned} \delta_{e,i\cdot} &= \frac{1}{n} \sum_{j=1}^{n} \delta_e(i, j) \\ &= I_e(i) + \frac{1}{n} \sum_{j=1}^{n} I_e(j) - 2 I_e(i) \frac{1}{n} \sum_{j=1}^{n} I_e(j) \\ &= p_e + (1 - 2p_e) I_e(i). \end{aligned}$$

Similarly,

$$\delta_{e,\cdot j} = p_e + (1 - 2p_e) I_e(j), \qquad \delta_{e,\cdot\cdot} = 2p_e(1 - p_e).$$

Therefore

$$\begin{aligned} &\delta_e(i, j) - \delta_{e,i\cdot} - \delta_{e,\cdot j} + \delta_{e,\cdot\cdot} \\ &\quad = -2\big(I_e(i) - p_e\big)\big(I_e(j) - p_e\big) = -2 z_e(i) z_e(j). \end{aligned}$$

By linearity of double-centering,

$$A_{ij} = -2 \sum_{e \in E(T_1)} w_e z_e(i) z_e(j).$$

The same argument for T_2 gives, for the split $\tau_f = Q_f \mid Q_f^c$,

$$J_f : X \to \{0,1\}, \qquad J_f(i) = \begin{cases} 1, & i \in Q_f, \\ 0, & i \in Q_f^c, \end{cases}$$

$$q_f = \frac{|Q_f|}{n}, \qquad y_f(i) = J_f(i) - q_f,$$

and

$$B_{ij} = -2 \sum_{f \in E(T_2)} v_f y_f(i) y_f(j).$$

Consequently,

$$\begin{aligned} V_n^2(T_1, T_2) &= \frac{1}{n^2} \sum_{i,j=1}^{n} A_{ij} B_{ij} \\ &= \frac{4}{n^2} \sum_{i,j=1}^{n} \left(\sum_{e \in E(T_1)} w_e z_e(i) z_e(j) \right) \left(\sum_{f \in E(T_2)} v_f y_f(i) y_f(j) \right) \\ &= 4 \sum_{e \in E(T_1)} \sum_{f \in E(T_2)} w_e v_f \left(\frac{1}{n} \sum_{i=1}^{n} z_e(i) y_f(i) \right)^2. \end{aligned}$$

It remains to identify the squared term with the split determinant square. Fix e and f, and define the normalized table

	Q_f	Q_f^c
P_e	a_{ef}	b_{ef}
P_e^c	c_{ef}	d_{ef}

by

$$a_{ef} = \frac{|P_e \cap Q_f|}{n}, \quad b_{ef} = \frac{|P_e \cap Q_f^c|}{n}, \quad c_{ef} = \frac{|P_e^c \cap Q_f|}{n}, \quad d_{ef} = \frac{|P_e^c \cap Q_f^c|}{n}.$$

Then

$$p_e = a_{ef} + b_{ef}, \qquad q_f = a_{ef} + c_{ef},$$

and hence

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^{n} z_e(i) y_f(i) &= \frac{1}{n} \sum_{i=1}^{n} (I_e(i) - p_e)(J_f(i) - q_f) \\ &= a_{ef} - p_e q_f \\ &= a_{ef} - (a_{ef} + b_{ef})(a_{ef} + c_{ef}) \\ &= a_{ef} d_{ef} - b_{ef} c_{ef}. \end{aligned}$$

Thus

$$\left(\frac{1}{n} \sum_{i=1}^{n} z_e(i) y_f(i) \right)^2 = (a_{ef} d_{ef} - b_{ef} c_{ef})^2 = D(\sigma_e, \tau_f).$$

Substituting this into the previous expression proves the identity and the nonnegativity. □

Proposition B.1 shows that the full split determinant statistic is the same positive semidefinite object as the double-centered tree-pseudometric inner product used in empirical distance covariance. This observation is separate from the quartet-distance extremal argument in the main text, but it explains why determinant squares form a natural nonnegative building block.

C An algorithmic example on two non-caterpillar binary trees

This appendix evaluates the proof on two concrete non-caterpillar unrooted binary phylogenetic trees with $N = 25$ labelled leaves. The same value of f_1 is obtained in three algebraically different ways:

(i) sum the four nonnegative families in (A.30) over all principal charts;

(ii) evaluate the four defining levels $U - \frac{1}{2}R - \frac{1}{2}C + \frac{1}{4}B$ directly;

(iii) enumerate the ordered-quadruple kernel on X^4.

Before presenting the global totals, we display one principal chart, one completed square, and its reversed bridge in full detail.

C.1 The two concrete non-caterpillar trees

Let $X = \{0, 1, \ldots, 24\}$. Take the two trees represented by the following Newick strings; line breaks inside each string are only for display.

```
T1 =
(((((((0,((19,20),23)),(14,24)),22),5),6),(9,10)),
(((((1,7),11),((2,((13,(16,18)),17)),12)),(8,15)),3),
(4,21));

T2 =
((((0,15),((2,7),24)),(10,20)),
(((((1,(((3,14),9),8)),19),11),5),18),
(((((4,(12,(13,21))),17),16),(22,23)),6));
```

Each tree has 25 leaves of degree one, 23 internal vertices of degree three, and 47 edges. To verify directly that neither graph is a caterpillar, delete the leaves and consider the induced internal skeleton. The numbers of skeleton vertices of degrees $1, 2, 3$ are

	deg 1	deg 2	deg 3
T_1	7	11	5
T_2	6	13	4

A binary caterpillar has a path as its internal skeleton and therefore has no degree-three skeleton vertex. The last column proves that both T_1 and T_2 are non-caterpillar binary trees.

Each tree has

$$|E_2(T_1)| = |E_2(T_2)| = 25 - 3 = 22$$

quartet-relevant internal edges. We use the following reproducible indexing. For an internal edge g, let $S(g)$ be the smaller side of its split; it is unique because $|X| = 25$ is odd. Order the internal edges first by $|S(g)|$ and then lexicographically by the increasing tuple of elements of $S(g)$. Write the resulting orders as

$$E_2(T_1) = \{e_0, \ldots, e_{21}\}, \qquad E_2(T_2) = \{f_0, \ldots, f_{21}\}.$$

Thus the principal charts are

$$\chi_{pq} = (e_p, f_q), \qquad 0 \le p, q \le 21,$$

and there are $22^2 = 484$ of them. As in equation (A.30), put

$$A_\chi := \frac{1}{4}\mathcal{S}_{1,\chi} + \frac{1}{2}\mathcal{S}_{2,\chi} + \frac{1}{2}\mathcal{S}_{3,\chi} + \mathcal{S}_{4,\chi}.$$

Then

$$f_1^{\text{A.30}}(T_1, T_2) = \sum_{p=0}^{21}\sum_{q=0}^{21} A_{\chi_{pq}}.$$

C.2 One principal chart in full detail

Consider

$$\chi_{7,6} = (e_7, f_6),$$

where the smaller sides of the two principal splits are

$$S(e_7) = \{1, 7, 11\}, \qquad S(f_6) = \{2, 7, 24\}.$$

Choose the endpoint v_- on the smaller side of each principal edge; equivalently, the displayed orientation points away from the smaller side. Endpoint atoms are ordered by the edge order above, followed by the residual atoms. The four row atoms are

index	sector	taxon block
0	a	$R_0 = \{1, 7\}$
1	b	$R_1 = \{2, 12, 13, 16, 17, 18\}$
2	b	$R_2 = X \setminus \{1, 2, 7, 11, 12, 13, 16, 17, 18\}$
3	c	$R_3 = \{11\}$

and the four column atoms are

index	sector	taxon block
0	b	$C_0 = \{0, 15\}$
1	a	$C_1 = \{2, 7\}$
2	b	$C_2 = X \setminus \{0, 2, 7, 15, 24\}$
3	c	$C_3 = \{24\}$

Consequently,

$$r = \left(\frac{2}{25}, \frac{6}{25}, \frac{16}{25}, \frac{1}{25}\right), \qquad c = \left(\frac{2}{25}, \frac{2}{25}, \frac{4}{5}, \frac{1}{25}\right),$$

$$(a_{i,k}) = \begin{pmatrix} 0 & \frac{1}{25} & \frac{1}{25} & 0 \\ 0 & \frac{1}{25} & \frac{1}{5} & 0 \\ \frac{2}{25} & 0 & \frac{13}{25} & \frac{1}{25} \\ 0 & 0 & \frac{1}{25} & 0 \end{pmatrix},$$

and

$$(x_{i,k}) = \begin{pmatrix} -\frac{4}{625} & \frac{21}{625} & -\frac{3}{125} & -\frac{2}{625} \\ -\frac{12}{625} & \frac{13}{625} & \frac{1}{125} & -\frac{6}{625} \\ \frac{18}{625} & -\frac{32}{625} & \frac{1}{125} & \frac{9}{625} \\ -\frac{2}{625} & -\frac{2}{625} & \frac{1}{125} & -\frac{1}{625} \end{pmatrix}.$$

The row-index classes used below are

$$D_{\chi_{7,6}} = \{(0,1), (0,2), (1,0), (1,3), (2,0), (2,3)\},$$
$$I_{\chi_{7,6}} = \{(0,3), (1,2), (2,1), (3,0), (3,1), (3,2), (3,3)\}.$$

C.3 One completed square

Take the term of $\mathcal{S}_{1,\chi_{7,6}}$ indexed by

$$(i,j,k) = (1,0,1).$$

In the definition of $\mathcal{S}_{1,\chi_{7,6}}$, the two arguments of this completed square are

$$X := x_{\bar{i},\bar{k}} + x_{i,\bar{k}} = -\frac{23}{625}, \qquad Y := x_{\bar{i},k} + x_{i,k} = \frac{32}{625}.$$

For this term,

$$c_k = \frac{2}{25}, \qquad c_{\bar{k}} = \frac{22}{25}, \qquad c_{k^+} = \frac{3}{25},$$

and

$$a_{i,\bar{k}} = \frac{1}{5}, \qquad a_{j,\bar{k}} = \frac{1}{25}, \qquad a_{i,k} = a_{j,k} = \frac{1}{25}.$$

The four completed-square terms in $\mathcal{P}_{1,0,1}(X,Y)$ are all nonzero and equal, in order, to

$$\frac{49}{1739375000}, \qquad \frac{11}{2343750}, \qquad \frac{81}{976562500}, \qquad \frac{81}{195312500}.$$

Therefore

$$\begin{aligned}\mathcal{P}_{1,0,1}\left(-\frac{23}{625}, \frac{32}{625}\right) &= \frac{49}{1739375000} + \frac{11}{2343750} + \frac{81}{976562500} + \frac{81}{195312500} \\ &= \frac{85107103}{16306640625000}.\end{aligned}$$

C.4 The four families in this chart

Evaluating every permitted target in the same chart gives

family	nonzero summands	exact sum
$\mathcal{S}_{1,\chi_{7,6}}$	16	1947947093/97839843750000
$\mathcal{S}_{2,\chi_{7,6}}$	16	366630403/48919921875000
$\mathcal{S}_{3,\chi_{7,6}}$	20	1912609/354492187500
$\mathcal{S}_{4,\chi_{7,6}}$	18	913296/29541015625

For one explicit Ψ-target term in $\mathcal{S}_{3,\chi_{7,6}}$, take $(i,j,u) = (0,1,0)$. Then

$$X := x_{\bar{i},\bar{u}} + x_{i,\bar{u}} = \frac{3}{625}, \qquad \Psi_{0,1}^{\chi_{7,6}}(0) = \frac{2}{375},$$

so that

$$\frac{1}{4}\Psi_{0,1}^{\chi_{7,6}}(0)X^2 = \frac{1}{4}\cdot\frac{2}{375}\left(\frac{3}{625}\right)^2 = \frac{3}{97656250}.$$

Substitution into equation (A.30) yields

$$\begin{aligned}A_{\chi_{7,6}} &= \frac{1}{4}\frac{1947947093}{97839843750000} + \frac{1}{2}\frac{366630403}{48919921875000} \\ &\quad + \frac{1}{2}\frac{1912609}{354492187500} + \frac{913296}{29541015625} \\ &= \boxed{\frac{1841063809}{43484375000000}}.\end{aligned}$$

C.5 The second square under bridge reversal

The endpoint column atom $C_1 = \{2, 7\}$ corresponds to the edge f_1 adjacent to the principal edge f_6. Hence the ordered endpoint chart is

$$\alpha = (f_6, f_1).$$

Its three blocks are

$$\begin{aligned} K &= \{2, 7\}, \\ H &= \{24\}, \\ L &= \{0, 1, 3, 4, 5, 6, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23\}. \end{aligned}$$

Bridge reversal gives

$$\rho\alpha = (f_1, f_6), \qquad (K, H, L) \longmapsto (L, H, K).$$

The reversed endpoint chart belongs to the principal chart

$$\chi_{7,1} = (e_7, f_1),$$

and not to $\chi_{7,6}$. The global sum in (A.30) therefore contains both orientations:

$$\begin{array}{lll} \chi_{7,6}, & \alpha = (f_6, f_1), & (K, H, L), \\ \chi_{7,1}, & \rho\alpha = (f_1, f_6), & (L, H, K). \end{array}$$

In the reversed chart, the endpoint block L has atom index 0. For the displayed completed-square form, the four square coefficients pair as follows:

second-square term in $\mathcal{P}_{1,0,1}^{\chi_{7,6}}$	coefficient	first-square term in $\mathcal{P}_{1,0,0}^{\chi_{7,1}}$
$(A.22),\ Y^2$	$1/5750$	$(A.23),\ X^2$
$(A.23),\ Y^2$	$33/5000$	$(A.22),\ X^2$
$(A.24),\ Y^2$	$1/2500$	$(A.25),\ X^2$
$(A.25),\ Y^2$	$1/500$	$(A.24),\ X^2$

Thus the sum of the four second-square coefficients in the left column and the sum of their four first-square partners in the right column are both $211/23000$. This equality relates two different principal charts and is precisely the $(A.22) \leftrightarrow (A.23)$ and $(A.24) \leftrightarrow (A.25)$ coefficient matching used by the symbolic proof. Reversal does not identify the subtotals of the two principal charts.

C.6 First computation: the global four-family sum

To display all 484 principal-chart contributions compactly, group the row edge indices into four disjoint ranges. Put

$$D_* := 88349873334030000000000$$

and

$$\begin{aligned} G_1 &= 509632929097785686741, \quad G_2 = 338603181900355216089, \\ G_3 &= 551304736571593597776, \quad G_4 = 697786193522003585794. \end{aligned}$$

The exact grouped sums are

principal row-edge range	number of charts	contribution
$0 \le p \le 4$	110	G_1/D_*
$5 \le p \le 10$	132	G_2/D_*
$11 \le p \le 15$	110	G_3/D_*
$16 \le p \le 21$	132	G_4/D_*

These ranges contain all 484 charts. Consequently,

$$\begin{aligned}\sum_{\chi} A_\chi &= \frac{G_1+G_2+G_3+G_4}{D_*}\\ &= \frac{2097327041091738086400}{88349873334030000000000}\\ &= \boxed{\frac{9273}{390625}}.\end{aligned}$$

Thus

$$\boxed{f_1^{\mathrm{A.30}}(T_1,T_2) = \frac{9273}{390625}}.$$

C.7 Second computation: the defining formula for f_1

For each pair (e,f), retain the notation

$$U_{e,f} = T[\sigma_e,\tau_f], \quad R_{e,f} = T[\sigma_e^+,\tau_f], \quad C_{e,f} = T[\sigma_e,\tau_f^+], \quad B_{e,f} = T[\sigma_e^+,\tau_f^+].$$

Direct exact enumeration of the four defining levels gives

$$\sum_{e\in E_2(T_1)}\sum_{f\in E_2(T_2)} U_{e,f} = \frac{32602}{78125} = \frac{163010}{390625}, \quad \sum_{e\in E_2(T_1)}\sum_{f\in E_2(T_2)} R_{e,f} = \frac{48424}{78125} = \frac{242120}{390625},$$
$$\sum_{e\in E_2(T_1)}\sum_{f\in E_2(T_2)} C_{e,f} = \frac{251002}{390625}, \qquad \sum_{e\in E_2(T_1)}\sum_{f\in E_2(T_2)} B_{e,f} = \frac{371296}{390625}.$$

Therefore

$$\begin{aligned}f_1^{\mathrm{direct}} &= \sum_{e\in E_2(T_1)}\sum_{f\in E_2(T_2)}\left(U_{e,f} - \frac{1}{2}R_{e,f} - \frac{1}{2}C_{e,f} + \frac{1}{4}B_{e,f}\right)\\ &= \frac{163010 - 121060 - 125501 + 92824}{390625}\\ &= \boxed{\frac{9273}{390625}}.\end{aligned}$$

Hence the two algebraically distinct accumulations agree exactly:

$$\boxed{f_1^{\mathrm{direct}}(T_1,T_2) = f_1^{\mathrm{A.30}}(T_1,T_2) = \frac{9273}{390625}}.$$

This agreement holds only after the global sum is taken. For the displayed chart, write $U_{76} := U_{e_7,f_6}$ and analogously for R_{76}, C_{76}, B_{76}. Direct enumeration gives

$$U_{76} = \frac{256}{390625}, \qquad R_{76} = \frac{502}{390625}, \qquad C_{76} = \frac{524}{390625}, \qquad B_{76} = \frac{989}{390625}.$$

Thus its fixed-chart inclusion–exclusion subtotal is

$$U_{76} - \frac{1}{2}R_{76} - \frac{1}{2}C_{76} + \frac{1}{4}B_{76} = -\frac{39}{1562500},$$

whereas

$$A_{\chi_{7,6}} = \frac{1841063809}{43484375000000} > 0.$$

Bridge reversal reallocates individual second-square coefficients across reversed principal charts. Equation (A.30) preserves the global sum, not each individual chart subtotal.

C.8 Third computation: the ordered-quadruple kernel

This calculation does not use the chart decomposition. We use the signed indexed sum $\mathfrak{O}_T$, anti-symmetric side-pair function ϕ_o, and kernel K_T from Section 5. For the two trees in this appendix,

$$|\overrightarrow{A}_2(T_1)| = 52, \qquad |\overrightarrow{A}_2(T_2)| = 50,$$

so $|\mathfrak{O}_{T_1}| = 74$ and $|\mathfrak{O}_{T_2}| = 72$, counting indexed occurrences.

For $\boldsymbol{q} = (x, y, z, w) \in X^4$, put

$$c(x, y, z, w) := \frac{1}{4}K_{T_1}(x, y; z, w)K_{T_2}(x, y; z, w).$$

The symbolic kernel identity (8) gives

$$N^4 f_1(T_1, T_2) = \sum_{(x,y,z,w)\in X^4} c(x, y, z, w).$$

Moreover, Lemma 5.3 proves $|c(x, y, z, w)| \le 1/4$ for every tuple, including every repeated-label pattern. We evaluate this expression separately for all

$$25^4 = 390625$$

ordered quadruples and classify each tuple by the number $d = |\{x, y, z, w\}|$ of distinct labels. The exact totals are

d	number of ordered quadruples	$\sum c(x, y, z, w)$
1	25	0
2	4200	287
3	82800	9098
4	303600	−112
all	390625	9273

For additional control of the repeated-label part, encode the equality pattern of (x, y, z, w) by assigning $0, 1, 2, 3$ in order of first appearance. The only nonzero pattern totals are

canonical equality pattern	exact contribution
$0101, 0110$	287/2 each
$0102, 0112, 0120, 0121$	4549/2 each
0123	−112

All other equality patterns contribute zero. Hence the repeated-label contribution is computed directly as

$$R_{\mathrm{rep}} = 0 + 287 + 9098 = 9385,$$

rather than being inferred by subtracting the distinct-label total from the answer.

For the $d = 4$ part, direct quartet enumeration gives

$$\mathrm{Same}(T_1, T_2) = 4198, \qquad \mathrm{Diff}(T_1, T_2) = 8452.$$

These values sum to $\binom{25}{4} = 12650$. The symbolic 24-ordering count in the proof of Proposition 5.4 gives subtotal 4 for every common displayed quartet and -2 for every different displayed quartet. The program also checks these 24-ordering subtotals separately for every four-subset in this example. Consequently,

$$4\,\mathrm{Same} - 2\,\mathrm{Diff} = 4 \cdot 4198 - 2 \cdot 8452 = -112,$$

which is exactly the $d = 4$ row of the table. Combining the distinct and repeated parts gives

$$\sum_{(x,y,z,w)\in X^4} c(x, y, z, w) = -112 + 9385 = 9273.$$

After normalization by $N^4 = 25^4 = 390625$,

$$f_1^{X^4\text{-enum}}(T_1, T_2) = \frac{1}{25^4} \sum_{(x,y,z,w)\in X^4} c(x, y, z, w) = \frac{9273}{390625}.$$

Therefore all three calculations agree exactly:

$$\boxed{f_1^{\mathrm{A.30}}(T_1, T_2) = f_1^{\mathrm{direct}}(T_1, T_2) = f_1^{X^4\text{-enum}}(T_1, T_2) = \frac{9273}{390625}}.$$

Among the 390625 ordered quadruples, 172324 have nonzero contribution. The largest observed absolute contribution of a single ordered quadruple is $1/4$, attaining the universal bound from Lemma 5.3.

All values in this appendix were recomputed in exact rational arithmetic by the accompanying program `verify_v048_appendix_c_n25_general.py`. The program evaluates the global A.30 chart sum, compares every paired square coefficient under bridge reversal, enumerates X^4, and verifies the stated distinct-count and equality-pattern totals. The A.30, defining-formula, and X^4 calculations use algebraically different formulas on the same deterministically generated tree data. The program also compares the canonical Newick serializations with the displayed strings, but the three calculations do not use independently parsed tree implementations.

Accordingly, these finite calculations provide a three-formula consistency check on shared parsed tree data. They do not replace the row decomposition with one column term fixed, the bridge-reversal argument, or the general same-side block coefficient identity in Appendix A; nor do they replace the full-domain kernel-to-path proof in Section 5.

Disclosure of AI Assistance and Prior Work

Section 5 of the main text and Lemmas A.4 and A.5 were developed with the assistance of GPT-5.6 Sol, and Appendix C was produced entirely by GPT-5.6 Sol; all remaining content was developed jointly by Stefan Grünewald and Libo Huang, including equations (25)–(28) and the final expression in equation (A.30), and this part of the work is also documented in Huang's doctoral dissertation, which was made available through the China National Knowledge Infrastructure (CNKI) in 2018.

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